分析力学

手写讲义合集

陈怡（弗恩曼） · ２０２０

按原文图片顺序整理

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整理说明

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原作者介绍讲稿共 １１５ 页；本文图片并非全套原稿。

原文：美妙无比、诚惶诚恐：首次讲授分析力学课程之旅

整理日期：２０２６-１１-１１

目录

CONTENTS / 01

		手稿	PDF页
一	分析力学引言	01	04
二	达朗贝尔原理	01	04
2．1	约束与自由度	01	04
2．2	广义坐标	02	05
2．3	虚功原理	02	05
	达朗贝尔原理的推导	04	07
	拉格朗日乘子法	06	09
三	拉格朗日力学	06	09
3．1	拉格朗日方程	06	09
3．2	运动积分	11	14
3．3	广义势能	13	16
3．4	小振动	15	18

点击目录条目即可跳转；正文保留原稿图片顺序。

右侧页码按当前 PDF 计算；手稿编号为 01—44。

目录

CONTENTS / 02

		手稿	PDF页
四	哈密顿力学	23	26
4．1	哈密顿正则方程	23	26
4．2	相空间	25	28
4．3	泊松括号	29	32
五	哈密顿原理	32	35
5．1	变分法初步	32	35
5．2	哈密顿原理	37	40
5．3	作用量原理的进一步形式	39	42
六	正则变换与哈密顿 - 雅可比方程	41	44
6．1	正则变换	41	44
6．2	哈密顿 - 雅可比方程	43	46

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2020年2月10日 19:00 分析力学(一)

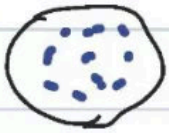
CHAPTER 1 分析力学引言
CHAPTER 2 达朗贝尔原理
CHAPTER 3 拉格朗日力学
CHAPTER 4 哈密顿力学
CHAPTER 5 哈密顿原理
CHAPTER 6 正则变换. 哈密顿-雅可比理论.

CHAPTER 2 达朗贝尔原理.

2.1 约束

2.1.1 约束及其分类

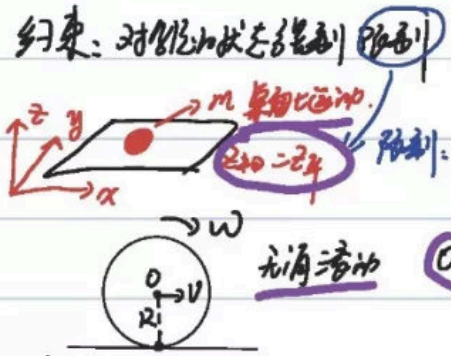
1. 质点: 质点加集合.



2. 质点状态: {位矢: 所有质点位置的集合. 速度: 速度. } 共同决定.

3. 自由: 自由粒子.

约束: 对质点加状态的限制 限制



约束分类:

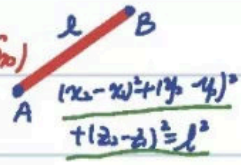
约束方程. $f(\vec{r}_1,\vec{r}_2,\cdots,\vec{r}_n;\dot{\vec{r}}_1,\dot{\vec{r}}_2,\cdots,\dot{\vec{r}}_n,t)=0$

4. 约束分类

按当前表达/初始要素/时间/对称

按当前表达.

几何约束: 仅限制位置. (先决限制速度)



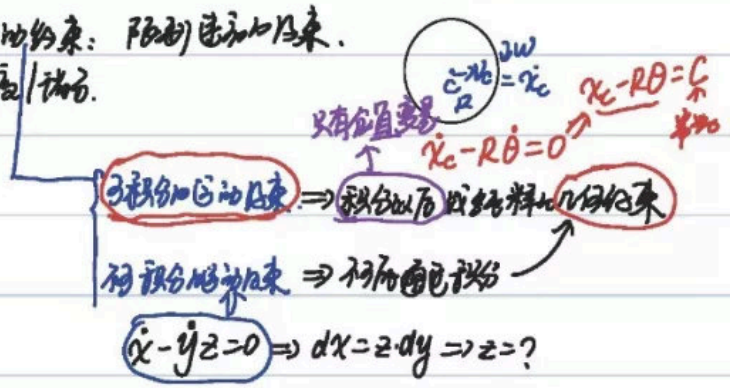
$f(\vec{r}_1,\vec{r}_2,\cdots,\vec{r}_n,t)=0$

$\frac{df}{dt}$ 限制位置 ⇒ 限制速度. $\dot{\vec{r}}_i$

直接导出 $\frac{df}{dt}=\frac{\partial f}{\partial t}+\sum_{i=1}^{n}\left(\frac{\partial f}{\partial x_i}\dot{x}_i+\frac{\partial f}{\partial y_i}\dot{y}_i+\frac{\partial f}{\partial z_i}\dot{z}_i\right)=0.$

运动约束: 限制速度的约束.

速度/动量.

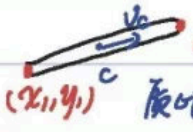


按初始要素分类

完整约束: 几何约束 + 可积分运动约束 研究重点

不完整约束: 不可积分运动约束.

例: 冰刀.



几何约束: $(x_2-x_1)^2+(y_2-y_1)^2=l^2$

运动约束: $\frac{x_2-x_1}{y_2-y_1}=\frac{(\dot{x}_2+\dot{x}_1)/2}{(\dot{y}_2+\dot{y}_1)/2}\Rightarrow\frac{dx_1+dx_2}{dy_1+dy_2}=\frac{x_2-x_1}{y_2-y_1}$ 不可积

显含时间 t. 分类.

{ 完整约束 $f(\vec{r}_1\cdots\vec{r}_n;\dot{\vec{r}}_1\cdots,\vec{r}_n)=0$ 不显含
非完整约束 $f(\vec{r}_1\cdots\vec{r}_n;\dot{\vec{r}}_1\cdots,\vec{r}_n,t)=0$ 显含.

力学特点分类

单侧约束. (可解约束) 比 杆

双侧 · · · (不 · · · ·)

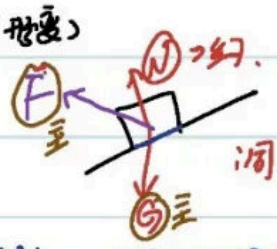
$(x_2-x_1)^2+(y_2-y_1)^2$
$+(z_2-z_1)^2\le l^2$
$=l^2$

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5. 约束存在. 主体为约束力(约束反力), 表现为.

弹力(排斥)

力: { 主动力 = G
约束力.



约束起到 → 阻碍物体. → 本质是提供位移.

2.2 广义坐标.

完整约束. = n个 + 0 部分运动

广义坐标: 唯一确定. 当且仅当 位形 被唯一确定.

单值性. 一组值 一个状态.

个数 = 自由度.

广义坐标为: n个质点组.

3n − 完整约束个数 = s

$q_1, q_2, \cdots, q_s$ 记为 q_α

广义速度: $\dot{q}_1, \dot{q}_2, \cdots, \dot{q}_s$ 记为 $\dot{q}_\alpha$

2.3 虚功原理.

目标: 消除约束力. 利用主动力.

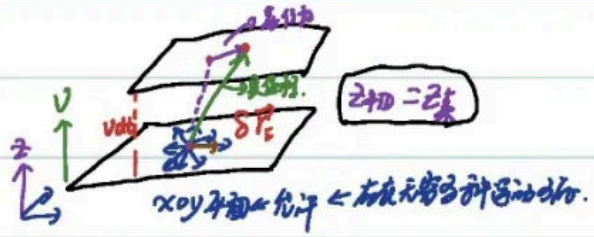
2.3.1 基本形式和各功名称.

1. 虚位移.

实位移. 实际发生位移. t: $\vec{r}_i$ } 实 $d\vec{r}_i$

$t+dt$. $\vec{r}_i+d\vec{r}_i$

虚位移: 表示在某时刻所允许的一切微小的运动情况. $\delta\vec{r}_i$



实位移 = $\vec{v}\cdot dt$ + 虚位移中的一个.

虚实关系. ✓

① 完整约束(约束随时间) $\frac{\partial f}{\partial t}=0$ 实位移是虚位移中一个

② 非完整约束(约束显含时间) $\frac{\partial f}{\partial t}\neq 0$ 实位移和虚位移 ---

记: dx 微分.
δx 变分. ⇒ { ① 几乎等同于"d"微分. ② $\delta t = 0$ (完整约束)

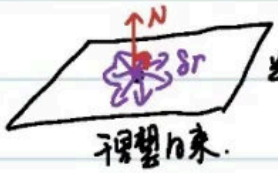
2. 理想约束 N个质点. { 主动力合 $\vec{F}_i$ 约束力合 $\vec{R}_i$

实功. 力·实位移 虚功: 力·虚位移.

$dW = \vec{F}\cdot d\vec{r}$ $\delta W = \sum_{i=1}^{N}(\vec{F}_i+\vec{R}_i)\cdot\delta\vec{r}_i$

$= \sum_{i=1}^{N}\vec{F}_i\delta\vec{r}_i + \sum_{i=1}^{N}\vec{R}_i\cdot\delta\vec{r}_i$

若约束中约束力的虚功之和为零. 理想约束 ← $\sum_{i=1}^{N}\vec{R}_i\cdot\delta\vec{r}_i = 0$



光滑 $\vec{N}\cdot\delta\vec{r}=0$

理想约束.

研究范畴: 完整 ⇒ 几何约束+可积分的运动约束

理想 ⇒ 光滑. 刚性.

质点

① 质点在光滑曲面.

② 无滑动.

③ 刚性连接.

④ 光滑铰链.

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3. 基本形式的动力学原理.

静力学平衡的必要 n个质点.

平衡: $\vec{F}_i + \vec{R}_i = 0$

$\delta W_i = (\vec{F}_i + \vec{R}_i)\cdot\delta\vec{r}_i = 0$

$\delta W = \sum_{i=1}^{n}\vec{F}_i\cdot\delta\vec{r}_i + \sum_{i=1}^{n}\vec{R}_i\cdot\delta\vec{r}_i = 0$

($\sum\vec{F}_i\cdot\delta\vec{r}_i = 0$; 理想约束 $\sum\vec{R}_i\cdot\delta\vec{r}_i = 0$)

虚功原理: 理想约束力的质点系平衡的必要

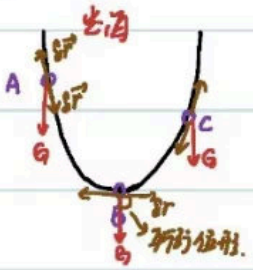
所有主动力的虚功之和为零. $\sum_{i=1}^{n}\vec{F}_i\cdot\delta\vec{r}_i = 0$

必要性: 平衡 $\Rightarrow \sum_{i=1}^{n}\vec{F}_i\cdot\delta\vec{r}_i = 0$

充分性: $\sum_{i=1}^{n}\vec{F}_i\cdot\delta\vec{r}_i = 0 \Rightarrow$ 平衡

$\sum_{i=1}^{n}(\vec{F}_i + \vec{R}_i)\cdot d\vec{r}_i = 0$

$dT = 0 \Rightarrow$ 平衡.



2.3.2 广义坐标下的虚功原理.

如: 自由质点. (无约束) 平衡的必要. 证明.

平衡: $F_x = F_y = F_z = 0$

虚功原理: $F_x\delta x + F_y\delta y + F_z\delta z = 0$ (各项独立)

有约束 如何处理? 各 $\delta\vec{r}_i$ 不独立. 怎么办?

约束方程. $f(\vec{r}_1, \cdots; \vec{r}_i \cdots) = 0$ 有不等.

n个质点. $\vec{r}_i$ $\Downarrow$ $\vec{r}_1, \vec{r}_2, \vec{r}_3$ --- 独立变量

广义坐标. 维数. $(q_1, q_2, \cdots, q_s)$

约束 $s < 3n$

① $\vec{r}_i = \vec{r}_i(q_1, \cdots q_s, t)$

② $\delta\vec{r}_i = \frac{\partial\vec{r}_i}{\partial t}\delta t + \sum_{\alpha=1}^{s}\left(\frac{\partial\vec{r}_i}{\partial q_\alpha}\cdot\delta q_\alpha\right)$ ($\delta t = 0$)

$\delta\vec{r}_i = \sum_{\alpha=1}^{s}\left(\frac{\partial\vec{r}_i}{\partial q_\alpha}\cdot\delta q_\alpha\right)$ $q_1, q_2, \cdots, q_s$ 独立; $\delta q_1, \delta q_2, \cdots, \delta q_s$ 独立

$\sum_{i=1}^{n}\vec{F}_i\cdot\delta\vec{r}_i = 0 \Rightarrow \sum_{i=1}^{n}\vec{F}_i\cdot\left(\sum_{\alpha=1}^{s}\frac{\partial\vec{r}_i}{\partial q_\alpha}\delta q_\alpha\right) = 0$

$\Rightarrow \sum_{\alpha=1}^{s}\left(\sum_{i=1}^{n}\vec{F}_i\cdot\frac{\partial\vec{r}_i}{\partial q_\alpha}\right)\delta q_\alpha = 0$

广义力 $Q_\alpha = \sum_{i=1}^{n}\vec{F}_i\cdot\frac{\partial\vec{r}_i}{\partial q_\alpha}$

$Q_1\delta q_1 + Q_2\delta q_2 + \cdots + Q_s\delta q_s = 0$ (独立 $\Downarrow$)

$Q_1 = 0,\ Q_2 = 0,\ \cdots,\ Q_s = 0$

普适: 满足 $Q_\alpha = \sum_{i=1}^{n}\vec{F}_i\cdot\frac{\partial\vec{r}_i}{\partial q_\alpha} = 0,\ \alpha = 1, 2, \cdots, s$ $\Leftrightarrow$ 各广义力都为零.

$\Uparrow$ 广义坐标下的虚功原理、结论. 各位理

广义力: 并一不是力. 是关于确定.

q_α 长度 Q_α 力

q_α 角度 Q_α 力矩

2.3.3 保守系的平衡原理.

$Q_\alpha = \sum_{i=1}^{n}\vec{F}_i\cdot\frac{\partial\vec{r}_i}{\partial q_\alpha} = 0,\ \alpha = 1, 2, \cdots, s$

$\vec{r}_i = x_i\vec{i} + y_i\vec{j} + z_i\vec{k}$

主动力 $\Rightarrow$ 保守力 $\vec{F}_i = -\nabla_i V \xrightarrow{\text{直角}} -\left(\frac{\partial V}{\partial x_i}\vec{i} + \frac{\partial V}{\partial y_i}\vec{j} + \frac{\partial V}{\partial z_i}\vec{k}\right)$

$Q_\alpha = -\sum_{i=1}^{n}\left(\frac{\partial V}{\partial x_i}\cdot\frac{\partial x_i}{\partial q_\alpha} + \frac{\partial V}{\partial y_i}\cdot\frac{\partial y_i}{\partial q_\alpha} + \frac{\partial V}{\partial z_i}\cdot\frac{\partial z_i}{\partial q_\alpha}\right)$

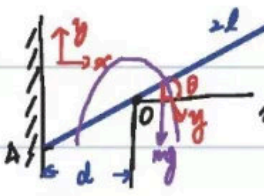
$= -\frac{\partial V}{\partial q_\alpha} = 0$

保守的平衡条件 $\frac{\partial V}{\partial q_\alpha} = 0$ 极值. $\alpha = 1, 2, \cdots, s$

$\delta W = -\delta V$

$\delta V = -\sum_{\alpha=1}^{s}Q_\alpha\delta q_\alpha$ $\Rightarrow Q_\alpha = -\frac{\partial V}{\partial q_\alpha} = 0$

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例1.  求：平衡位形.（A端）

解：①坐标、理想.

②自由度 1 $q=\theta$

③主动力 $\vec{G}=-mg\vec{j}$

$\sum \vec{F}_i\cdot\delta\vec{r}_i=0 \Rightarrow -mg\vec{j}(\delta x\vec{i}+\delta y\vec{j})=0$ ⊗

仅是投影. $-mg\delta y=0 \Rightarrow mg=0$ ×

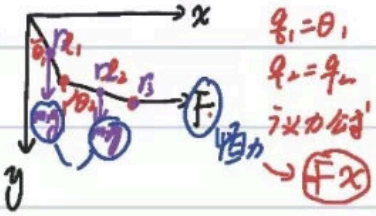
$y=l\sin\theta-d\tan\theta \Rightarrow \delta y=l\cos\theta\delta\theta-\frac{d}{\cos^2\theta}\delta\theta$

$mg\left(l\cos\theta-\frac{d}{\cos^2\theta}\right)\delta\theta=0$

$Q_\theta=0 \Rightarrow \cos^3\theta=d/l \Rightarrow \cos\theta=\sqrt[3]{d/l}$

机构仪 ⇒ 奇点.

$\frac{\partial V}{\partial\theta}=0 \Rightarrow \frac{\partial(mgl\sin\theta-mgd\tan\theta)}{\partial\theta}=0$

例2. 

$q_1=\theta_1$

$q_2=\theta_2$

广义力合计

阻力 → F_x

$Q_\alpha=\sum_{i=1}^{n}\vec{F}_i\cdot\frac{\partial\vec{r}_i}{\partial q_\alpha}=0,\ \alpha=1,2,\cdots,s$

保守系统

$mgV \Rightarrow m_1gy_1$

$mgV \Rightarrow m_2gy_2$

$F \Rightarrow -Fx_3$

$V=--$

$\frac{\partial V}{\partial\theta_1}=0 \Big/ \frac{\partial V}{\partial\theta_2}=0$

$Q_1=m_1\vec{g}\cdot\frac{\partial\vec{r}_1}{\partial\theta_1}+m_2\vec{g}\cdot\frac{\partial\vec{r}_2}{\partial\theta_1}+\vec{F}\cdot\frac{\partial\vec{r}_3}{\partial\theta_1}=0$

$Q_2=m_1\vec{g}\cdot\frac{\partial\vec{r}_1}{\partial\theta_2}+m_2\vec{g}\cdot\frac{\partial\vec{r}_2}{\partial\theta_2}+\vec{F}\cdot\frac{\partial\vec{r}_3}{\partial\theta_2}=0$

2.3.4 达朗贝尔原理.

质点原理：静力学（平衡）

书二：$\vec{F}_i+\vec{R}_i=m_i\vec{a}_i$

达朗贝尔原理：$\vec{F}_i+\vec{R}_i-m_i\vec{a}_i=0$ 惯性力.

动力学问题 ⇓ 三力作用下的平衡问题.

主动力 约束力 达朗贝尔力 ← 假想的力.

惯性力和质点的加速度方向相反.

虚功原理

$\sum_{i=1}^{n}(\vec{F}_i+\vec{R}_i)\cdot\delta\vec{r}_i=0$

↓ 理想约束

$\sum_{i=1}^{n}\vec{F}_i\cdot\delta\vec{r}_i=0$

↓ 广义坐标

$\sum_{\alpha=1}^{s}Q_\alpha\cdot\delta q_\alpha=0$

$Q_\alpha=\sum_{i=1}^{n}\vec{F}_i\cdot\frac{\partial\vec{r}_i}{\partial q_\alpha}=0$

达朗贝尔原理

$\sum_{i=1}^{n}(\vec{F}_i+\vec{R}_i-m_i\vec{a}_i)\cdot\delta\vec{r}_i=0$

↓ 理想约束

$\sum_{i=1}^{n}(\vec{F}_i-m_i\vec{a}_i)\cdot\delta\vec{r}_i=0$

出发点 ↓ 广义坐标

动力学普遍方程 — 解决出发点

$\sum_{\alpha=1}^{s}\square\cdot\delta q_\alpha=0$

$\square=Q_\alpha-\sum_{i=1}^{n}m_i\ddot{\vec{r}}_i\cdot\frac{\partial\vec{r}_i}{\partial q_\alpha}=0$

$\vec{r}_i=\vec{r}_i(q_1,q_2,\cdots,q_s,t)$

$\delta\vec{r}_i=\sum_{\alpha=1}^{s}\frac{\partial\vec{r}_i}{\partial q_\alpha}\cdot\delta q_\alpha$

$\sum_{i=1}^{n}(\vec{F}_i-m_i\ddot{\vec{r}}_i)\cdot\delta\vec{r}_i=\sum_{i=1}^{n}(\vec{F}_i-m_i\ddot{\vec{r}}_i)\cdot\left(\sum_{\alpha=1}^{s}\frac{\partial\vec{r}_i}{\partial q_\alpha}\cdot\delta q_\alpha\right)$

$=\sum_{\alpha}\left(\sum_{i=1}^{n}\vec{F}_i\cdot\frac{\partial\vec{r}_i}{\partial q_\alpha}-\sum_{i=1}^{n}m_i\ddot{\vec{r}}_i\cdot\frac{\partial\vec{r}_i}{\partial q_\alpha}\right)\cdot\delta q_\alpha=0$

$\sum_{i=1}^{n}\vec{F}_i\cdot\frac{\partial\vec{r}_i}{\partial q_\alpha}=Q_\alpha$

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2020年2月11日 19:00 分析力学(二)

CHAPTER 1 分析力学引言 ✓

CHAPTER 2 达朗贝尔原理 ✓

CHAPTER 3 拉格朗日力学

CHAPTER 4 哈密顿力学

CHAPTER 5 哈密顿原理

CHAPTER 6 正则变换. 哈密顿-雅可比理论.

复习.

约束方程: $f(\vec{r}_1, \vec{r}_2, \cdots \vec{r}_n; \vec{r}_1, \vec{r}_2, \cdots, \vec{r}_n) = 0$

自由度: 位形 ⑤

广义坐标: $q_1, q_2 \cdots q_s$, q_α $\alpha = 1, 2, \cdots, s$

广义速度: $\dot{q}_1, \dot{q}_2 \cdots \dot{q}_s$, $\dot{q}_\alpha = \frac{dq_\alpha}{dt}$

主动力:

约束力: ← 约束

基本形式的虚功原理: $\sum_{i=1}^{n} (\vec{F}_i + \vec{R}_i) \cdot \delta \vec{r}_i = 0$

理想 $\sum_{i=1}^{n} \vec{F}_i \cdot \delta \vec{r}_i = 0$

广义坐标的虚功原理: $\sum_{\alpha=1}^{s} Q_\alpha \cdot \delta q_\alpha = 0$

$Q_\alpha = \sum_{i=1}^{n} \vec{F}_i \cdot \frac{\partial \vec{r}_i}{\partial q_\alpha} = 0 \quad \alpha = 1, 2, \cdots, s$

保守体系的虚功原理: $Q_\alpha = -\frac{\partial V}{\partial q_\alpha} = 0, \ \alpha = 1, 2, \cdots, s$

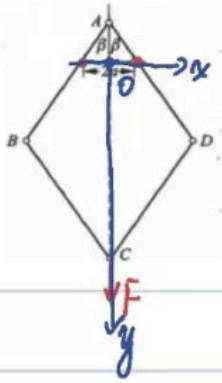
达朗贝尔原理: $\sum_i (\vec{F}_i + \vec{R}_i - m_i \ddot{\vec{r}}_i) \cdot \delta \vec{r}_i = 0$

动力学普遍方程 ↓理想.

$\sum_i (\vec{F}_i - m_i \ddot{\vec{r}}_i) \cdot \delta \vec{r}_i = 0$

设坐标: $\sum_{\alpha=1}^{s} \square \cdot \delta q_\alpha = 0, \ \alpha = 1, 2, \cdots, s$ $\square = Q_\alpha - \sum_i m_i \ddot{\vec{r}}_i \cdot \frac{\partial \vec{r}_i}{\partial q_\alpha} = 0$

长为 l 的四根轻杆，用光滑铰链连成菱形 $ABCD$，如图所示. AB 和 AD 支于相距 $2a$ 且在同一水平线上的两根钉子上，C 点受竖直恒力 F. 求平衡时 A 处半顶角 β.



余弦定理

① 理. 生 ② $\theta = \beta$

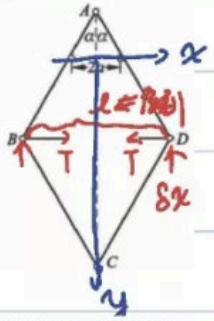
③ $F \cdot \delta y_C = 0$

$y_C = 2l\cos\beta - a\cot\beta$

$\delta y_C = (-2l\sin\beta + a\csc^2\beta) \cdot \delta\beta$

$= 0 \Rightarrow \sin^3\beta = a/2l$

在 B 与 D 间用不可伸长轻绳连接，A 处半顶角为 α，如 BL5.2(1) 图所示. 求绳的张力.

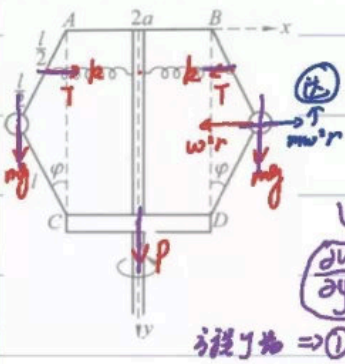


附加约束: $\theta = \alpha$

$F\delta y_C + T_B \delta x_B - T_C \delta x_C = 0$

$T = F\left(\frac{a}{2l}\csc^2\alpha - 1\right)\tan\alpha$

例 图 2-3(a) 是离心调速器的简图. AB 边长 $2a$，固定于转轴. 其余各斜臂的长度都是 l，质量甚轻. 两个小球的质量各为 m. 套管 CD 重 P，套在转轴上，可以上下滑动. 劲度系数为 k 的弹簧两端分别系于两个上臂的中点. 在不转动时，弹簧恰好既不伸长也不缩短. 各铰链都是光滑的. 试求匀速转动的角速度 ω 与张角 φ 的关系.



$q = \varphi$ 惯性力.

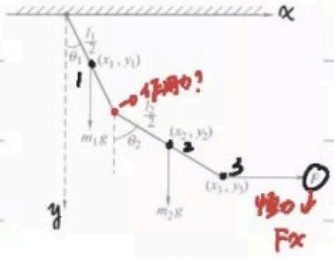
P38.

$2T, 2mg, 2m\omega^2 r$

V = 重力 + 弹性 + 常量.

$\frac{\partial V}{\partial \varphi} = 0 \Rightarrow \omega \sim \varphi$

方程个数 ⇒ ①, 未知 ⇒ ⑫



$\sum_{i=1}^{n} \vec{F}_i \cdot \delta \vec{r}_i = 0$

$q_1 = \theta_1, \ q_2 = \theta_2$

$Q_\alpha = \sum \vec{F}_i \cdot \frac{\partial \vec{r}_i}{\partial q_\alpha}$

$Q_1 = m_1\vec{g} \cdot \frac{\partial \vec{r}_1}{\partial \theta_1} + m_2\vec{g} \cdot \frac{\partial \vec{r}_2}{\partial \theta_2} + \vec{F} \cdot \frac{\partial \vec{r}_2}{\partial \theta_1} = 0$

$Q_2 = m_1\vec{g} \cdot \frac{\partial \vec{r}_1}{\partial \theta_2} + m_2\vec{g} \cdot \frac{\partial \vec{r}_2}{\partial \theta_2} + \vec{F} \cdot \frac{\partial \vec{r}_2}{\partial \theta_2} = 0$

$\vec{r}_1 = \frac{1}{2}l_1(\sin\theta_1 \vec{i} + \cos\theta_1 \vec{j}) \quad \vec{r}_2 = \cdots \quad \vec{r}_3 = \cdots$

弗恩曼

2.3.5 约束力求解（拉格朗日未定乘子法）

广义坐标 ⇔ 约束条件. 约束力 × 外力 ✓.

广义坐标数 ⇒ 自由度数.

S 个广义坐标 = $(S-l)$ 个自由度 + l 个约束条件.

（必须满足的约束条件 / 额外的）

约束条件：$f_j(q_1, q_2, \cdots, q_S, t) = 0$，$j = 1, 2, \cdots, S$

$$\Rightarrow \sum_{\alpha=1}^{S} \frac{\partial f_j}{\partial q_\alpha} \delta q_\alpha + \frac{\partial f_j}{\partial t}\delta t = 0 \Rightarrow \sum_{\alpha=1}^{S} \frac{\partial f_j}{\partial q_\alpha} \delta q_\alpha = 0 \quad ①$$

（$\delta t = 0$；$j = 1, 2, \cdots, l$；l 个方程）

理想约束条件：$\sum_{\alpha=1}^{S} Q_\alpha \delta q_\alpha = 0$，$\alpha = 1, 2, \cdots, S$ ②

（$Q_\alpha = 0$；$\delta q_1, \delta q_2, \cdots \delta q_S$ 相互独立；S > 自由度数；S 个方程）

约束和

① × λ_j + ②

$$\sum_{\alpha=1}^{S} \left(Q_\alpha + \sum_{j=1}^{l} \lambda_j \frac{\partial f_j}{\partial q_\alpha} \right) \delta q_\alpha = 0$$

（合为 S 项）

$$\left(Q_1 + \sum_{j=1}^{l} \lambda_j \frac{\partial f_j}{\partial q_1} \right) \delta q_1 + \cdots + \left(Q_S + \sum_{j=1}^{l} \lambda_j \frac{\partial f_j}{\partial q_S} \right) \delta q_S = 0$$

（$\times S$ 次）

适当特定 $\lambda_1, \cdots, \lambda_l$

希望
$$\begin{cases} Q_1 + \frac{\partial f_1}{\partial q_1}\lambda_1 + \cdots + \frac{\partial f_l}{\partial q_1}\lambda_l = 0 \\ \vdots \\ Q_l + \frac{\partial f_1}{\partial q_l}\lambda_1 + \cdots + \frac{\partial f_l}{\partial q_l}\lambda_l = 0 \end{cases}$$

$q_1, \cdots, q_l \Rightarrow \lambda_1, \lambda_2 \cdots, \lambda_l$.

前面为 $S-l$ 项.

为 0（因为 $\delta q = 0$）

参考 P56. 例 4

CHAPTER 3 拉格朗日力学

3.1 拉格朗日方程

问题：矢位移 ⇒ 不便用牛顿. 矢位移各自相关. 不便.

目标：~~矢位移~~ 相互 ⇒ 广义坐标描述力学规律的等式.

3.1.1 拉格朗日关系

关系 1. $\frac{\partial \dot{\vec{r}}_i}{\partial \dot{q}_\alpha} = \frac{\partial \vec{r}_i}{\partial q_\alpha}$

关系 2. $\frac{d}{dt}\left(\frac{\partial \vec{r}_i}{\partial q_\alpha}\right) = \frac{\partial \dot{\vec{r}}_i}{\partial q_\alpha}$

证明 1. $\vec{r}_i = \vec{r}_i(q_1, \cdots, q_S, t)$ $i = 1, 2, \cdots, n$

$$\dot{\vec{r}}_i = \frac{\partial \vec{r}_i}{\partial t} + \sum_{\beta=1}^{S} \left(\frac{\partial \vec{r}_i}{\partial q_\beta} \cdot \dot{q}_\beta \right)$$

$$\frac{\partial \dot{\vec{r}}_i}{\partial \dot{q}_\alpha} = \frac{\partial}{\partial \dot{q}_\alpha}\left(\frac{\partial \vec{r}_i}{\partial t} + \sum_{\beta=1}^{S} \left(\frac{\partial \vec{r}_i}{\partial q_\beta} \cdot \dot{q}_\beta \right) \right)$$

（$\frac{\partial \vec{r}_i}{\partial t} = 0$；只与 $q_1, \cdots q_S$ 线性，与 $\dot{q}_\alpha$ 无关.）

$$= \frac{\partial}{\partial \dot{q}_\alpha}\left(\sum_{\beta=1}^{S} \left(\frac{\partial \vec{r}_i}{\partial q_\beta} \cdot \dot{q}_\beta \right) \right) = \frac{\partial \vec{r}_i}{\partial q_\alpha}$$

证明 2. $\vec{r}_i = \vec{r}_i(q_1, \cdots, q_S, t)$ $i = 1, 2, \cdots, n$

$$\dot{\vec{r}}_i = \frac{\partial \vec{r}_i}{\partial t} + \sum_{\beta=1}^{S} \left(\frac{\partial \vec{r}_i}{\partial q_\beta} \cdot \dot{q}_\beta \right) \Big| \frac{\partial \dot{\vec{r}}_i}{\partial \dot{q}_\alpha} = \frac{\partial \vec{r}_i}{\partial q_\alpha}(q_1 \cdots, q_S, t)$$

$$\frac{d}{dt}\left(\frac{\partial \vec{r}_i}{\partial q_\alpha}\right) = \frac{\partial}{\partial t}\left(\frac{\partial \vec{r}_i}{\partial q_\alpha}\right) + \sum_{\beta=1}^{S} \left[\frac{\partial}{\partial q_\beta}\left(\frac{\partial \vec{r}_i}{\partial q_\alpha}\right) \dot{q}_\beta \right]$$

（↓ 特定值）

$$= \frac{\partial}{\partial q_\alpha}\left(\frac{\partial \vec{r}_i}{\partial t} + \sum_{\beta=1}^{S} \frac{\partial \vec{r}_i}{\partial q_\beta} \cdot \dot{q}_\beta \right) = \frac{\partial \dot{\vec{r}}_i}{\partial q_\alpha}$$

$$\frac{d}{dt}\left(\frac{\partial \vec{r}_i}{\partial q_\alpha}\right) = \frac{\partial \dot{\vec{r}}_i}{\partial q_\alpha} = \frac{\partial}{\partial q_\alpha}\left(\frac{d\vec{r}_i}{dt}\right)$$

弗恩曼

3.1.2 基本形式拉格朗日方程.

达朗贝尔原理 $\sum_{i=1}^{n}(\vec{F}_i - m_i\ddot{\vec{r}}_i)\cdot\delta\vec{r}_i = 0$

$$\Rightarrow \sum_{\alpha=1}^{s}\left(\sum_{i=1}^{n}\vec{F}_i\cdot\frac{\partial\vec{r}_i}{\partial q_\alpha} - \sum_{i=1}^{n}m_i\ddot{\vec{r}}_i\frac{\partial\vec{r}_i}{\partial q_\alpha}\right)\delta q_\alpha = 0$$

$$\Downarrow \quad Q_\alpha - \sum_{i=1}^{n}m_i\ddot{\vec{r}}_i\frac{\partial\vec{r}_i}{\partial q_\alpha} = 0,\ \alpha = 1,2,\cdots,s$$

$$Q_\alpha - \sum_{i=1}^{n}m_i\ddot{\vec{r}}_i\frac{\partial\vec{r}_i}{\partial q_\alpha} = 0,\ \alpha = 1,2,\cdots,s$$

$$\Rightarrow Q_\alpha - \frac{d}{dt}\left(\sum_{i=1}^{n}m_i\dot{\vec{r}}_i\frac{\partial\vec{r}_i}{\partial q_\alpha}\right) + \sum_{i=1}^{n}\left[m_i\dot{\vec{r}}_i\frac{d}{dt}\left(\frac{\partial\vec{r}_i}{\partial q_\alpha}\right)\right] = 0$$

$$\Rightarrow Q_\alpha - \frac{d}{dt}\left(\sum_{i=1}^{n}m_i\dot{\vec{r}}_i\frac{\partial\dot{\vec{r}}_i}{\partial\dot{q}_\alpha}\right) + \sum_{i=1}^{n}\left[m_i\dot{\vec{r}}_i\cdot\frac{\partial\dot{\vec{r}}_i}{\partial q_\alpha}\right] = 0$$

$$\Rightarrow Q_\alpha - \frac{d}{dt}\left(\frac{\partial\sum\frac{1}{2}m_i\dot{\vec{r}}_i\cdot\dot{\vec{r}}_i}{\partial\dot{q}_\alpha}\right) + \frac{\partial\left(\sum_{i=1}^{n}\frac{1}{2}m_i\dot{\vec{r}}_i\cdot\dot{\vec{r}}_i\right)}{\partial q_\alpha} = 0$$

$$T = \sum\frac{1}{2}m_i\dot{\vec{r}}_i\cdot\dot{\vec{r}}_i$$

$$\Rightarrow Q_\alpha - \frac{d}{dt}\left(\frac{\partial T}{\partial\dot{q}_\alpha}\right) + \frac{\partial T}{\partial q_\alpha} = 0,\ \alpha = 1,2,\cdots,s$$

$$\Rightarrow \boxed{Q_\alpha = \frac{d}{dt}\left(\frac{\partial T}{\partial\dot{q}_\alpha}\right) - \frac{\partial T}{\partial q_\alpha},\ \alpha = 1,2,\cdots,s}$$

基本形式拉格朗日方程.

注意事项

① 约束必为. 自由度 s 须为. 方程为 s 个.

② 标量方程

③ 一劳永逸 矢量方程

小结: 利用 $\boxed{Q_\alpha = \frac{d}{dt}\left(\frac{\partial T}{\partial\dot{q}_\alpha}\right) - \frac{\partial T}{\partial q_\alpha},\ \alpha = 1,2,\cdots,s}$ 求. 诸各标下加相体加速度表达式.

例: 质点 m. ① 自由度 3. $q_1 = r$, $q_2 = \theta$, $q_3 = \varphi$.

② $\vec{F} = F_r\vec{e}_r + F_\theta\vec{e}_\theta + F_\varphi\vec{e}_\varphi$

$$\vec{v} = \dot{r}\vec{e}_r + r\dot{\theta}\vec{e}_\theta + r\sin\theta\dot{\varphi}\vec{e}_\varphi$$

$$d\vec{r} = dr\vec{e}_r + rd\theta\vec{e}_\theta + r\sin\theta d\varphi\vec{e}_\varphi$$

③ $Q_\alpha = \vec{F}\cdot\frac{\partial\vec{r}}{\partial q_\alpha}$ 定义

$$\delta W = \vec{F}\cdot\delta\vec{r} = (F_r\vec{e}_r + F_\theta\vec{e}_\theta + F_\varphi\vec{e}_\varphi)\cdot(\delta r\vec{e}_r + r\delta\theta\vec{e}_\theta + r\sin\theta\delta\varphi\vec{e}_\varphi)$$

$$= F_r\delta r + F_\theta r\delta\theta + F_\varphi r\sin\theta\delta\varphi$$

$$\Downarrow Q_r \qquad \Downarrow Q_\theta \qquad \Downarrow Q_\varphi$$

$$T = \frac{1}{2}m(\dot{r}^2 + r^2\dot{\theta}^2 + r^2\sin^2\theta\dot{\varphi}^2)$$

$$Q_\varphi = F_\varphi r\sin\theta = \frac{d}{dt}\left(\frac{\partial T}{\partial\dot{\varphi}}\right) - \frac{\partial T}{\partial\varphi} = \frac{d}{dt}(mr^2\sin^2\theta\dot{\varphi}) - 0$$

$$= 2mr\cdot\dot{r}\sin^2\theta\dot{\varphi} + mr^2\cdot 2\sin\theta\cos\theta\cdot\dot{\theta}\dot{\varphi} + mr^2\sin^2\theta\cdot\ddot{\varphi}$$

$$a_\varphi = F_\varphi/m = 2\dot{r}\sin\theta\dot{\varphi} + 2r\cos\theta\cdot\dot{\theta}\dot{\varphi} + r\sin\theta\cdot\ddot{\varphi}$$

$$Q_\theta = rF_\theta = \frac{d}{dt}\left(\frac{\partial T}{\partial\dot{\theta}}\right) - \frac{\partial T}{\partial\theta} = \frac{d}{dt}(mr^2\dot{\theta}) - (mr^2\sin\theta\cos\theta\dot{\varphi}^2)$$

$$= m\cdot 2r\cdot\dot{r}\dot{\theta} + mr^2\ddot{\theta} - mr^2\sin\theta\cos\theta\dot{\varphi}^2$$

$$a_\theta = F_\theta/m = 2\dot{r}\dot{\theta} + r\ddot{\theta} - r\sin\theta\cos\theta\dot{\varphi}^2$$

$$Q_r = F_r = \frac{d}{dt}\left(\frac{\partial T}{\partial\dot{r}}\right) - \frac{\partial T}{\partial r} = \frac{d}{dt}(m\dot{r}) - (mr\dot{\theta}^2 + mr\sin^2\theta\dot{\varphi}^2)$$

$$= m\ddot{r} - mr\dot{\theta}^2 - mr\sin^2\theta\dot{\varphi}^2$$

$$a_r = F_r/m = \ddot{r} - r\dot{\theta}^2 - r\sin^2\theta\dot{\varphi}^2$$

弗恩曼

3.1.3 保守系的拉格朗日方程（拉氏方程）

$$Q_\alpha=\frac{d}{dt}\left(\frac{\partial T}{\partial \dot q_\alpha}\right)-\frac{\partial T}{\partial q_\alpha},\quad \alpha=1,2,\cdots,s$$

基本形式的拉格朗日方程

主动力、保守力 $\Rightarrow q_1,q_2,\cdots,q_s$

$$\underline{V}=(\vec r_1,\vec r_2,\cdots,\vec r_n,t)$$

$$\vec F_i=-\nabla_i V$$

$$Q_\alpha=\sum_{i=1}^n \vec F_i\cdot\frac{\partial \vec r_i}{\partial q_\alpha}=-\sum_{i=1}^n \nabla_i V\cdot\frac{\partial \vec r_i}{\partial q_\alpha}=-\frac{\partial V}{\partial q_\alpha}\quad(\alpha=1,2,\cdots,s)$$

$$-\frac{\partial V}{\partial q_\alpha}=\frac{d}{dt}\left(\frac{\partial T}{\partial \dot q_\alpha}\right)-\frac{\partial T}{\partial q_\alpha}$$

$$\Rightarrow \frac{d}{dt}\left(\frac{\partial (T-V)}{\partial \dot q_\alpha}\right)-\frac{\partial (T-V)}{\partial q_\alpha}=0,\quad \alpha=1,2,\cdots,s$$

由于 $\frac{\partial V}{\partial \dot q_\alpha}=0$ 令 $L=T-V$ 拉格朗日函数

$L=L(q,\dot q,t)$ 拉氏量

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot q_\alpha}\right)-\frac{\partial L}{\partial q_\alpha}=0,\quad \alpha=1,2,\cdots,s$$ 拉格朗日方程

$$L=L(q_1,q_2,\cdots,q_s,\dot q_1,\dot q_2,\cdots,\dot q_s,t)=L(q,\dot q,t)$$

注：① 能拉成的为不唯一 $L'=L+C$ / $L\cdot C$ ✓

不同的拉氏函数 $\Rightarrow$ 相同的动力学方程

② 拉氏量不唯一 不可测量 $L=T-V$

③ n个质点 力学系统 $\begin{cases}\text{平衡：}3n\text{个}\\ \text{拉氏：}3n-k\text{个}\end{cases}$ 方程

约束越多 自由度越少

④ 标量 矢量方程 物理学范畴

平衡 ⑪ 矢量

⑤ 拉氏方程 各阶加全 + 广义坐标

⑥ 广义速度 $\dot q_\alpha$，

广义动量 ~~$m\dot q_\alpha$~~ $p_\alpha=\frac{\partial L}{\partial \dot q_\alpha}$ $(\alpha=1,2,\cdots,s)$

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot q_\alpha}\right)-\frac{\partial L}{\partial q_\alpha}=0\Rightarrow \dot p_\alpha=\frac{\partial L}{\partial q_\alpha}$$

广义动量 $p_\alpha=\frac{\partial L}{\partial \dot q_\alpha}$

本质：$q\sim[L]$ $[p_\alpha]\sim[\text{动量}]$ $[\dot p_\alpha]\sim$ 力

$q\sim[\theta]$ $[p_\alpha]\sim[\text{角动量}]$ $[\dot p_\alpha]\sim$ 力矩

⑦ 力学成为分析的力学. → 第零定律

⑧ 总力 = 保守力 + 非保守力 + 约束力

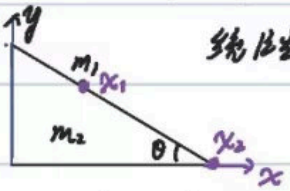
$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot q_\alpha}\right)-\frac{\partial L}{\partial q_\alpha}=Q_\alpha(\text{非保守力})$$

⑨ $L=T-V\Rightarrow$ L分离组成 $\Rightarrow \frac{d}{dt}\left(\frac{\partial L}{\partial \dot q_\alpha}\right)-\frac{\partial L}{\partial q_\alpha}=0$

L先算出来，$q_\alpha,\dot q_\alpha$ 独立坐标，独立的广义坐标

⑩ 不同的组合有不同的拉氏函数.

新 $\Rightarrow$ "L" $\begin{cases}\text{力保守：}L=T-V\\ \text{第四类：}L=T-(q\psi-q\vec v\cdot\vec A)\end{cases}$



统一出发点 求质点的加速度 斜面加速度.

① 主动力 m_1g，m_2g 保守力

解：① 完整、理想 ② 自由度2. $q_1=x_1$，$q_2=x_2$

质点 (x_1,y_1) 斜面 x_2 $y_1=(x_2-x_1)\cdot\tan\theta$

$$T=\frac{1}{2}m_1(\dot x_1^2+\dot y_1^2)+\frac{1}{2}m_2\dot x_2^2=\frac{1}{2}m_1(\dot x_1^2+(\dot x_2-\dot x_1)^2\tan^2\theta)+\frac{1}{2}m_2\dot x_2^2$$

$$V=m_1g(x_2-x_1)\tan\theta+m_2g\cdot 0$$

$$L=T-V=\frac{1}{2}m_1(\dot x_1^2+(\dot x_2-\dot x_1)^2\tan^2\theta)+\frac{1}{2}m_2\dot x_2^2-m_1g(x_2-x_1)\tan\theta$$

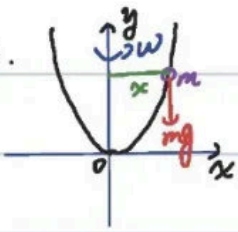
保守组 $\frac{d}{dt}\left(\frac{\partial L}{\partial \dot q_\alpha}\right)=\frac{\partial L}{\partial q_\alpha}$ $\alpha=1,2$

① $q_1=x_1$ $\frac{d}{dt}(m_1\dot x_1-m_1(\dot x_2-\dot x_1)\tan^2\theta)=m_1g\tan\theta$

$(\ddot x_1)\Rightarrow$ $m_1\ddot x_1-m_1(\ddot x_2-\ddot x_1)\tan^2\theta=m_1g\tan\theta$ ①

② $q_2=x_2$ $\frac{d}{dt}(m_1(\dot x_2-\dot x_1)\tan^2\theta+m_2\dot x_2)=-m_1g\tan\theta$

$(\ddot x_2)\Rightarrow$ $m_1(\ddot x_2-\ddot x_1)\tan^2\theta+m_2\ddot x_2=-m_1g\tan\theta$

例2. 

抛物线 $x^2=4ay$ 光滑

求：x 随时间变化规律

解：自由度 1. $q=x$, $y=\frac{x^2}{4a}$.

主动力 mg.

$$T=\frac{1}{2}m(\dot{x}^2+\dot{y}^2+w^2x^2)=\frac{1}{2}m(\dot{x}^2+(\frac{x}{2a}\cdot\dot{x})^2+w^2x^2)$$

$$V=mgy=mg\frac{x^2}{4a}$$

$$L=T-V=\frac{1}{2}m(1+\frac{x^2}{4a^2})\dot{x}^2+w^2x^2)-mg\frac{x^2}{4a}$$

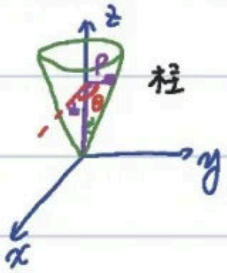
$$\frac{d}{dt}(\frac{\partial L}{\partial\dot{x}})=\frac{\partial L}{\partial x}$$

$$\frac{d}{dt}(m(1+\frac{x^2}{4a^2})\dot{x})=(m\cdot\frac{x}{4a^2}\cdot\dot{x}^2+mw^2x)-\frac{mg}{2a}x$$

$$\Rightarrow m\ddot{x}+\frac{m}{4a^2}(\ddot{x}\cdot x^2+2x\cdot\dot{x}\cdot\dot{x})=(m\cdot\frac{x}{4a^2}\cdot\dot{x}^2+mw^2x)-\frac{mg}{2a}x$$

$$\Rightarrow (1+\frac{x^2}{4a^2})\ddot{x}+\frac{x}{4a^2}\dot{x}^2+(\frac{g}{2a}-w^2)x=0$$

例3. 质点在圆锥面运动. 求其 Lagrange 方程. 光滑



圆锥

解：① 坐标与约束

② 自由度 2. $q_1=\rho$, $q_2=\theta$

约束 $z=\rho\cdot\cot\alpha$

$\dot{z}=\dot{\rho}\cdot\cot\alpha$

③ $$T=\frac{1}{2}m(\dot{\rho}^2+(\rho\dot{\theta})^2+\dot{z}^2)=\frac{1}{2}m(\dot{\rho}^2+\rho^2\dot{\theta}^2+\dot{\rho}^2\cot^2\alpha)$$

$$V=mgz=mg\rho\cot\alpha$$

$$L=T-V=\frac{1}{2}m(\dot{\rho}^2+\rho^2\dot{\theta}^2+\dot{\rho}^2\cot^2\alpha)-mg\rho\cot\alpha$$

④ $$\frac{d}{dt}(\frac{\partial L}{\partial\dot{q}_k})=\frac{\partial L}{\partial q_k}$$

① $$\frac{d}{dt}(\frac{\partial L}{\partial\dot{\rho}})=\frac{\partial L}{\partial\rho}\Rightarrow\frac{d}{dt}(m\dot{\rho}+m\dot{\rho}\cot^2\alpha)=m\rho\dot{\theta}^2-mg\cot\alpha$$

$$m(\ddot{\rho}+\ddot{\rho}\cot^2\alpha)=m(\rho\dot{\theta}^2-g\cot\alpha)$$

$$\Rightarrow (1+\cot^2\alpha)\ddot{\rho}-\rho\dot{\theta}^2+g\cot\alpha=0$$

② $$\frac{d}{dt}(\frac{\partial L}{\partial\dot{\theta}})=\frac{\partial L}{\partial\theta}\Rightarrow\frac{d}{dt}(m\rho^2\dot{\theta})=0$$

$$\rho^2\ddot{\theta}+2\rho\cdot\dot{\rho}\dot{\theta}=0\Rightarrow\rho\ddot{\theta}+2\dot{\rho}\dot{\theta}=0$$

机械 查路 计算机

2020年2月12日 19:00 分析力学(三)

CHAPTER 3 拉格朗日力学

3.1 拉格朗日方程

3.1.1 拉格朗日关系: ① $\frac{\partial \vec{r}_i}{\partial \dot{q}_\alpha} = \frac{\partial \vec{r}_i}{\partial q_\alpha}$

② $\frac{d}{dt}\left(\frac{\partial \vec{r}_i}{\partial \dot{q}_\alpha}\right) = \frac{\partial \vec{r}_i}{\partial q_\alpha}$

3.1.2 基本形式的拉格朗日方程

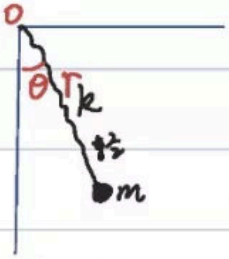
广义力 $Q_\alpha = \frac{d}{dt}\left(\frac{\partial T}{\partial \dot{q}_\alpha}\right) - \frac{\partial T}{\partial q_\alpha}$ $(\alpha = 1, 2, \cdots, s)$

3.1.3 (保守系统的拉格朗日方程)

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}_\alpha}\right) - \frac{\partial L}{\partial q_\alpha} = 0, \quad \alpha = 1, 2, \cdots, s$$

拉氏量 $L = T - V$ 广义动量 $p_\alpha = \frac{\partial L}{\partial \dot{q}_\alpha}$

例: 质点自动力学方程. k, 静长 l, m.



解: ① 完. 理

② $q_1 = r$ $q_2 = \theta$ $s = 2$

③ $T = \frac{1}{2}m(\dot{r}^2 + r^2\dot{\theta}^2)$

$V = -mgr\cos\theta + \frac{1}{2}k(r-l_0)^2$

$mg = k(l-l_0)$

$\Rightarrow l_0 = l - \frac{mg}{k}$

$L = T - V = \frac{1}{2}m(\dot{r}^2 + r^2\dot{\theta}^2) + mgr\cos\theta - \frac{1}{2}k(r-l_0)^2$

$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{r}}\right) = \frac{\partial L}{\partial r} \Rightarrow \frac{d}{dt}(m\dot{r}) = mr\dot{\theta}^2 + mg\cos\theta - k(r-l_0)$

$$m\ddot{r} - mr\dot{\theta}^2 - mg\cos\theta + k(r-l_0) = 0$$

$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{\theta}}\right) = \frac{\partial L}{\partial \theta} \Rightarrow \frac{d}{dt}(mr^2\dot{\theta}) = -mgr\sin\theta$

$$mr^2\ddot{\theta} + 2mr\dot{r}\dot{\theta} = -mgr\sin\theta$$

$$m\ddot{r} - mr\dot{\theta}^2 - mg\cos\theta + k(r-l_0) = 0 \quad ①$$

非线性方程.

$$mr^2\ddot{\theta} + 2mr\dot{r}\dot{\theta} + mgr\sin\theta = 0 \quad ②$$

作小振动近似. $\sin\theta \approx \theta$ $\cos\theta \approx 1$

$$\left.\begin{aligned} &m\ddot{r} - mr\dot{\theta}^2 - mg + k(r-l_0) = 0 \\ &r^2\ddot{\theta} + k\dot{r}\dot{\theta} + gr\theta = 0 \end{aligned}\right\} \Rightarrow$$

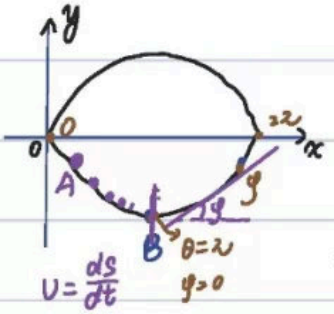
设 ρ 为相对平衡位置的一阶小量. $\rho = \frac{r-l}{l} \Rightarrow r = l(\rho+1)$

$(\rho+1)l\ddot{\theta} + \dot{\rho}l\dot{\theta} + g\theta = 0 \Rightarrow \dot{\rho}\cdot\dot{\theta}$ 二阶小量.

$\ddot{r} - r\dot{\theta}^2 - g + \frac{k}{m}(r-l_0) = 0$ $\rho \ll 1$

$$\left\{\begin{aligned} &\ddot{\theta} + \frac{g}{l}\theta = 0 \\ &\ddot{\rho} + \frac{k}{m}\rho = 0 \end{aligned}\right. \Rightarrow$$

例: 半圆弦线. $\Rightarrow$ 摆线



$U = \frac{ds}{dt}$

无论 A 在何处 $t_{AB} = C$.

上: $\begin{cases} x = a(\theta - \sin\theta) \\ y = a(1-\cos\theta) \end{cases}$

上: $\begin{cases} x = a(\theta - \sin\theta) \\ y = -a(1-\cos\theta) \end{cases}$

$ds = \sqrt{(dx)^2 + (dy)^2} = \sqrt{a^2(1-\cos\theta)^2 + a^2(\sin\theta)^2}\cdot d\theta$

$= ad\theta\cdot\sqrt{2-2\cos\theta} = \sqrt{2}a\,d\theta\cdot\sqrt{1-(1-2\sin^2\frac{\theta}{2})}$

$ds = 2a\sin\frac{\theta}{2}d\theta = 4a\sin\frac{\theta}{2}d\frac{\theta}{2}$

$\tan\varphi = \frac{dy}{dx} = \frac{-\sin\theta}{1-\cos\theta} = -\frac{2\sin\frac{\theta}{2}\cos\frac{\theta}{2}}{2\sin^2\frac{\theta}{2}} = -\cot\frac{\theta}{2}$

$\tan\varphi = -\cot\frac{\theta}{2} \Rightarrow \tan\frac{\theta}{2} = -\cot\varphi = \cot(-\varphi) = \tan(\frac{\pi}{2} - \varphi)$

$\frac{\theta}{2} = \frac{\pi}{2} - \varphi$ 代替.

$ds = 4a\cdot\sin(\frac{\pi}{2} - \varphi)(-d\varphi) = -4a\cos\varphi\,d\varphi$

$S = \int_0^\varphi -4a\cos\varphi\,d\varphi = 4a\sin\varphi\Big|_0^\varphi = 4a\sin\varphi$

弗恩曼



$T=\frac{1}{2}m\dot{s}^2$

$V=mgh=mg\cdot\frac{s^2}{8a}$

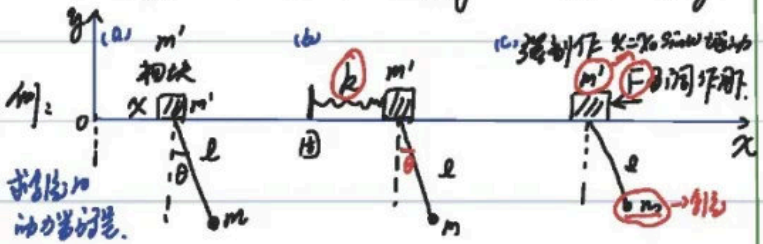
$\tan\varphi=\frac{dy}{dx}=\frac{dh}{dx}\Rightarrow\sin\varphi=\frac{dh}{ds}=\frac{s}{4a}\Rightarrow\int_0^h dh=\int_0^s\frac{1}{4a}s\,ds\Rightarrow h=\frac{s^2}{8a}$

$L=\frac{1}{2}m\dot{s}^2-mg\cdot\frac{s^2}{8a}$

$\frac{d}{dt}\left(\frac{\partial L}{\partial\dot{s}}\right)=\frac{\partial L}{\partial s}\Rightarrow\frac{d}{dt}(m\dot{s})=-\frac{mg}{4a}s\Rightarrow m\ddot{s}=-\frac{mg}{4a}s$

$\Rightarrow\ddot{s}+\frac{g}{4a}s=0\Rightarrow T=4\pi\sqrt{\frac{a}{g}}\Rightarrow t_{AB}=\pi\sqrt{\frac{a}{g}}$

例2



m'

(a) $m'+m$ $q_1=x\to\dot{x}$

$q_2=\theta\to m(\dot{x}+l\cos\theta\cdot\dot{\theta},\ -l\sin\theta\cdot\dot{\theta})\Rightarrow(\dot{x}+l\cos\theta\cdot\dot{\theta},\ l\sin\theta\cdot\dot{\theta})$

$L=T-V=\frac{1}{2}m'\dot{x}^2+\frac{1}{2}m\left[(\dot{x}+l\cos\theta\cdot\dot{\theta})^2+(l\sin\theta\cdot\dot{\theta})^2\right]-(-mgl\cos\theta)$

$L=\frac{1}{2}(m+m')\dot{x}^2+\frac{1}{2}ml^2\dot{\theta}^2+ml\cos\theta\cdot\dot{x}\dot{\theta}+mgl\cos\theta$

$\frac{d}{dt}\left(\frac{\partial L}{\partial\dot{x}}\right)=\frac{\partial L}{\partial x}\Rightarrow\frac{d}{dt}\left((m+m')\dot{x}+ml\cos\theta\cdot\dot{\theta}\right)=0$

$\Rightarrow(m+m')\ddot{x}+ml\left(-\sin\theta\cdot\dot{\theta}\cdot\dot{\theta}+\cos\theta\cdot\ddot{\theta}\right)=0$

$\frac{d}{dt}\left(\frac{\partial L}{\partial\dot{\theta}}\right)=\frac{\partial L}{\partial\theta}\Rightarrow\frac{d}{dt}\left(ml^2\dot{\theta}+ml\cos\theta\cdot\dot{x}\right)=-ml\sin\theta\dot{x}\dot{\theta}-mgl\sin\theta$

$\Rightarrow ml^2\ddot{\theta}+ml\left(-\sin\theta\cdot\dot{\theta}\dot{x}+\cos\theta\ddot{x}\right)=-ml\sin\theta\dot{x}\dot{\theta}-mgl\sin\theta$

$l\ddot{\theta}+\ddot{x}\cos\theta+g\sin\theta=0$

$-\frac{1}{2}k(x-x_0)^2$

(b) $L=\frac{1}{2}(m+m')\dot{x}^2+\frac{1}{2}ml^2\dot{\theta}^2+ml\cos\theta\cdot\dot{x}\dot{\theta}+mgl\cos\theta$

(c) $L=$

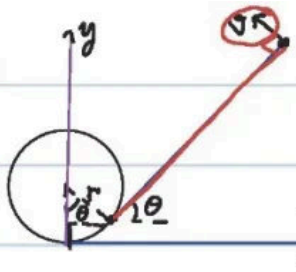
$L=T-V=\frac{1}{2}m\left[(\dot{x}+l\cos\theta\cdot\dot{\theta})^2+(l\sin\theta\cdot\dot{\theta})^2\right]-(-mgl\cos\theta)$

$x_0\sin\omega t$

自由度1. $q=\theta$

$\frac{d}{dt}\left(\frac{\partial L}{\partial\dot{\theta}}\right)=\frac{\partial L}{\partial\theta}$

例：



出滑

(小球 initial 为起始)

自由度1. $q=\theta$

$x=r\sin\theta+(l-r\theta)\cos\theta\Rightarrow\dot{x}=-(l-r\theta)\sin\theta\cdot\dot{\theta}$

$y=r(1-\cos\theta)+(l-r\theta)\sin\theta\Rightarrow\dot{y}=(l-r\theta)\cos\theta\cdot\dot{\theta}$

$L=\frac{1}{2}m(\dot{x}^2+\dot{y}^2)-0=\frac{1}{2}m(l-r\theta)^2\cdot\dot{\theta}^2$

$\frac{d}{dt}\left(\frac{\partial L}{\partial\dot{\theta}}\right)=\frac{\partial L}{\partial\theta}\Rightarrow\frac{d}{dt}\left(m(l-r\theta)^2\dot{\theta}\right)=m(l-r\theta)\cdot(-r)\dot{\theta}^2$

$\ddot{\theta}(l-r\theta)-r\dot{\theta}^2=0\Rightarrow\frac{d}{dt}\left[(l-r\theta)\cdot\dot{\theta}\right]=0$

$(l-r\theta)\cdot\dot{\theta}=v_0\Rightarrow\theta\sim t$ （C, v_0）

3.2 运动积分.

守恒律

积分 / 对称

积分

解题方法：$\vec{F}=m\ddot{\vec{r}}\xrightarrow{\text{积分}}f(\vec{r},\vec{v})=C$

$\begin{cases}m\vec{v}=C\\ \vec{r}\times m\vec{v}=C\\ T+V=C\Rightarrow\text{机械能}\end{cases}\Rightarrow$ 广义动量

↓ 推广

$\frac{d}{dt}\left(\frac{\partial L}{\partial\dot{q}_\alpha}\right)=\frac{\partial L}{\partial q_\alpha}\Rightarrow$ 广义动量守恒. $\left(\frac{\partial L}{\partial\dot{q}_\alpha}\right)$

等式右边为 $f(q_\alpha,\dot{q}_\alpha)=C$ 不随时间变化. 也为 $\Rightarrow$ 运动积分

⇓ 守恒律 $\Rightarrow$ 运动积分.

3.2.1 广义动量积分（$p_\alpha=C$ 不随时间变化）

$\frac{d}{dt}\left(\frac{\partial L}{\partial\dot{q}_\alpha}\right)=\frac{\partial L}{\partial q_\alpha}$ ⇓ $p_\alpha=\frac{\partial L}{\partial\dot{q}_\alpha}$

$\Rightarrow 0=\frac{dp_\alpha}{dt}=\frac{\partial L}{\partial q_\alpha}=0\Rightarrow L(q_1,q_2,\cdots q_n,\dot{q}_1,\dot{q}_2,\cdots\dot{q}_n,t)$ 缺 q_α

不随时间变化. $p_\alpha=C$

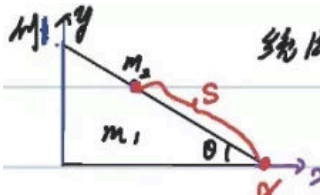
若 $L(q,\dot{q},t)$ 不显含广义坐标 q_β，那么 $\frac{\partial L}{\partial q_\beta}=0$

$\Rightarrow\frac{dp_\beta}{dt}=0\Rightarrow p_\beta=\frac{\partial L}{\partial\dot{q}_\beta}=C$（是守恒量）

L 不显含 q_β. q_β 循环坐标.（可遗忘坐标）

⇓ p_β 广义动量守恒（广义动量积分）

弗恩曼

例1: 

统一出发点. 求质点相对加速度
斜面加速度.
③ 主动力 mg, mg 保守力

解: ① 完整. 理想. ② 自由度2. $q_1=x$ $q_2=s$

$L=T-V=\frac{1}{2}m_1\dot{x}^2+\frac{1}{2}m_2[(\dot{x}+\dot{s}\cos\theta)^2+\dot{s}^2\sin^2\theta]-mgs\sin\theta$

L含s. L不含x. x是循环坐标. ⇒ x 方向动量守恒

$p_x=\frac{\partial L}{\partial\dot{x}}=m_1\dot{x}+m_2(\dot{x}+\dot{s}\cos\theta)=C$

例: 有心力作用. (r,θ)

$L=\frac{1}{2}m(\dot{r}^2+r^2\dot{\theta}^2)-G\frac{mm}{r}$

L不含θ. θ是循环坐标. ⇒ $p_\theta=C$ ✓ → 角动量

$p_\theta=\frac{\partial L}{\partial\dot{\theta}}=mr^2\dot{\theta}=C$

大小运动角动量.

3.2.2 广义动量积分(分离变比积分)

广义动量积分 ⇒ 动量/角动量守恒 空间 ⇒ q_α

广义动量积分 ⇒ 类似机械(能)守恒 ⇒ 是同一个量

$\frac{d}{dt}(p_\alpha)=\frac{\partial L}{\partial q_\alpha}$ 目标 $\frac{\partial L}{\partial t}=0$ → □ 动量守恒 □ 守恒量

拉氏方程 空间→时间

$L=L(q,\dot{q},t)$

$\frac{dL}{dt}=\frac{\partial L}{\partial t}+\sum_{\alpha=1}^{s}\frac{\partial L}{\partial q_\alpha}\dot{q}_\alpha+\sum_{\alpha=1}^{s}\frac{\partial L}{\partial\dot{q}_\alpha}\frac{d\dot{q}_\alpha}{dt}$

$=\frac{\partial L}{\partial t}+\sum_{\alpha=1}^{s}\frac{d}{dt}\left(\frac{\partial L}{\partial\dot{q}_\alpha}\right)\dot{q}_\alpha+\sum_{\alpha=1}^{s}\frac{\partial L}{\partial\dot{q}_\alpha}\frac{d\dot{q}_\alpha}{dt}$

$=\frac{\partial L}{\partial t}+\sum_{\alpha=1}^{s}\left(\frac{d}{dt}\left(\frac{\partial L}{\partial\dot{q}_\alpha}\dot{q}_\alpha\right)-\frac{\partial L}{\partial\dot{q}_\alpha}\cdot\frac{d\dot{q}_\alpha}{dt}\right)+\sum_{\alpha=1}^{s}\frac{\partial L}{\partial q_\alpha}\frac{dq_\alpha}{dt}$

$=\frac{\partial L}{\partial t}+\sum_{\alpha=1}^{s}\frac{d}{dt}\left(\frac{\partial L}{\partial\dot{q}_\alpha}\cdot\dot{q}_\alpha\right)$

$=\frac{\partial L}{\partial t}+\sum_{\alpha=1}^{s}\frac{d}{dt}(p_\alpha\cdot\dot{q}_\alpha)$

$\frac{dL}{dt}=\frac{\partial L}{\partial t}+\sum_{\alpha=1}^{s}\frac{d}{dt}(p_\alpha\cdot\dot{q}_\alpha)$

$\Rightarrow\frac{d}{dt}\left(\sum_{\alpha=1}^{s}p_\alpha\cdot\dot{q}_\alpha-L\right)=-\frac{\partial L}{\partial t}\Rightarrow\frac{dH}{dt}=-\frac{\partial L}{\partial t}$

令 $H=\left(\sum_{\alpha=1}^{s}p_\alpha\cdot\dot{q}_\alpha\right)-L$ ⇒ 广义能量.

$\frac{dH}{dt}=-\frac{\partial L}{\partial t}$ ⇒ 若L不显含t, $\frac{\partial L}{\partial t}=0$.

$\frac{dH}{dt}=0\Rightarrow H=C$

若L不显含t, $L=L(q,\dot{q})$, H=C 守恒量. → 广义能量积

H相当于什么为何? H为何加? $H=\left(\sum_{\alpha=1}^{s}p_\alpha\cdot\dot{q}_\alpha\right)-L$

(1) 定常约束. $\vec{r}_i=\vec{r}_i(q)\Rightarrow\frac{\partial\vec{r}_i}{\partial t}=0$

$\dot{\vec{r}}_i=\frac{\partial\vec{r}_i}{\partial t}+\sum_{\alpha=1}^{s}\frac{\partial\vec{r}_i}{\partial q_\alpha}\cdot\dot{q}_\alpha=\sum_{\alpha=1}^{s}\frac{\partial\vec{r}_i}{\partial q_\alpha}\cdot\dot{q}_\alpha$

$T=\sum_{i=1}^{n}\frac{1}{2}m_i\dot{\vec{r}}_i\cdot\dot{\vec{r}}_i=\sum_{i=1}^{n}\frac{1}{2}m_i\left(\sum_{\alpha=1}^{s}\frac{\partial\vec{r}_i}{\partial q_\alpha}\dot{q}_\alpha\right)\left(\sum_{\beta=1}^{s}\frac{\partial\vec{r}_i}{\partial q_\beta}\dot{q}_\beta\right)$

$=\sum_{i=1}^{n}\sum_{\alpha=1}^{s}\sum_{\beta=1}^{s}\left(\frac{1}{2}m_i\frac{\partial\vec{r}_i}{\partial q_\alpha}\cdot\frac{\partial\vec{r}_i}{\partial q_\beta}\dot{q}_\alpha\dot{q}_\beta\right)$ → 广义速度二次齐次函数.

□ $\dot{q}_\alpha\dot{q}_\beta$

T-V, $V=V(q,t)$

$\sum_{\alpha=1}^{s}p_\alpha\dot{q}_\alpha=\sum_{\alpha=1}^{s}\frac{\partial L}{\partial\dot{q}_\alpha}\dot{q}_\alpha=\sum_{\alpha=1}^{s}\frac{\partial T}{\partial\dot{q}_\alpha}\dot{q}_\alpha=2T$

$H=\sum_{\alpha=1}^{s}p_\alpha\cdot\dot{q}_\alpha-L=2T-(T-V)=T+V$ 机械能.

若 $\vec{r}_i=\vec{r}_i(q)$, $H=T+V$. 不含时间 机械能守恒.

若 $\vec{r}_i=\vec{r}_i(q,t)$

$\dot{\vec{r}}_i=\frac{\partial\vec{r}_i}{\partial t}+\sum_{\alpha=1}^{s}\frac{\partial\vec{r}_i}{\partial q_\alpha}\cdot\dot{q}_\alpha$

$T=\sum_{i=1}^{n}\frac{1}{2}m_i\left(\frac{\partial\vec{r}_i}{\partial t}+\sum_{\alpha=1}^{s}\frac{\partial\vec{r}_i}{\partial q_\alpha}\cdot\dot{q}_\alpha\right)\left(\frac{\partial\vec{r}_i}{\partial t}+\sum_{\beta=1}^{s}\frac{\partial\vec{r}_i}{\partial q_\beta}\cdot\dot{q}_\beta\right)$

$=\sum_{i=1}^{n}\left(\frac{1}{2}m_i\left(\frac{\partial\vec{r}_i}{\partial t}\right)^2+\sum_{\alpha=1}^{s}m_i\frac{\partial\vec{r}_i}{\partial t}\frac{\partial\vec{r}_i}{\partial q_\alpha}\dot{q}_\alpha+\right.$ $T=T_0+T_1+T_2$

$(\quad)\dot{q}_\alpha^0+(\quad)\dot{q}_\alpha$

T_0 T_1

$\left.+\sum_{\alpha=1}^{s}\sum_{\beta=1}^{s}\frac{1}{2}m_i\frac{\partial\vec{r}_i}{\partial q_\alpha}\cdot\frac{\partial\vec{r}_i}{\partial q_\beta}\dot{q}_\alpha\dot{q}_\beta\right)$

$(\quad)\dot{q}_\alpha\dot{q}_\beta$ T_2

$\sum_{\alpha=1}^{s}\frac{\partial T}{\partial\dot{q}_\alpha}\dot{q}_\alpha=$

$0\cdot T_0+1\cdot T_1+2T_2$

$T = T_0 + T_1 + T_2$

$\sum_{\alpha=1}^{s} \frac{\partial T}{\partial \dot{q}_\alpha}\dot{q}_\alpha = 0\cdot T_0 + 1\cdot T_1 + 2T_2 = T_1 + 2T_2$

$H = \sum_{\alpha=1}^{s} \frac{\partial T}{\partial \dot{q}_\alpha}\dot{q}_\alpha - L = (T_1+2T_2) - (T_0+T_1+T_2-V)$

$H = T_2 - T_0 + V$ 并非广义机械能.(动能+势能)

T_2: $\frac{1}{2}\cdot a_{\alpha\beta}\dot{q}_\alpha\dot{q}_\beta$ T_0: $\left(\frac{d\vec{r}_i}{dt}\right)^2$

$\vec{r}_i(q_i,t)$ 显含时间. H 守恒. 机械能不守恒

$\vec{r}_i(q_i)$ 稳定约束 $H = T+V$

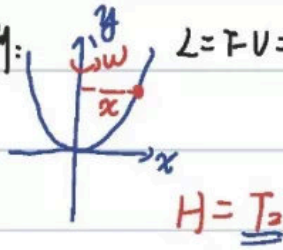
结论.

若 L 稳定约束. $L = L(q,\dot{q})$. $H = \sum_{\alpha=1}^{s} B_\alpha\dot{q}_\alpha - L$ 守恒是

$\Downarrow$ $\vec{r}_i = \vec{r}_i(q) \Rightarrow H = T+V = T_2+V$ 机械能守恒

$\vec{r}_i = \vec{r}_i(q,t) \Rightarrow H = T_2 - T_0 + V$ 广义能量守恒 机械能不守恒

-随动系.

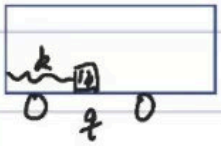
例: 

$L = T-V = \frac{1}{2}m\left(1+\frac{x^2}{4a^2}\right)\dot{x}^2 + \omega^2x^2 - mg\frac{x}{4a}$

$\dot{q}\cdot\dot{q}$ T_2 $\dot{q}\cdot q$ T_0

$H = T_2 - T_0 + V$ 广义能量守恒 $\Rightarrow$

$H = \frac{1}{2}m\left(1+\frac{x^2}{4a^2}\right)\dot{x}^2 - (-\omega^2x^2) - mg\frac{x}{4a}$.

例: 

(1) 静 $v_0 \to$ 静系. $L = \frac{1}{2}m\dot{q}^2 - \frac{1}{2}kq^2$ $\frac{\partial L}{\partial t} = 0$

$H = T_2 + V = \frac{1}{2}m\dot{q}^2 + \frac{1}{2}kq^2$ 机守.

(2) 静 $v_0 \to$ 动系.

$L = \frac{1}{2}m(\dot{q}+v_0)^2 - \frac{1}{2}kq^2 = \frac{1}{2}m\dot{q}^2 + \frac{1}{2}mv_0^2 + m\dot{q}v_0 - \frac{1}{2}kq^2$

T_2 T_0 T_1

$\frac{\partial L}{\partial t} = 0 \Rightarrow H = C$

$H = T_2 - T_0 + V = \frac{1}{2}m\dot{q}^2 - \frac{1}{2}mv_0^2 + \frac{1}{2}kq^2 = C$

$\Rightarrow \frac{1}{2}m\dot{q}^2 + \frac{1}{2}kq^2 = C + \frac{1}{2}mv_0^2$

(2) $\xrightarrow{a_t}$ 动系.

$L = \frac{1}{2}m(\dot{q}+at)^2 - \frac{1}{2}kq^2 \Rightarrow \frac{\partial L}{\partial t} \neq 0$

$H \pm C$. 平动参考系.

3.3 广义势能.

保守势能: $V = V(q,t)$ 与 $\dot{q}$ 无关

$\Downarrow$ $Q_\alpha = -\frac{\partial V}{\partial q_\alpha} \Rightarrow L = T-V \Rightarrow \frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}_\alpha}\right) - \frac{\partial L}{\partial q_\alpha} = 0$ $\alpha=1,2,\cdots,s$

广义势能: $U = U(q,\dot{q},t)$

如果. 满足 $Q_\alpha = -\frac{\partial U}{\partial q_\alpha} + \frac{d}{dt}\left(\frac{\partial U}{\partial \dot{q}_\alpha}\right)$. 仍然成立.

$\frac{d}{dt}\left(\frac{\partial T}{\partial \dot{q}_\alpha}\right) - \frac{\partial T}{\partial q_\alpha} = Q_\alpha = -\frac{\partial U}{\partial q_\alpha} + \frac{d}{dt}\left(\frac{\partial U}{\partial \dot{q}_\alpha}\right)$

$\Rightarrow \frac{d}{dt}\left(\frac{\partial (T-U)}{\partial \dot{q}_\alpha}\right) - \frac{\partial (T-U)}{\partial q_\alpha} = 0 \Rightarrow \frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}_\alpha}\right) - \frac{\partial L}{\partial q_\alpha} = 0$

$\Rightarrow$ 广义势能. 满足 $Q_\alpha = \cdots$; 满足拉氏方程.

例: 导线单电荷 q 在电磁场中的拉格朗日函数.

$\vec{F} = q\vec{E} + q\vec{v}\times\vec{B}$

$\nabla\times\vec{E} = -\frac{\partial \vec{B}}{\partial t}$ $\nabla\times\vec{E} = -\frac{\partial}{\partial t}(\nabla\times\vec{A}) = -\nabla\times\left(\frac{\partial \vec{A}}{\partial t}\right)$

$\nabla\cdot\vec{E} = \rho/\varepsilon_0$ $\nabla\times\left(\vec{E}+\frac{\partial \vec{A}}{\partial t}\right) = 0$

$\nabla\times\vec{B} = \mu_0\varepsilon_0\frac{\partial \vec{E}}{\partial t} + \mu_0\vec{j}$ 标势函数 $-\nabla\varphi = \vec{E} + \frac{\partial \vec{A}}{\partial t}$

$\nabla\cdot\vec{B} = 0 \Rightarrow \vec{B} = \nabla\times\vec{A}$ $\Rightarrow \begin{cases} \vec{E} = -\nabla\varphi - \frac{\partial \vec{A}}{\partial t} \\ \vec{B} = \nabla\times\vec{A} \end{cases}$

$A_x = A_x(x,y,z,t)$

$\vec{F} = q\left(-\nabla\varphi - \frac{\partial \vec{A}}{\partial t}\right) + q\vec{v}\times(\nabla\times\vec{A})$ 直角分量.

$\Rightarrow F_x = q\left(-\frac{\partial \varphi}{\partial x} - \frac{\partial A_x}{\partial t} + v_y\left(\frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y}\right) - v_z\left(\frac{\partial A_x}{\partial z} - \frac{\partial A_z}{\partial x}\right)\right)$

$-\frac{dA_x}{dt} + \frac{\partial A_x}{\partial x}\cdot v_x + \frac{\partial A_x}{\partial y}\cdot v_y + \frac{\partial A_x}{\partial z}v_z = -\frac{\partial A_x}{\partial t}$

弗恩曼

$$F_x = q\left(-\frac{\partial\varphi}{\partial x} - \frac{dA_x}{dt} + v_x\frac{\partial A_x}{\partial x} + v_y\frac{\partial A_y}{\partial x} + v_z\frac{\partial A_z}{\partial x}\right)$$

$$\frac{\partial}{\partial x}(\vec{A}\cdot\vec{v}) = \frac{d\vec{A}}{dx}\cdot\vec{v} + \frac{\partial\vec{v}}{\partial x}\vec{A}$$

$$-\frac{dA_x}{dt} = \frac{d}{dt}\left[\frac{\partial}{\partial v_x}(\vec{A}\cdot\vec{v})\right]$$

$$\frac{\partial(\vec{A}\cdot\vec{v})}{\partial v_x} = \vec{A}\frac{\partial\vec{v}}{\partial v_x} + \vec{v}\frac{\partial\vec{A}}{\partial v_x} = 0 = A_x.$$

$$F_x = q\left(-\frac{\partial}{\partial x}(\varphi - \vec{A}\cdot\vec{v}) + \frac{d}{dt}\left[\frac{\partial}{\partial v_x}(\varphi - \vec{A}\cdot\vec{v})\right]\right)$$

$$\vec{F} = -\nabla(q\varphi - q\vec{A}\cdot\vec{v}) + \frac{d}{dt}\left(\nabla_v(\varphi - \vec{A}\cdot\vec{v})\right)$$

$$Q_\alpha = -\frac{\partial U}{\partial q_\alpha} + \frac{d}{dt}\left(\frac{\partial U}{\partial \dot{q}_\alpha}\right)$$

$$U = q\varphi - q\vec{A}\cdot\vec{v} = q(\varphi - \vec{A}\cdot\vec{v})$$

$$L = \frac{1}{2}mv^2 - q\varphi + q\vec{A}\cdot\vec{v}$$

$$H = \vec{p}\cdot\vec{v} - L = mv^2 + q\vec{A}\cdot\vec{v} - \left(\frac{1}{2}mv^2 - q\varphi + q\vec{A}\cdot\vec{v}\right)$$

$$\vec{p} = m\vec{v} + q\vec{A} \Rightarrow \frac{1}{2}mv^2 + q\varphi \Rightarrow \text{守恒}$$

$$H = \frac{1}{2}m(\vec{p} - e\vec{A})^2 + q\varphi$$

$L = \frac{1}{2}mv^2 - q\varphi + q\vec{A}\cdot\vec{v}$

$U = q\varphi - q\vec{A}\cdot\vec{v}$

$\vec{p} = m\vec{v} + q\vec{A}$ → 机械动量

↓ 动量

拉氏？

电荷 q 在 $\vec{E} = E\vec{j}$

$\vec{B} = B\vec{k}$

动力学？

2020年2月13日 19:00 分析力学(四)

3.1 拉格朗日方程 $\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}_\alpha}\right)-\frac{\partial L}{\partial q_\alpha}=0$, $\alpha=1,2,\cdots,s$

3.2 运动积分：守恒量

广义动量积分：$L(q,\dot{q},t)$ 程序 q_β. $\frac{\partial L}{\partial q_\beta}=0$

$\Rightarrow p_\beta=\frac{\partial L}{\partial \dot{q}_\beta}=C$

广义能量积分：$L(q,\dot{q},t)$ 程序 t. $\frac{dH}{dt}=-\frac{\partial L}{\partial t}=0$

$H=\left(\sum_{\alpha=1}^{s} p_\alpha\dot{q}_\alpha\right)-L \Rightarrow H$ 是守恒量.

$T_2=\frac{1}{2}mv^2=\frac{1}{2}m\vec{r}\cdot\vec{r}$ $H=T_2-T_0+V$ ($\square\cdot\vec{r}^2$)

3.3 广义势能：$Q_\alpha=-\frac{\partial V}{\partial q_\alpha}$ 普通势

$Q_\alpha=-\frac{\partial U}{\partial q_\alpha}+\frac{d}{dt}\left(\frac{\partial U}{\partial \dot{q}_\alpha}\right)$ 广义势.

电磁场：$U=q\varphi-q\vec{A}\cdot\vec{v}$ $\vec{P}=m\vec{v}+q\vec{A}$

(φ 标势, $\vec{A}$ 磁矢势)

例：q. $\vec{E}=E\vec{j}$, $\vec{B}=B\vec{k}$ 拉氏方程

$\vec{B}=\nabla\times\vec{A}$

$\varphi=-Ey$

$\vec{A}=-\frac{1}{2}By\vec{i}+\frac{1}{2}Bx\vec{j}$

$\vec{v}=\dot{x}\vec{i}+\dot{y}\vec{j}+\dot{z}\vec{k}$

$L=\frac{1}{2}m(\dot{x}^2+\dot{y}^2+\dot{z}^2)-q\cdot(-Ey)+q\vec{A}\cdot\vec{v}$

$=\frac{1}{2}m(\dot{x}^2+\dot{y}^2+\dot{z}^2)+qEy+q(x\dot{y}-y\dot{x})\cdot\frac{1}{2}B$

(x). $\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{x}}\right)=\frac{\partial L}{\partial x} \Rightarrow \frac{d}{dt}\left(m\dot{x}-\frac{1}{2}qBy\right)=\frac{1}{2}qB\dot{y}$

$\Rightarrow m\ddot{x}-\frac{1}{2}qB\dot{y}-\frac{1}{2}qB\dot{y}=0$

$\Rightarrow m\ddot{x}-qB\dot{y}=0$

(y) $\frac{d}{dt}\left(m\dot{y}+\frac{1}{2}qBx\right)=qE-\frac{1}{2}qB\dot{x}$

$m\ddot{y}+qB\dot{x}=qE$

(z) L 程序 z, 对应的动量守恒. $m\dot{z}=C$ $m\ddot{z}=0$

相对论力学中. $L=?$ 经典情况 $L=T-V$

相对论 $L\neq T-V$

广义能量积分 $L=?$

守恒量：$H=T+mc^2+V=$ 守恒量. $\sum_{\alpha=1}^{s} p_\alpha\dot{q}_\alpha$

假设 $H=\vec{p}\cdot\vec{v}-L$

$-L=H-\vec{p}\cdot\vec{v}=T+mc^2+V-\vec{p}\cdot\vec{v}$

$=\frac{m_0}{\sqrt{1-\beta^2}}\cdot c^2+V-\frac{m_0v}{\sqrt{1-\beta^2}}\cdot v$

$-L=\frac{m_0(c^2-v^2)}{\sqrt{1-\beta^2}}+V=m_0c^2\frac{1-v^2/c^2}{\sqrt{1-\beta^2}}+V$

$=m_0c^2\sqrt{1-v^2/c^2}+V$

相对论力学中 $L=-m_0c^2\sqrt{1-v^2/c^2}-V(x,y,z)$ 已证明.

$\Downarrow$ x方向动力

$F_x=\frac{d}{dt}\left(\frac{\partial L}{\partial v_x}\right)-\frac{\partial L}{\partial x}$ $\frac{\partial L}{\partial v_x}=+m_0c^2\cdot\frac{+\frac{1}{2}\cdot+v_x}{+\sqrt{1-v^2/c^2}}$

$=\frac{d}{dt}\left(\frac{m_0v_x}{\sqrt{1-v^2/c^2}}\right)-0=\frac{d}{dt}(mv_x)-0=\frac{d}{dt}p_x\vec{i}=\frac{d\vec{p}}{dt}$

(m)

3.4 小振动

小振动：保守的无约束系统的平衡位置附近的微小振动

3.4.1 小振动拉氏函数一般形式.

S 自由度. 广义坐标 $q_1,q_2,\cdots,q_S$.

平衡位置. ① $q_1=q_2=\cdots=q_S=0$ ② 平衡位置. 势能取极小.

拉氏函数：$L=T-V$.

$T=\frac{1}{2}\sum_{i=1}^{n}m_i\vec{r}_i\cdot\vec{r}_i=\frac{1}{2}\sum_{i=1}^{n}m_i\left(\sum_{\alpha=1}^{s}\frac{\partial\vec{r}_i}{\partial q_\alpha}\dot{q}_\alpha\right)\left(\sum_{\beta=1}^{s}\frac{\partial\vec{r}_i}{\partial q_\beta}\dot{q}_\beta\right)$

$=\frac{1}{2}\sum_{\alpha=1}^{s}\sum_{\beta=1}^{s}\left(\sum_{i=1}^{n}m_i\frac{\partial\vec{r}_i}{\partial q_\alpha}\cdot\frac{\partial\vec{r}_i}{\partial q_\beta}\right)\dot{q}_\alpha\dot{q}_\beta$

$\Downarrow m_{\alpha\beta}$

$\simeq\frac{1}{2}\sum_{\alpha=1}^{s}\sum_{\beta=1}^{s}m_{\alpha\beta}\dot{q}_\alpha\dot{q}_\beta$

$= \frac{1}{2}\sum_{\alpha=1}^{s}\sum_{\beta=1}^{s}\left(\sum_{i=1}^{n} m_i \frac{\partial \vec{r}_i}{\partial q_\alpha}\cdot\frac{\partial \vec{r}_i}{\partial q_\beta}\right)\dot{q}_\alpha\dot{q}_\beta$ → 广义质量 → 广义坐标

$m_{\alpha\beta} = m_{\beta\alpha} = \sum_{i=1}^{n} m_i \left.\frac{\partial \vec{r}_i}{\partial q_\alpha}\right|_{q_\alpha=0}\cdot\left.\frac{\partial \vec{r}_i}{\partial q_\beta}\right|_{q_\beta=0}$ 质心是. 质心是

$T \approx \frac{1}{2}\sum_{\alpha=1}^{s}\sum_{\beta=1}^{s} m_{\alpha\beta}\dot{q}_\alpha\dot{q}_\beta$; $m_{\alpha\beta} = \sum_{i=1}^{n} m_i \left.\frac{\partial \vec{r}_i}{\partial q_\alpha}\right|_{q_\alpha=0}\cdot\left.\frac{\partial \vec{r}_i}{\partial q_\beta}\right|_{q_\beta=0}$

质是参数.

$V = V(q_1, q_2, \cdots, q_s)$ $V(0)$ 极值.

$V = V(0,0,\cdots,0) + \sum_{\alpha=1}^{s}\left(\frac{\partial V}{\partial q_\alpha}\right|_{q_\alpha=0}) + \frac{1}{2}\sum_{\alpha=1}^{s}\sum_{\beta=1}^{s}\left(\frac{\partial^2 V}{\partial q_\alpha \partial q_\beta}\right|_{\substack{q_\alpha=0\\ q_\beta=0}}) + \cdots$

$V(0)=0$ $\frac{\partial V}{\partial q_\alpha}|_{q_\alpha=0}=0$ $k_{\alpha\beta}=k_{\beta\alpha}$ 线性齐次

$L = T - V = \frac{1}{2}\sum_{\alpha=1}^{s}\sum_{\beta=1}^{s}(m_{\alpha\beta}\dot{q}_\alpha\dot{q}_\beta - k_{\alpha\beta}q_\alpha q_\beta)$

$\frac{\partial L}{\partial \dot{q}_\alpha} = \frac{\partial}{\partial \dot{q}_\alpha}\left(\frac{1}{2}\sum_{\alpha=1}^{s}\sum_{\beta=1}^{s} m_{\alpha\beta}\dot{q}_\alpha\dot{q}_\beta\right) = \frac{1}{2}\sum_{\beta=1}^{s} m_{\beta\alpha}\dot{q}_\beta + \frac{1}{2}\sum_{\alpha=1}^{s} m_{\alpha\beta}\dot{q}_\alpha$

$= \sum_{\alpha=1}^{s} m_{\alpha\beta}\dot{q}_\alpha$

$\frac{\partial L}{\partial q_\alpha} = -\sum_{\beta=1}^{s} k_{\alpha\beta}q_\beta$

$\Rightarrow \sum_{\beta=1}^{s}(m_{\alpha\beta}\ddot{q}_\beta + k_{\alpha\beta}q_\beta) = 0$ $\alpha = 1,2,\cdots,s$

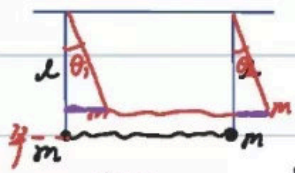
例1. $L = \frac{1}{2}ml^2\dot{\theta}^2 - mgl(1-\cos\theta)$

$\theta \ll 1$ $\cos\theta = 1 - \frac{1}{2}\theta^2$

$L = \frac{1}{2}ml^2\dot{\theta}^2 - mgl\cdot\frac{1}{2}\theta^2 = \frac{1}{2}ml^2\dot{\theta}^2 - \frac{1}{2}mgl\theta^2$

3.1.2 2自由度. 小振动.

1. 单摆合摆.



自由度2. θ_1, θ_2

$T = \frac{1}{2}ml^2\dot{\theta}_1^2 + \frac{1}{2}ml^2\dot{\theta}_2^2$

$V_{重} = mgl(1-\cos\theta_1) + mgl(1-\cos\theta_2)$

$V_{重} \approx mgl\cdot\frac{1}{2}\theta_1^2 + mgl\cdot\frac{1}{2}\theta_2^2$

$V_{弹} = \frac{1}{2}k(l\sin\theta_1 - l\sin\theta_2)^2 \approx \frac{1}{2}kl^2(\theta_1-\theta_2)^2$

$L = T - V = \frac{1}{2}ml^2(\dot{\theta}_1^2+\dot{\theta}_2^2) - \frac{1}{2}mgl(\theta_1^2+\theta_2^2) - \frac{1}{2}kl^2(\theta_2-\theta_1)^2$

$\theta_2^2+\theta_1^2 - 2\theta_1\theta_2$

$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{\theta}_1}\right) = \frac{\partial L}{\partial \theta_1} \Rightarrow ml^2\ddot{\theta}_1 = -mgl\theta_1 + kl^2(\theta_2-\theta_1)$

$\Rightarrow ml^2\ddot{\theta}_1 + mgl\theta_1 + kl^2(\theta_1-\theta_2) = 0$ ①

$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{\theta}_2}\right) = \frac{\partial L}{\partial \theta_2} \Rightarrow ml^2\ddot{\theta}_2 + mgl\theta_2 + kl^2(\theta_2-\theta_1) = 0$ ②

方法一. 试解法. 设 $\theta_1 = A_1\cos(\omega t+\psi)$ $\theta_2 = A_2\cos(\omega t+\psi)$

① $\Rightarrow \left(-ml^2A_1\omega^2 + mgl\cdot A_1 + kl^2(A_1-A_2)\right)\cdot\cos(\omega t+\psi) = 0$

$\Rightarrow (-ml\omega^2 + mg + kl)A_1 - klA_2 = 0$ ③

③ $\Rightarrow -klA_1 + (-ml\omega^2+mg+kl)A_2 = 0$ ④

$A_1=0=A_2$. 都是. 小振动. A_1, A_2 至少一个非零.

设 $\begin{vmatrix} (-ml\omega^2+mg+kl) & -kl \\ -kl & +(-ml\omega^2+mg+kl) \end{vmatrix} = 0$ A_1, A_2 非零解

$(ml\omega^2 - mg - kl)^2 - (kl)^2 = 0$ ⑤ $\Rightarrow \omega_1^2 = g/l$

$ml\omega^2 - mg - kl = \pm kl$ $\Rightarrow$ ⑤ $\omega_2^2 = g/l + 2k/m$

若 $\omega_1 = \sqrt{g/l} \Rightarrow \omega_1$ 代入③. ④ 成立. $\Rightarrow A_1/A_2 = 1$

若 $\omega_2 = \sqrt{g/l+2k/m} \Rightarrow \omega_2$ 代入③. ④ $\Rightarrow A_1/A_2 = -1$

$\theta_1 = A\cos(\omega_1 t+\psi_{01}) + B\cos(\omega_2 t+\psi_{02})$

$\theta_2 = A\cos(\omega_1 t+\psi_{01}) - B\cos(\omega_2 t+\psi_{02})$

通解. 非同频 非0.

相位差. 两个自由度. q_1, q_2 非零.

重新选. $q_1 = \theta_1+\theta_2 = 2A\cos(\omega_1 t+\psi_{01})$ ①

$q_2 = \theta_1-\theta_2 = 2B\cos(\omega_2 t+\psi_{02})$ ②

变量分离 两个简谐振动 耦合

通解. 非同频. 不动的同频率

特解: $\begin{cases} q_1 = 2A\cos(\omega_1 t+\psi_{01}) \\ q_2 = 0 \end{cases}$ (ω_1) $\begin{cases} q_1 = 0 \\ q_2 = 2B\cos(\omega_2 t+\psi_{02}) \end{cases}$ (ω_2)

第一组 ω_1 $q_2 = 0$

$\omega_1 \Rightarrow$ 整体振动. θ_1, θ_2 同相位. 弹簧不起作用.

$\Rightarrow$ θ_1 相对 θ_2 静止. 质心不动. 两个单摆. 同时摆动.

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S个自由度. 进行小振动. ⇒ w 必然有 S 个结果.

$\omega_1, \omega_2, \cdots, \omega_s$

一般: $(q_1, q_2, \cdots, q_s)$

$$\begin{cases} q_1 = \sum_{i=1}^{s} A_{i1}\cos(\omega_i t+\varphi_i) & \text{通解.} \\ \vdots \\ q_s = \sum_{i=1}^{s} A_{is}\cos(\omega_i t+\varphi_i) \end{cases}$$

S块 $\cos(\omega_i t+\varphi_i)$ 叠加.

特殊: 广义坐标 $(\xi_1, \xi_2, \cdots, \xi_s)$ 简正坐标.

$$\begin{cases} \xi_1 = A_1\cos(\omega t+\varphi_1) + 0\cdot\cos(\omega_2 t+\varphi_2)+\cdots \\ \xi_2 = \quad + A_2\cos(\omega t+\varphi_2) \\ \vdots \\ \xi_s = \quad + A_s\cos(\omega t+\varphi_s) \end{cases}$$

简正频率.

ω_1 振动 $\xi_1 \neq 0$, $\xi_2 \cdots \xi_s = 0$ ⇒ 简正模.

$\omega_2 \cdots$ $\xi_2 \neq 0$, 余下 $=0$

寻找简正模方法

⇒ 找到广义坐标. 用任意时刻. 随时广义坐标.

① 配分方法. 从出发 ⇒ 主动力势. $\dot\theta_1\dot\theta_2$, $\theta_1\theta_2$ 动能

$$L = T - V = \frac{1}{2}ml^2(\dot\theta_1^2+\dot\theta_2^2) - \frac{1}{2}mgl(\theta_1^2+\theta_2^2) - \frac{1}{2}kl^2(\theta_2-\theta_1)^2$$

(22) $a^2+b^2 = \frac{1}{2}[(a+b)^2+(a-b)^2]$ $ab = \frac{1}{4}[(a+b)^2-(a-b)^2]$

$$L = \frac{1}{4}ml^2[(\dot\theta_1+\dot\theta_2)^2+(\dot\theta_1-\dot\theta_2)^2] - \frac{1}{4}mgl[(\theta_1+\theta_2)^2+(\theta_1-\theta_2)^2] - \frac{1}{2}kl^2(\theta_2-\theta_1)^2$$

令. $\xi_1 = \theta_1+\theta_2$, $\xi_2 = \theta_1-\theta_2$

$$L = \frac{1}{4}ml^2(\dot\xi_1^2+\dot\xi_2^2) - \frac{1}{4}mgl(\xi_1^2+\xi_2^2) - \frac{1}{2}kl^2\xi_2^2$$

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot\xi_1}\right) = \frac{\partial L}{\partial \xi_1} \Rightarrow \frac{1}{2}ml^2\ddot\xi_1 = -\frac{1}{2}mgl\xi_1$$

$$\Rightarrow \ddot\xi_1 + \frac{g}{l}\xi_1 = 0 \Rightarrow \omega_1^2 = \frac{g}{l}$$

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot\xi_2}\right) = \frac{\partial L}{\partial \xi_2} \Rightarrow \frac{1}{2}ml^2\ddot\xi_2 = -\frac{1}{2}mgl\xi_2 - kl^2\xi_2$$

$$\Rightarrow \ddot\xi_2 + \left(\frac{2k}{m}+\frac{g}{l}\right)\xi_2 = 0 \Rightarrow \omega_2^2 = \frac{2k}{m}+\frac{g}{l}$$

$$\begin{cases} \xi_1 = A\cos(\omega_1 t+\varphi_{01}) + 0\cdot B\cos(\omega_2 t+\varphi_{02}) \\ \xi_2 = 0 \quad + B\cos(\omega_2 t+\varphi_{02}) \end{cases}$$

② 耦合. 消除交叉项. 2自由度.

$$\begin{cases} ml^2\ddot\theta_1 + mgl\theta_1 + kl^2(\theta_1-\theta_2) = 0 & ① \\ ml^2\ddot\theta_2 + mgl\theta_2 + kl^2(\theta_2-\theta_1) = 0 & ② \end{cases}$$

① ⇒ $\ddot\xi_1 + \square\xi_1 = 0$ ② $\ddot\xi_2 + \square\xi_2 = 0$

$$① + ②: ml^2(\ddot\theta_1+\ddot\theta_2) + mgl(\theta_1+\theta_2) = 0$$

$$① - ②: ml^2(\ddot\theta_1-\ddot\theta_2) + mgl(\theta_1-\theta_2) + 2kl^2(\theta_1-\theta_2) = 0$$

③ 通解:

$$\begin{cases} x_1 = A_{11}\cos(\omega_1 t+\varphi_1) + A_{21}\cos(\omega_2 t+\varphi_2) + \cdots + A_{s1}\cos(\omega_s t+\varphi_s) \\ \vdots \\ x_s = A_{1s}\cos(\omega_1 t+\varphi_1) + A_{2s}\cos(\omega_2 t+\varphi_2) + \cdots + A_{ss}\cos(\omega_s t+\varphi_s) \end{cases}$$

简正坐标. $\xi_1 \sim \omega_1$, $\xi_2 \sim \omega_2$

互相导出: $\xi_1 \sim (x_1, \cdots, x_n)$

$\xi_1 \sim (x_1, \cdots, x_n)$

例: 直角边. 两端点连接. k_1, l 斜边. $k_2, \sqrt{2}l$

自由放落. 平行于直角

问. 小振动. 振动频率 m. m.

解: 自由度. 2. (x, y) ⇒ 偏离平行位置坐标.

$$T = \frac{1}{2}m(\dot x^2+\dot y^2) \qquad V = \frac{1}{2}\sum_{\alpha=1}^{s}\sum_{\beta=1}^{s} k_{\alpha\beta}\cdot q_\alpha q_\beta$$

$$V = \frac{1}{2}k_1x^2 + \frac{1}{2}k_1y^2 + \frac{1}{2}k_2\left[\sqrt{(l+x)^2+(l+y)^2} - \sqrt{2}l\right]^2$$

$$\left.\frac{\partial^2 V}{\partial x^2}\right|_{x=0} = k_1+\frac{1}{2}k_2 \qquad V = \frac{1}{2}(k_1+\frac{1}{2}k_2)x^2 + \frac{1}{2}k_2xy + \frac{1}{2}k_2yx + \frac{1}{2}(k_1+\frac{1}{2}k_2)y^2$$

$$\left.\frac{\partial^2 V}{\partial y^2}\right|_{y=0} = k_1+\frac{1}{2}k_2$$

$$\left.\frac{\partial^2 V}{\partial y\partial x}\right|_{\substack{x=0\\y=0}} = \left.\frac{\partial^2 V}{\partial x\partial y}\right|_{\substack{x=0\\y=0}} = \frac{1}{2}k_2$$

$L=\frac{1}{2}m(\dot{x}^2+\dot{y}^2)-\frac{1}{2}(k_1+\frac{1}{2}k_2)x^2-\frac{1}{2}(k_1+\frac{1}{2}k_2)y^2-\frac{1}{2}k_2xy$

$=\frac{1}{2}m(\dot{x}^2+\dot{y}^2)-\frac{1}{2}(k_1+\frac{1}{2}k_2)(x^2+y^2)-\frac{1}{2}k_2(xy)$

$x:\ m\ddot{x}+(k_1+\frac{1}{2}k_2)x+\frac{1}{2}k_2y=0$ ①

$y:\ m\ddot{y}+(k_1+\frac{1}{2}k_2)y+\frac{1}{2}k_2x=0$ ②

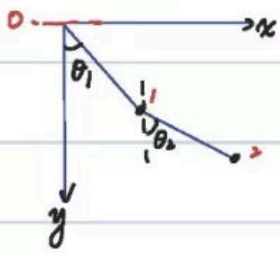
① + ② $\Rightarrow m(\ddot{x}+\ddot{y})+(k_1+\frac{1}{2}k_2+\frac{1}{2}k_2)(x+y)=0$ ③ (q_1, Q_1)

① − ② $\Rightarrow m(\ddot{x}-\ddot{y})+k_1(x-y)=0$ ④ (q_2, Q_2)

$q_1 \Rightarrow \omega_1=\sqrt{k_1+k_2/m}$

$q_2 \Rightarrow \omega_2=\sqrt{k_1/m}$

补充习题.



$\theta_1,\ \theta_2$

$T_1=\frac{1}{2}ml^2\dot{\theta}_1^2$

$T_2=\frac{1}{2}ml^2(\dot{\theta}_1+\dot{\theta}_2)^2$

$V=-mgl\cos\theta_1-mgl(\cos\theta_1+\cos\theta_2)$

$=-mgl(1-\frac{1}{2}\theta_1^2)-mgl(1-\frac{1}{2}\theta_1^2+1-\frac{1}{2}\theta_2^2)$

取初始位置为零点, 即 $\theta_1=0,\ \theta_2=0$ 时 $V=0$

$\Leftarrow V=-2mgl+\frac{1}{2}mgl(2\theta_1^2+\theta_2^2)$

$V=\frac{1}{2}mgl(2\theta_1^2+\theta_2^2)$

$L=\frac{1}{2}ml^2(2\dot{\theta}_1^2+\dot{\theta}_2^2+2\dot{\theta}_1\dot{\theta}_2)-\frac{1}{2}mgl(2\theta_1^2+\theta_2^2)$

$=\frac{1}{2}ml^2\left(\frac{1}{2}[(\sqrt{2}\dot{\theta}_1+\dot{\theta}_2)^2+(\sqrt{2}\dot{\theta}_1-\dot{\theta}_2)^2]\right)$

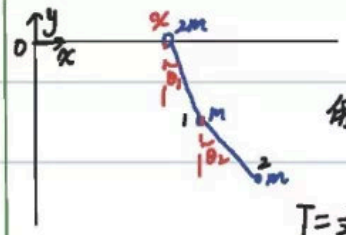
$+\frac{1}{2}ml^2\left(\frac{\sqrt{2}}{4}[(\sqrt{2}\dot{\theta}_1+\dot{\theta}_2)^2-(\sqrt{2}\dot{\theta}_1-\dot{\theta}_2)^2]\right)$

$-\frac{1}{2}mgl\left(\frac{1}{2}[(\sqrt{2}\theta_1+\theta_2)^2+(\sqrt{2}\theta_1-\theta_2)^2]\right)$

$q_1=\sqrt{2}\theta_1+\theta_2$

$q_2=\sqrt{2}\theta_1-\theta_2$

$\begin{cases}\omega_1^2=2+\sqrt{2}\,g/l\\ \omega_2^2=2-\sqrt{2}\,g/l\end{cases}$



求出振动的解.

解: $x,\ \theta_1,\ \theta_2$

$T=\frac{1}{2}\cdot 2m\cdot\dot{x}^2+\frac{1}{2}m(\dot{x}_1^2+\dot{y}_1^2)+\frac{1}{2}m(\dot{x}_2^2+\dot{y}_2^2)$

$=\frac{1}{2}2m\dot{x}^2+\frac{1}{2}m(\dot{x}+l\dot{\theta}_1)^2+\frac{1}{2}m(\dot{x}+l\dot{\theta}_1+l\dot{\theta}_2)^2$

$V=-mgl_1\cos\theta_1-mg(l_1\cos\theta_1+l_2\cos\theta_2)$

$=-2mgl+\frac{1}{2}mgl(2\theta_1^2+\theta_2^2)$

$L=m\dot{x}^2+\frac{1}{2}m(\dot{x}+l\dot{\theta}_1)^2+\frac{1}{2}m(\dot{x}+l\dot{\theta}_1+l\dot{\theta}_2)^2-\frac{1}{2}mgl(2\theta_1^2+\theta_2^2)$

$=m\dot{x}^2+\frac{1}{2}m(\dot{x}^2+2\dot{x}l\dot{\theta}_1+l^2\dot{\theta}_1^2)+\frac{1}{2}m(\dot{x}^2+l^2\dot{\theta}_1^2+l^2\dot{\theta}_2^2$

$+2l\dot{x}\dot{\theta}_1+2l\dot{x}\dot{\theta}_2+2l^2\dot{\theta}_1\dot{\theta}_2)$

$-\frac{1}{2}mgl(2\theta_1^2+\theta_2^2)$

$=2m\dot{x}^2+ml^2\dot{\theta}_1^2+2m\dot{x}l\dot{\theta}_1+ml\dot{x}\dot{\theta}_2+ml^2\dot{\theta}_1\dot{\theta}_2+\frac{1}{2}ml^2\dot{\theta}_2^2-$ 势.

化简合. x. P_x 守恒. $P_x=\frac{\partial L}{\partial\dot{x}}\Rightarrow$ ----

θ_1,θ_2 作拉格朗日.

$\begin{cases}2l\ddot{\theta}_1+2\ddot{x}+l\ddot{\theta}_2+2g\theta_1=0\\ l\ddot{\theta}_2+\ddot{x}+l\ddot{\theta}+g\theta_2=0\\ 4\ddot{x}+2l\ddot{\theta}_1+l\ddot{\theta}_2=0\Rightarrow\ddot{x}=-\frac{1}{2}l\ddot{\theta}_1-\frac{1}{4}l\ddot{\theta}_2\end{cases}$

① ③ $\ddot{x}\Rightarrow 2l\ddot{\theta}_1+l\ddot{\theta}_2+4g\theta_1=0\Rightarrow 2\ddot{\theta}_1+\ddot{\theta}_2+4\frac{g}{l}\theta_1=0$

② ③ $\ddot{x}\Rightarrow 2l\ddot{\theta}_1+2l\ddot{\theta}_2+4g\theta_2=0\Rightarrow 2\ddot{\theta}_1+3\ddot{\theta}_2+4\frac{g}{l}\theta_2=0$ ④

$\theta_1=A_1\cos(\omega_1 t+\varphi_1)\qquad\theta_2=A_2\cos(\omega_2 t+\varphi_2)$

$\begin{cases}(4g-2l\omega_1^2)A_1-\omega^2lA_2=0\\ -2l\omega^2A_1+(4g-3l\omega^2)=0\end{cases}\Rightarrow\begin{vmatrix}4g-2l\omega_1^2 & -\omega^2l\\ -2\omega^2l & 4g-3l\omega^2\end{vmatrix}=0$

$\omega_1=\sqrt{g/l}\qquad\omega_2=\sqrt{4g/l}$ 代回④式

$\frac{A_{11}}{A_{21}}=\frac{1}{2}\qquad\frac{A_{12}}{A_{22}}=\frac{1}{-1}$

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$\theta_1 = A_1\cos(\omega_1 t+\varphi_1) + B_1\cos(\omega_2 t+\varphi_2)$

$\theta_2 = 2A_1\cos(\omega_1 t+\varphi_1) - B_1\cos(\omega_2 t+\varphi_2)$

$\theta_1 = A\beta_1 + B\beta_2$ $\Rightarrow$ $3A\beta_1 = \theta_1+\theta_2$ 线性-组成关系.

$\theta_2 = 2A\beta_1 - B\beta_2$ $3B\beta_2 = \theta_1-\theta_2$

$\ddot{x} = -\frac{l}{2}\ddot{\theta}_1 - \frac{l}{4}\ddot{\theta}_2$ $\Rightarrow$ 找公式.

$x = C_1+C_2 t - \frac{1}{2}l\theta_1 - \frac{1}{4}l\theta_2$

$t=0$ 时刻. m_1 m_2 x 满足关系.

2020年2月14日 19:00 分析力学(五)

小振动的基本理论. S自由度. $q_1, \cdots, q_s$ 平衡 $V=0$ $q_\beta=0$

$T=\frac{1}{2}\sum_{\alpha=1}^{s}\sum_{\beta=1}^{s}\left(m_{\alpha\beta}\dot{q}_\alpha\dot{q}_\beta\right) \Rightarrow (T_2) \Rightarrow H=T_2+V$

质量系数 $m_{\alpha\beta}=m_{\beta\alpha}=\sum_{i=1}^{n} m_i \frac{\partial \vec{r}_i}{\partial q_\alpha}\cdot\frac{\partial \vec{r}_i}{\partial q_\beta}\Big|_{q_\alpha=0}$

$V=\frac{1}{2}\sum_{\alpha=1}^{s}\sum_{\beta=1}^{s}\left(k_{\alpha\beta}q_\alpha q_\beta\right)$

弹性系数 $k_{\alpha\beta}=k_{\beta\alpha}=\frac{\partial^2 V}{\partial q_\alpha \partial q_\beta}\Big|_{q_\alpha=0}$

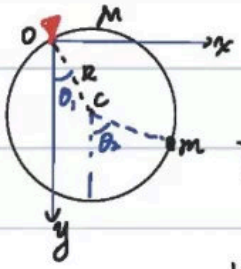
$L=T-V \Rightarrow$ 拉氏方程 $\Rightarrow$ S个微分方程.

S自由 $\Rightarrow$ S个 $\omega \Rightarrow$ S个简谐振.

$q_\gamma=\sum_{j=1}^{s}A_{j\gamma}\cos(\omega_j t+\varphi_j)$ $\gamma=1,2,\cdots,s$

↓ 主坐标 → 实际坐标(简正坐标)

$\xi_\gamma=B_\gamma\cos(\omega_\gamma t+\varphi_\gamma)$

例: 

m, m 在平面上小振动. 求简正振.

$T=\frac{1}{2}mR^2(\dot{\theta}_1+\dot{\theta}_2)^2+\frac{1}{2}(MR^2+mR^2)\cdot\dot{\theta}_1^2$

$V=\frac{1}{2}mg\theta_1^2+\frac{1}{2}mgR(\theta_1^2+\theta_2^2)$

$L=\frac{1}{2}mR^2(\dot{\theta}_1+\dot{\theta}_2)^2+(MR^2+mR^2)\cdot\dot{\theta}_1^2-\frac{1}{2}mg\theta_1^2-\frac{1}{2}mgR(\theta_1^2+\theta_2^2)$

$\xi_1=\theta_1-\theta_2$ $\xi_2=\frac{M+m}{m}\theta_1+\theta_2$

$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{\theta}}\right)=\frac{\partial L}{\partial \theta}$, $\theta=\theta_1,\theta_2$

$\omega_1=\sqrt{g/R}$ $\omega_2=\sqrt{\frac{(M+m)g}{MR}}$

$\theta_1=A_{11}\cos(\omega_1 t+\varphi_1)+A_{12}\cos(\omega_2 t+\varphi_2)$

$\theta_2=A_{11}\cos(\omega_1 t+\varphi_1)-\frac{M+m}{m}A_{12}\cos(\omega_2 t+\varphi_2)$

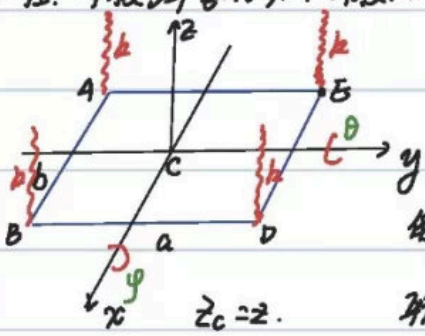
$\xi_1=A\cos(\omega_1 t+\varphi_1)$ 单独振动

$\xi_2=B\cos(\omega_2 t+\varphi_2)$

S个自由度小振动 ($S\geq 3$)

例1. 平板弹簧振动. 平板 m. a×b. 边和顶点C处各有弹簧 k 上下振动.

问平板小振动模式.



C 指x 指y

解: 自由度3. z, φ, θ.

$z_C=z$. 平衡位置 $z_C=z=0$.

$z_A=z-\frac{1}{2}a\varphi+\frac{1}{2}b\theta$ $z_B=z-\frac{1}{2}a\varphi-\frac{1}{2}b\theta$

$z_D=z+\frac{1}{2}a\varphi-\frac{1}{2}b\theta$ $z_E=z+\frac{1}{2}a\varphi+\frac{1}{2}b\theta$

$T=\frac{1}{2}m\dot{z}^2+\frac{1}{2}\left(\frac{1}{12}ma^2\right)\dot{\varphi}^2+\frac{1}{2}\left(\frac{1}{12}mb^2\right)\dot{\theta}^2$

$V=\frac{1}{2}kz^2+\frac{1}{2}k(z_A^2+z_B^2+z_C^2+z_D^2) \Rightarrow$

$L=\frac{1}{2}m\dot{z}^2+\frac{1}{24}ma^2\dot{\varphi}^2+\frac{1}{24}mb^2\dot{\theta}^2-\frac{1}{2}k(4z^2+a^2\varphi^2+b^2\theta^2)-mgz$ (重力势)

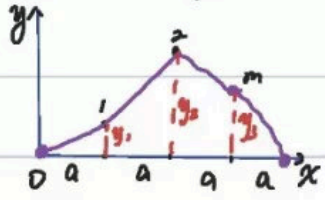
$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{z}}\right)=\frac{\partial L}{\partial z} \Rightarrow m\ddot{z}+4kz=0 \Rightarrow \omega_1=\sqrt{\frac{4k}{m}}$

$\varphi \Rightarrow m\ddot{\varphi}+12k\varphi=0$ $\omega_2=\sqrt{\frac{12k}{m}}$

$\theta \Rightarrow m\ddot{\theta}+12k\theta=0$ $\omega_3=\sqrt{\frac{12k}{m}}$

(三个振动互相独立) → 正振.

例2. 弦振动(连续振动)



三个质点. 四段弦. 原长a.

m

质点均匀分布.

弦张力恒为 F. 振动模式.

$\Delta l=\sqrt{y_1^2+a^2}+\sqrt{(y_2-y_1)^2+a^2}+\sqrt{(y_3-y_2)^2+a^2}+\sqrt{y_3^2+a^2}-4a$

$\approx\frac{1}{2}a\left[\left(\frac{y_1}{a}\right)^2+\left(\frac{y_2-y_1}{a}\right)^2+\left(\frac{y_3-y_2}{a}\right)^2+\left(\frac{y_3}{a}\right)^2\right]$

$V=F\cdot\Delta l=\frac{F}{2a}\left[y_1^2+(y_2-y_1)^2+(y_3-y_2)^2+y_3^2\right]$

$T=\frac{1}{2}m\left(\dot{y}_1^2+\dot{y}_2^2+\dot{y}_3^2\right)$

$L=\frac{1}{2}m\left(\dot{y}_1^2+\dot{y}_2^2+\dot{y}_3^2\right)-\frac{F}{2a}\left[y_1^2+(y_2-y_1)^2+(y_3-y_2)^2+y_3^2\right]$

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$L=\frac{1}{2}m(\dot{y}_1^2+\dot{y}_2^2+\dot{y}_3^2)-\frac{F}{2a}[y_1^2+(y_2-y_1)^2+(y_3-y_2)^2+y_3^2]$

$\frac{d}{dt}(\frac{\partial L}{\partial \dot{y}})=\frac{\partial L}{\partial y}$

① $\Rightarrow m\ddot{y}_1+\frac{F}{2a}[2y_1+2(y_2-y_1)\cdot(-1)]=0$

$\Rightarrow \ddot{y}_1+\frac{F}{ma}(2y_1-y_2)=0$

② $\Rightarrow m\ddot{y}_2+\frac{F}{2a}[2(y_2-y_1)+2(y_3-y_2)\cdot(-1)]=0$

$\Rightarrow \ddot{y}_2+\frac{F}{ma}[2y_2-y_1-y_3]=0$

③ $\Rightarrow m\ddot{y}_3+\frac{F}{2a}[2(y_3-y_2)+2y_3]=0$

$\Rightarrow \ddot{y}_3+\frac{F}{ma}(2y_3-y_2)=0$

$\beta=\frac{F}{ma}$

$\ddot{y}_1+\beta(2y_1-y_2)=0$

$\ddot{y}_2+\beta(2y_2-y_1-y_3)=0$

$\ddot{y}_3+\beta(2y_3-y_2)=0$

设 $y_1=A_1\cos(\omega t+\varphi)$

$y_2=A_2\cos(\omega t+\varphi)$

$y_3=A_3\cos(\omega t+\varphi)$

代入.

$(\omega^2-2\beta)A_1+\beta A_2=0$

$\beta A_1+(\omega^2-2\beta)A_2+\beta A_3=0$

$\beta A_2+(\omega^2-2\beta)A_3=0$

$A_1A_2A_3$ 非零解

$$\begin{vmatrix}\omega^2-2\beta & \beta & 0\\ \beta & \omega^2-2\beta & \beta\\ 0 & \beta & \omega^2-2\beta\end{vmatrix}=0$$

$\Rightarrow(\omega^2-2\beta)(\omega^4-4\beta\omega^2+2\beta^2)=0\Rightarrow$

$\omega_1^2=(2-\sqrt{2})\beta$ 代① $A_1:A_2:A_3=1:\sqrt{2}:1$

$\omega_2^2=2\beta$ 代② $A_1:A_2:A_3=1:0:-1$

$\omega_3^2=(2+\sqrt{2})\beta$ 代③ $A_1:A_2:A_3=1:-\sqrt{2}:1$

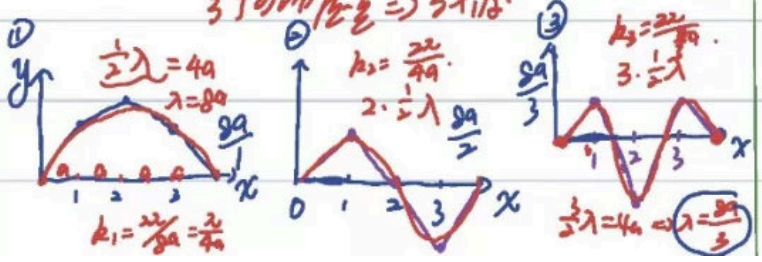
$y_1=A_{11}\cos(\omega_1 t+\varphi_1)+A_{21}\cos(\omega_2 t+\varphi_2)+A_{31}\cos(\omega_3 t+\varphi_3)$

$y_2=\sqrt{2}A_{11}\cos(\omega_1 t+\varphi_1)-0\cdot\cos(\omega_2 t+\varphi_2)-\sqrt{2}A_{31}\cos(\omega_3 t+\varphi_3)$

$y_3=A_{11}\cos(\omega_1 t+\varphi_1)-A_{21}\cos(\omega_2 t+\varphi_2)+A_{31}\cos(\omega_3 t+\varphi_3)$

① ② ③

3个初始量 $\Rightarrow$ 3个待定



q_1,q_2,q_3

$y_1=q_1+q_2+q_3$

$y_2=\sqrt{2}q_1-0+\sqrt{2}q_3$

$y_3=q_1-q_2+q_3$

$\Rightarrow$

$q_1=\frac{1}{4}(y_1+\sqrt{2}y_2+y_3)$

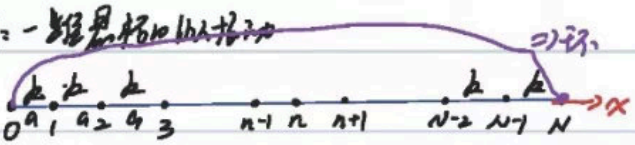
$q_2=\frac{1}{2}(y_1-y_3)$

$q_3=\frac{1}{4}(y_1-\sqrt{2}y_2+y_3)$

$q_1=y_1+y_3+\sqrt{2}y_2$

$q_2=y_1-y_3$

$q_3=y_1+y_3-\sqrt{2}y_2$



平衡位置 $x_n^0=na$ 自由度 N $\Rightarrow$ u_n 代表偏离平衡位置.

$u_n=x_n-x_n^0\Rightarrow\ddot{u}_n=\ddot{x}_n$

$n-1$ 与 n 的弹簧 $=-k(u_n-u_{n-1})$

$n+1$ 与 n $=+k(u_{n+1}-u_n)$

⑭ $m\ddot{u}_n=k(u_{n+1}+u_{n-1}-2u_n)$ $n=1,2,\cdots,N-1$

自由度 $N-1$. ω 有 $N-1$ 个. $u_n=A_n\cos(\omega t+\varphi)$

$N-1$ 个方程 $\begin{vmatrix}N-1\text{行}\\ N-1\text{列}\end{vmatrix}=0$ 三次方 β

$u_n=A\cos(\beta\cdot na)\cos(\omega t+\varphi)$

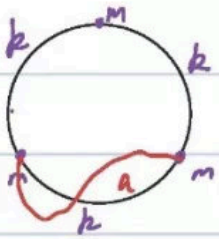
代回

$\Rightarrow\omega^2=\frac{4k}{m}\sin^2(\frac{1}{2}\beta a)$ $\Rightarrow\omega_m=\omega=\sqrt{\frac{4k}{m}}\cdot\sin(\frac{1}{2}\cdot\frac{2\pi}{a}\cdot a)$

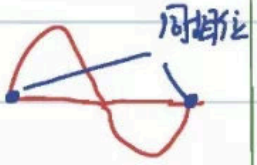
边界条件. $\beta Na=m\pi\Rightarrow\beta=\frac{m\pi}{Na}$, $m=1,2,\cdots,N-1$

$N\to\infty$. β 连续. $\beta_{max}=\frac{N-1}{Na}\pi=\frac{\pi}{a}$.

为大数量



质点一个以很大.
以他作为圆周运动.
圆上形成驻波.



$u_n = A\cos(\beta \cdot na)\cos(\omega t+\varphi)$
时间

$\Rightarrow \omega^2 = \frac{4k}{m}\sin^2(\frac{1}{2}\beta a)$

边界条件. $na = \lambda_\alpha \cdot (\alpha-1)$ $\alpha = 2, \cdots, n$

圆周长. $\omega_{max} = \sqrt{\frac{4k}{m}}$. 正为第2个边界条件.



利用最早一维振动模型. 求有质量连接的振动周期.
(5.16) 多次解.

CHAPTER 4 哈密顿力学.

拉格朗日方程: $q_\alpha, \dot{q}_\alpha \Rightarrow \frac{d}{dt}(\frac{\partial L}{\partial \dot{q}_\alpha}) - \frac{\partial L}{\partial q_\alpha} = 0$

注意方程 ⇓ 方程组 ⇑ $L = T - V = L(q_\alpha, \dot{q}_\alpha, t)$

哈密顿力学: $q_\alpha, p_\alpha \Rightarrow H = ---$

①量纲: $[q_\alpha][\dot{q}_\alpha] = [q_\alpha \frac{dq_\alpha}{dt}]$ 量纲相同 不确定

$[q_\alpha][p_\alpha] = [q_\alpha \frac{\partial L}{\partial \dot{q}_\alpha}] = [\frac{L}{\dot{q}_\alpha/t}] = [Lt]$ 作用量

对照关系: $\Delta p_x \cdot \Delta x \sim \Delta E \cdot \Delta t \sim h \sim$ 同量纲

原子 量子力学 宏观 ⇒ 微观 相对论 吸收质量 等效

4.1 哈密顿正则方程

4.1.1 勒让德变换

出发 $F = F(x_1, x_2, \cdots, x_n)$ ⇒ 相空间

$dF = \sum_{i=1}^n \frac{\partial F}{\partial x_i} dx_i$ 设 $u_i = \frac{\partial F}{\partial x_i}$ ⇒ $dF = \sum_{i=1}^n u_i dx_i$

2d证明 $\begin{vmatrix} \frac{\partial u_1}{\partial x_1} & \cdots & \frac{\partial u_1}{\partial x_n} \\ \vdots & & \vdots \\ \frac{\partial u_n}{\partial x_1} & \cdots & \frac{\partial u_n}{\partial x_n} \end{vmatrix} = \left|\frac{\partial(u_1, u_2, \cdots, u_n)}{\partial(x_1, x_2, \cdots, x_n)}\right| \neq 0$ 那 $u_1, u_2, \cdots u_n$ 相互也是独立的.

构造出 $G = G(u_1, \cdots, u_n) = (\sum_{i=1}^n x_i u_i) - F$

$dG = \sum_{i=1}^n \frac{\partial G}{\partial u_i} du_i$ $\sum_{i=1}^n u_i dx_i$

$dG = \sum_{i=1}^n x_i du_i + \sum_{i=1}^n u_i dx_i - dF = \sum_{i=1}^n x_i du_i$

又因 $F + G = \sum_{i=1}^n x_i u_i$ ⇒ $x_i = \frac{\partial G}{\partial u_i}$ 勒让德变换.

$u_i = \frac{\partial F}{\partial x_i}$

自变量 $x_1, \cdots x_n$ ↓ $u_1, \cdots, u_n$ F ↓ G

推广证明过程

$dF = \sum_{i=1}^n \frac{\partial F}{\partial x_i} dx_i + \sum_{i=1}^n \frac{\partial F}{\partial y_i} dy_i$

$dG = \sum_{i=1}^n \frac{\partial G}{\partial u_i} du_i + \sum_{i=1}^n \frac{\partial G}{\partial y_i} dy_i$

$dG = \sum_{i=1}^n x_i du_i + \sum_{i=1}^n u_i dx_i - \sum_{i=1}^n u_i dx_i - \sum_{i=1}^n \frac{\partial F}{\partial y_i} dy_i$

推广

$F = F(x_1, \cdots, x_n, y_1, \cdots, y_m)$ $u_i = \frac{\partial F}{\partial x_i}$ ✓

$G = G(u_1, \cdots, u_n, y_1, \cdots, y_m)$ ⇒ $G = \sum_{i=1}^n x_i u_i - F$ ⇒ $x_i = \frac{\partial G}{\partial u_i}$, $\frac{\partial G}{\partial y_i} = \frac{\partial F}{\partial y_i}$

4.1.2 哈密顿正则方程

s个自由度.

拉: $\frac{d}{dt}(\frac{\partial L}{\partial \dot{q}_\alpha}) = \frac{\partial L}{\partial q_\alpha}$, $\alpha = 1, \cdots, s$ s个二阶微分方程 ⇒ $\dot{q}_\alpha$

拉格朗日函数 $L = L(q_1, \cdots, q_s; \dot{q}_1, \dot{q}_2 \cdots, \dot{q}_s, t)$

目标. y_i 对称 x_i

构造. 哈密顿函数 $H = H(q_1, \cdots, q_s; p_1, p_2, \cdots, p_s, t)$ 对动量 ⇒ u_i

对动量: $p_\alpha = \frac{\partial L}{\partial \dot{q}_\alpha}$ ⇐ $u_i = \frac{\partial F}{\partial x_i}$ (u_i, x_i)

构造. $H = \sum_{\alpha=1}^s \dot{q}_\alpha p_\alpha - L = \sum_{\alpha=1}^s p_\alpha \dot{q}_\alpha - L$

$\dot{q}_\alpha = \frac{\partial H}{\partial p_\alpha}$ 对速度. ⇐ 勒让德变换

$p_\alpha = \frac{\partial L}{\partial \dot{q}_\alpha}$ 对动量 $\frac{\partial L}{\partial q_\alpha} = -\frac{\partial H}{\partial q_\alpha}$ ⇒ $\dot{p}_\alpha = -\frac{\partial H}{\partial q_\alpha}$

$\frac{d}{dt}(\frac{\partial L}{\partial \dot{q}_\alpha}) = \frac{\partial L}{\partial q_\alpha}$ ⇒ $\dot{p}_\alpha = \frac{\partial L}{\partial q_\alpha}$ ($\frac{\partial L}{\partial \dot{q}_\alpha} = p_\alpha$)

哈密顿正则方程 $H(q_\alpha, p_\alpha, t)$

$\frac{dq_\alpha}{dt} = \frac{\partial H}{\partial p_\alpha} \Rightarrow \dot{q}_\alpha = \frac{\partial H}{\partial p_\alpha}$, $\alpha = 1, \cdots, s$

$\frac{dp_\alpha}{dt} = -\frac{\partial H}{\partial q_\alpha} \Rightarrow \dot{p}_\alpha = -\frac{\partial H}{\partial q_\alpha}$, $\alpha = 1, \cdots, s$

①2s个一阶微分方程.

②形式简单对称, p_α, q_α 正则变量 正则坐标.

4.1.3 运动积分

对动量守恒: H不显含 q_β, $\frac{\partial H}{\partial q_\beta} = 0 \Rightarrow \dot{p}_\beta = 0 \Rightarrow p_\beta = C$

勒让德变换 $\frac{\partial L}{\partial \dot{q}_\beta} = -\frac{\partial H}{\partial q_\beta} = 0 \Rightarrow L$ 不显含 $\dot{q}_\beta$.

对动量积分

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设拉氏量不显含 t（积分）$H(p,q,t)$

则 $\frac{dH}{dt}=\frac{\partial H}{\partial t}+\sum_{i=1}^{s}\frac{\partial H}{\partial q_i}\dot{q}_i+\sum_{i=1}^{s}\frac{\partial H}{\partial p_i}\dot{p}_i=0$

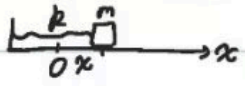
$\dot{p}_i=-\frac{\partial H}{\partial q_i}$，$\frac{\partial H}{\partial q_i}$ 为 $-\dot{p}_i$；$\frac{\partial H}{\partial p_i}$ 为 $\dot{q}_i$

若 H 不显含 t，那么，H 是运动积分。

$\frac{dH}{dt}=0=\frac{\partial H}{\partial t}=-\frac{\partial L}{\partial t}\Rightarrow L$ 不显含时间。

$$\dot{q}_i=\frac{\partial H}{\partial p_i}\quad;\quad \dot{p}_i=-\frac{\partial H}{\partial q_i}$$ 正则方程

例1. 谐振子。



$H=\sum_i p_i\dot{q}_i-L$

$L=T-V=\frac{1}{2}m\dot{x}^2-\frac{1}{2}kx^2$

$H(q,p,t)$

$p_x=\frac{\partial L}{\partial \dot{x}}=m\dot{x}\Rightarrow \dot{x}=\frac{p_x}{m}$

$H=p_x\cdot\dot{x}-L=\frac{p_x^2}{m}-\left(\frac{1}{2}m\cdot\frac{p_x^2}{m^2}-\frac{1}{2}kx^2\right)=\frac{p_x^2}{2m}+\frac{1}{2}kx^2$

$\dot{x}=\frac{\partial H}{\partial p_x}=\frac{p_x}{m}\Rightarrow \dot{p}_x=m\ddot{x}$

$\dot{p}_x=-\frac{\partial H}{\partial x}=-kx$

$\Rightarrow m\ddot{x}+kx=0$

$\frac{d}{dt}p_x=+\frac{\partial L}{\partial x}$，$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{x}}\right)=\frac{\partial L}{\partial x}$

目前为止，哈密顿正则方程 $\Leftrightarrow$ 拉格朗日方程等价。

相空间

我们动力学量。

有心力 $L=T-V=\frac{1}{2}m(\dot{r}^2+r^2\dot{\theta}^2)-V$

拉：$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{r}}\right)=\frac{\partial L}{\partial r}\Rightarrow m(\ddot{r}-r\dot{\theta}^2)=-\frac{dV}{dr}$.

$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{\theta}}\right)=\frac{\partial L}{\partial \theta}\Rightarrow r^2\dot{\theta}=C$ 角动量守恒 L 显含 θ.

哈：$\begin{cases}p_r=\frac{\partial L}{\partial \dot{r}}=m\dot{r}\\ p_\theta=\frac{\partial L}{\partial \dot{\theta}}=mr^2\dot{\theta}\end{cases}$

$H=\dot{r}p_r+\dot{\theta}p_\theta-L=\frac{1}{2}\frac{p_r^2}{m}+\frac{1}{2}\frac{p_\theta^2}{mr^2}+V$

H 不显含 θ，角动量守恒；H 不显含 t，能量守恒

证：$\begin{cases}\dot{r}=\frac{\partial H}{\partial p_r}=\frac{p_r}{m}\Rightarrow p_r=m\ddot{r}\\ \dot{p}_r=-\frac{\partial H}{\partial r}=mr\dot{\theta}^2-\frac{dV}{dr}\end{cases}\Rightarrow m\ddot{r}-mr\dot{\theta}^2=-\frac{dV}{dr}$.

$\begin{cases}\dot{\theta}=\frac{\partial H}{\partial p_\theta}=\frac{p_\theta}{mr^2}\\ \dot{p}_\theta=-\frac{\partial H}{\partial \theta}=0\end{cases}\Rightarrow mr^2\dot{\theta}=C$

相对论力学 $H=mc^2+V=m_0c^2+T+V$

带电粒子. $H=\frac{1}{2}mv^2+e\varphi=?$

$L=\frac{1}{2}mv^2-e\varphi+e\vec{v}\cdot\vec{A}$

$\vec{p}=\frac{\partial L}{\partial \vec{v}}=m\vec{v}+e\vec{A}$ 正则动量.

$H=p\cdot\dot{q}-L=\vec{p}\vec{v}-L=mv^2+e\vec{A}\cdot\vec{v}-\frac{1}{2}mv^2-e\varphi+e\vec{A}\cdot\vec{v}$

$=\frac{1}{2}mv^2+e\varphi$

经典力学. ② 等于 ⑪ 能量守恒

⇓ 能量1

⑪ 等于 L 能量守恒.

新的体系. 能量守恒 $\Rightarrow$ H 守恒

弗恩曼

2020年2月15日 19:00 分析力学(六)

CHAPTER 4 哈密顿力学

4.1 哈密顿正则方程

4.1.1 勒让德变换

$$\left.\begin{aligned} F&=F(x_1,\cdots,x_n;y_1\cdots y_n)\\ G&=F(u_1\cdots,u_n,y_1\cdots y_n)\\ G+F&=\sum_{i=1}^{n}x_iu_i \end{aligned}\right\}\Rightarrow \begin{aligned} u_i&=\frac{\partial F}{\partial x_i}\\ x_i&=\frac{\partial G}{\partial u_i}\\ \frac{\partial F}{\partial y_j}&=-\frac{\partial G}{\partial y_j} \end{aligned}$$

4.1.2 正则方程

$$H=\sum_{\alpha=1}^{s}p_\alpha\dot q_\alpha-L\Rightarrow\begin{cases}\dot q_\alpha=\dfrac{\partial H}{\partial p_\alpha}\\ \dot p_\alpha=-\dfrac{\partial H}{\partial q_\alpha}\end{cases}$$ 正则方程.

4.1.3 运动积分.

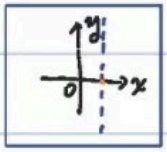
$H(p,q,t)$ 不含 q_β. $p_\beta=C$ 运动积分.

$H(p,q,t)$ 不含 t $H=C$ 广义能量积分.

保守机械系统 $T=\square\cdot\dot q_\alpha\dot q_\beta\Rightarrow T_2$ V 动能 势能

$H=T+V$

4.2 相空间



x,y 相互独立 $y=f(x)$ → 曲线

对称性: x → 方向多了 y.

$L=L(q,\dot q,t)=L(q_1,\cdots,q_s,\dot q_1,\cdots\dot q_s,t)$

q 广义坐标, 独立 → s 个广义坐标, 独立

↓ 相空间.

↓ 张成 s 维的空间(位形空间)

找到一个守恒量, s 个运动积分 → 对 s 个一阶微分方程.

$H=H(q_1,\cdots,q_s,p_1,\cdots,p_s,t)$ q,p 互相独立吗?

↓ 相空间

↓ $2s$ 维空间(相空间)

找到 $2s$ 个守恒量. $2s$ 个运动积分 → 方程求解了 $2s$ 个一阶微分方程.

q,p 相互独立. 困难? → 解决问题.

$p_\alpha=\dfrac{\partial L}{\partial\dot q_\alpha}$ → 能拉格朗日 L

$\dot q_\alpha\to\dot q_\alpha\to p_\alpha$ 比要方便多了

一一对应关系?

方法. $L=?$ 参照 L 方便多. $p_\alpha=\dfrac{\partial L}{\partial\dot q_\alpha}$ 相同 L 不同 无论 $\dot q\to\dot q\leftarrow p_\alpha$

求 L, $\Rightarrow\dfrac{d}{dt}\left(\dfrac{\partial L}{\partial\dot q_\alpha}\right)=\dfrac{\partial L}{\partial q}\Rightarrow$ 动力学方程 $\Rightarrow$ 验证是不是对的.

证明: $f=f(q_1,\cdots,q_s,t)$

$$\dot f=\frac{df}{dt}=\frac{\partial f}{\partial t}+\sum_{\beta=1}^{s}\frac{\partial f}{\partial q_\beta}\dot q_\beta$$

$$\frac{\partial\dot f}{\partial\dot q_\alpha}=\frac{\partial}{\partial\dot q_\alpha}\left(\frac{\partial f}{\partial t}+\sum_{\beta=1}^{s}\frac{\partial f}{\partial q_\beta}\dot q_\beta\right)=0+\frac{\partial f}{\partial q_\alpha}$$

$\left(\frac{\partial f}{\partial t}\right)$: (q,t); $\left(\frac{\partial f}{\partial q_\beta}\right)$: (q,t)

$$\Rightarrow\frac{d}{dt}\left(\frac{\partial\dot f}{\partial\dot q_\alpha}\right)=\frac{d}{dt}\left(\frac{\partial f}{\partial q_\alpha}\right)=\frac{\partial^2 f}{\partial q_\alpha\partial t}+\sum_{\beta=1}^{s}\frac{\partial^2 f}{\partial q_\alpha\partial q_\beta}\cdot\dot q_\beta$$

$$\frac{\partial\dot f}{\partial q_\alpha}=\frac{\partial}{\partial q_\alpha}\left(\frac{\partial f}{\partial t}+\sum_{\beta=1}^{s}\frac{\partial f}{\partial q_\beta}\dot q_\beta\right)=\frac{\partial^2 f}{\partial q_\alpha\partial t}+\sum_{\beta=1}^{s}\frac{\partial^2 f}{\partial q_\alpha\partial q_\beta}\cdot\dot q_\beta$$

$\left(\frac{\partial f}{\partial q_\beta}\right)$: (q,t)

$$\boxed{\frac{d}{dt}\left(\frac{\partial\dot f}{\partial\dot q_\alpha}\right)-\frac{\partial\dot f}{\partial q_\alpha}=0}$$

$f=f(q,t)$ → 总能令 $L=f$

$$\frac{d}{dt}\left(\frac{\partial L}{\partial\dot q_\alpha}\right)-\frac{\partial L}{\partial q_\alpha}=0$$

结论. ① L 拉格朗日. $L'=L+f(q,t)$ 也拉格朗日.

② $\dot q_\alpha\longrightarrow p_\alpha=\dfrac{\partial L}{\partial\dot q_\alpha}$ 广义动量.

$f=f(q,t)$ 任意的 f 方向多了

$$p'_\alpha=\frac{\partial L'}{\partial\dot q_\alpha}=\frac{\partial}{\partial\dot q_\alpha}(L+f)=p_\alpha+\frac{\partial f}{\partial\dot q_\alpha}$$

一个广义坐标 → 方向多了广义动量.

q: 正则坐标

p 正则动量

正则共轭变量(辛结构) (p,q) $2s$ 维相空间

谐振动: $L=\frac{1}{2}m\dot{x}^2-\frac{1}{2}kx^2$ $H=\frac{1}{2}m\dot{x}^2+\frac{1}{2}kx^2$

$P_x=\frac{\partial L}{\partial \dot{x}}=m\dot{x}\Rightarrow \dot{x}=\frac{P_x}{m}\Rightarrow H=\frac{P_x^2}{2m}+\frac{1}{2}kx^2$

运动方程: $\dot{P}_x=-\frac{\partial H}{\partial x}=-kx$ (P_x, x) 相轨迹

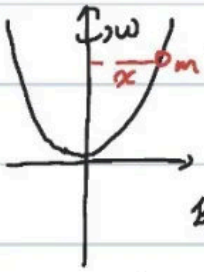
$kx=-\dot{P}_x=-\frac{dP_x}{dx}\cdot\dot{x}=-\frac{dP_x}{dx}\cdot\frac{P_x}{m}$

$\Rightarrow mkx\,dx=-p\,dp\Rightarrow mkx^2+p^2=C\Rightarrow$ 相轨迹. 椭圆轮廓.

↑ 动量

相轨迹.

周期性运动 $\Rightarrow$ 闭合曲线. 初始条件. 重要. $\Rightarrow$ 不依赖于. 相轨迹. 强依赖于.



例: ① 平衡点. 与 x 无关. (此系统 ω 恒定)

② 平衡. 稳定/不稳定.

解: $q=x$

$L=T-V=\frac{1}{2}m\left(\left(1+\frac{x^2}{4a^2}\right)\dot{x}^2+\omega^2x^2\right)-mg\frac{x^2}{4a^2}$

下次说

$P_x=\frac{\partial L}{\partial \dot{x}}=m\left(1+\frac{x^2}{4a^2}\right)\dot{x}$

$H=P_x\cdot\dot{x}-L=m\left(1+\frac{x^2}{4a^2}\right)\dot{x}^2-\frac{1}{2}m\left(\left(1+\frac{x^2}{4a^2}\right)\dot{x}^2+\omega^2x^2\right)+mg\frac{x^2}{4a^2}$

$\Downarrow$ $H=H(q,P,t)$ $q\ \dot{x}$

$H=\frac{P_x^2}{2m\left(1+\frac{x^2}{4a^2}\right)}-\left(\frac{1}{2}m\omega^2-\frac{mg}{4a}\right)x^2$

$\dot{x}=\frac{\partial H}{\partial P_x}=\frac{P_x}{m\left(1+\frac{x^2}{4a^2}\right)}\Rightarrow \dot{P}_x=\frac{d}{dt}\left(\dot{x}m\left(1+\frac{x^2}{4a^2}\right)\right)=\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{x}}\right)$

$\dot{P}_x=-\frac{\partial H}{\partial x}=\cdots=\frac{\partial L}{\partial x}$

$\Rightarrow \frac{d}{dt}\left(\frac{\partial L}{\partial \dot{x}}\right)=\frac{\partial L}{\partial x}$

② 稳定平衡 $\Rightarrow$ 小振动 $\Rightarrow$ 闭合曲线.

$H=\frac{P_x^2}{2m\left(1+\frac{x^2}{4a^2}\right)}-\left(\frac{1}{2}m\omega^2-\frac{mg}{4a}\right)x^2$

$\Downarrow$ 平衡位置 $x=0$, $P_x=0\Rightarrow$ 坐标原点

$x\ll 1$ $H\approx\frac{P_x^2}{2m}-\left(\frac{1}{2}m\omega^2-\frac{mg}{4a}\right)x^2$

$H\approx\frac{P_x^2}{2m}-\left(\frac{1}{2}m\omega^2-\frac{mg}{4a}\right)x^2=C\Rightarrow$ 椭圆相轨迹.

① H不显含 t. H 为守恒量. $=C$

② 稳定平衡 $\Rightarrow$ 小振动 $\Rightarrow$ 闭合相轨迹.

相图: $-\left(\frac{1}{2}m\omega^2-\frac{mg}{4a}\right)>0\Rightarrow 0<\omega^2<g/2a$

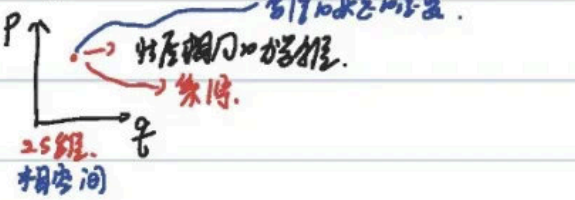
不稳定平衡 $\rightarrow$ 运动 $\rightarrow$ 开放轨道.

双曲线 $-\left(\frac{1}{2}m\omega^2-\frac{mg}{4a}\right)<0\Rightarrow \omega^2>g/2a$

刘维尔定理:

相体积 S很大. $S\to\infty$.

方法: 给定初始时刻 → 状态随时间的演化 → 轨迹.



$dV\Rightarrow dN=\rho\cdot dq_1\cdots dq_s\,dp_1\cdots dp_s$

刘维尔定理: 在相空间运动过程中. 密度不变.

$\rho(p,q,t)$

$\frac{d\rho}{dt}=\frac{\partial\rho}{\partial t}+\sum_{i=1}^{s}\left[\frac{\partial\rho}{\partial p_i}\cdot\dot{p}_i+\frac{\partial\rho}{\partial q_i}\cdot\dot{q}_i\right]$

对时间变化 相空间中运动

$dA_1=dq_2\cdots dq_s\,dp_1\cdots dp_s$

进: $\rho\cdot\dot{q}_1\,dA_1|_{q_1}$

出: $\rho\,\dot{q}_1\,dA_1|_{q_1+dq_1}$

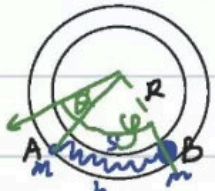
$\frac{\partial N_1}{\partial t}=\rho\dot{q}_1\,dA_1|_{q_1}-\rho\dot{q}_1\,dA_1|_{q_1+dq_1}$

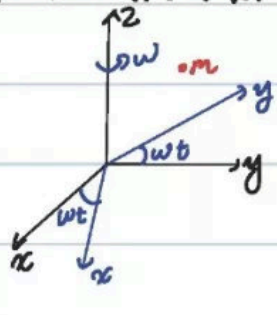
$=-\frac{\partial(\rho\dot{q}_1)}{\partial q_1}\cdot dq_1\,dA_1=-\frac{\partial(\rho\dot{q}_1)}{\partial q_1}\cdot dV$

密度变化. $\frac{\partial\rho_1}{\partial t}=\frac{1}{dV}\cdot\frac{\partial N_1}{\partial t}=-\frac{\partial(\rho\dot{q}_1)}{\partial q_1}$

$\frac{\partial\rho}{\partial t}=-\sum_{i=1}^{s}\left[\frac{\partial(\rho\dot{q}_i)}{\partial q_i}+\frac{\partial(\rho\dot{p}_i)}{\partial p_i}\right]$

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R
θ
φ
A
B
m
m
k


z
ω
m
y
ωt
y
ωt
x
x

例：力→找 H. 运动轨道 ① 有心力. 目标：U, L, H

有限 中心力 单位制 ① $F(r)=\frac{1}{r^2}\left(1-\frac{\dot r^2-2r\ddot r}{c^2}\right)$

解：二维有心力 (r,θ) $\vec F(r)=-\frac{1}{r^2}\left(1-\frac{\dot r^2-2r\ddot r}{c^2}\right)\hat r$ 验证

$\theta_r=-\frac{1}{r^2}\left(1-\frac{\dot r^2-2r\ddot r}{c^2}\right)\hat r$, $Q_\theta=0$

$Q_r=\frac{d}{dt}\left(\frac{\partial U}{\partial \dot r}\right)-\frac{\partial U}{\partial r}=-\frac{1}{r^2}\left(1-\frac{\dot r^2-2r\ddot r}{c^2}\right)$

$Q_\theta=\frac{d}{dt}\left(\frac{\partial U}{\partial \dot\theta}\right)-\frac{\partial U}{\partial \theta}=0\Rightarrow U(r,\dot r)$ 与 θ 无关

$\frac{d}{dt}\left(\frac{\partial U}{\partial \dot r}\right)=\frac{\partial^2 U}{\partial \dot r^2}\cdot\ddot r+\frac{\partial^2 U}{\partial \dot r\partial r}\cdot\dot r=\frac{\partial U}{\partial r}-\frac{1}{r^2}\left(1-\frac{\dot r^2-2r\ddot r}{c^2}\right)$ （$\frac{\partial^2 U}{\partial \dot r\partial r}$ 不含 $\ddot r$）

$=-\frac{1}{r^2}+\frac{\dot r}{c^2r^2}-\frac{2r\ddot r}{c^2r^2}+\frac{\partial U}{\partial r}$

$\frac{\partial^2 U}{\partial \dot r^2}\cdot\ddot r=-\frac{2r\cdot\dot r}{c^2r^2}\Rightarrow\frac{\partial U}{\partial \dot r}=-\frac{2}{c^2r}$

积 $\Rightarrow\frac{\partial U}{\partial \dot r}=-\frac{2\dot r}{c^2r}+f(r)$

再积 $\Rightarrow U=-\frac{\dot r}{c^2r}+f(r)\cdot\dot r+g(r)$

$\frac{\partial^2 U}{\partial r\partial \dot r}\cdot\dot r-\frac{\partial U}{\partial r}=-\frac{1}{r^2}\left(1-\frac{\dot r^2}{c^2}\right)$

$\Rightarrow\frac{dg(r)}{dr}=\frac{1}{r^2}\Rightarrow g(r)=-\frac{1}{r}$

$U=-\frac{1}{r}\left(1+\frac{\dot r^2}{c^2}\right)+f(r)\dot r$ 无法确定.

$L=T-U=\frac{1}{2}m(\dot r^2+r^2\dot\theta^2)+\frac{1}{r}\left(1+\frac{\dot r^2}{c^2}\right)-f(r)$

↓ P_r, P_θ $H=\dot\theta P_\theta+\dot r P_r-L$

$\ddot x+i\ddot y=i\omega_c(\dot x+\frac{1}{2}\dot y)+\frac{1}{2}\omega_z^2(x+iy)$

$\ddot u=-i\omega_c\dot u+\frac{1}{2}\omega_z^2 u$

$\Rightarrow\ddot u-i\omega_c\dot u+\frac{1}{2}\omega_z^2u=0\Rightarrow u=e^{-i\omega t}$ 匀速运动

$\Rightarrow\omega_0^2-\omega_c\omega+\frac{1}{2}\omega_z^2=0\Rightarrow\omega_\pm=\frac{1}{2}\left(\omega_c\pm\sqrt{\omega_c^2-2\omega_z^2}\right)$

$\omega_c^2>2\omega_z^2\Rightarrow$ 简谐运动 $\Rightarrow$ 叠加 $\Rightarrow$ 椭圆

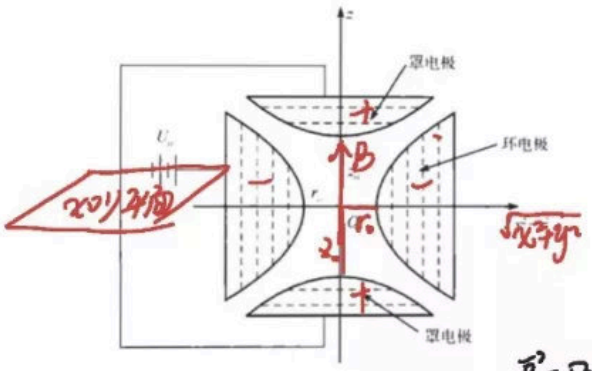
$x=A\cos(\omega_+t+\varphi_+)+B\cos(\omega_-t+\varphi_-)$

$y=A\sin(\omega_+t+\varphi_+)+B\sin(\omega_-t+\varphi_-)$

彭宁阱 (Penning trap) 是一种用来将带电粒子、离子等俘陷在空间的一个特定区域中进而研究它们的物理性质或进行其他相关高精度实验的一种装置，有许多非常重要的应用⑤. 彭宁阱由一对罩电极和一个环电极组成，电极表面均为关于 z 轴的旋转双曲面 (题 8.25 图). 对带正电荷的粒子，在环电极和罩电极上分别加正负电压并在 z 方向施加磁场，带电粒子将在电磁场的作用下局限在阱中的某一区域内运动. 设磁场是均匀的，且 $\boldsymbol{B}=B\boldsymbol{k}$，其中 B 为常数. 电场强度为 $\boldsymbol{E}=-\nabla\Phi$，其中电势 Φ 由下式给出：

$$\Phi(x,y,z)=\frac{U_0}{2z_0^2+r_0^2}\left(2z^2-x^2-y^2\right)\tag{1}$$

其中，U_0 为罩电极和环电极之间的电势差，r_0，z_0 分别为电极表面与径向轴 $\sqrt{x^2+y^2}$ 和 z 轴的交点到坐标原点 O 的距离 (题 8.25 图). 试求在电磁场作用下，电量为 $q>0$，质量为 m 的带电粒子的运动规律.



$\vec B=\nabla\times\vec A$

$T=\frac{1}{2}m(\dot x^2+\dot y^2+\dot z^2)$ $U=q\varphi-q\vec v\cdot\vec A$ $\vec A=\frac{1}{2}B(-y\vec i+x\vec j)$

$L=\frac{1}{2}m(\dot x^2+\dot y^2+\dot z^2)-\frac{1}{2}m\omega_z^2(2z^2-x^2-y^2)-\frac{1}{2}m\omega_c(\dot x y-\dot y x)$

$\omega_z^2=\frac{4qU_0}{m(2z_0^2+r_0^2)}$ $\omega_c^2=\frac{q^2B^2}{m^2}$

$P_x=\frac{\partial L}{\partial \dot x}=m\dot x+qA_x=m\dot x+q\cdot(-\frac{1}{2}By)$

$P_y=\frac{\partial L}{\partial \dot y}=m\dot y+qA_y=m\dot y+q\cdot(\frac{1}{2}Bx)$

$P_z=\frac{\partial L}{\partial \dot z}=m\dot z+0$

$H=\dot xP_x+\dot yP_y+\dot zP_z-L=$

$=\frac{1}{2m}(P_x^2+P_y^2+P_z^2)-\frac{1}{2}\omega_c(xP_y-yP_x)+\frac{1}{8}m(\omega_c^2-2\omega_z^2)(x^2+y^2)+\frac{1}{2}m\omega_z^2z^2$

正则运动 $\dot x=\frac{P_x}{m}+\frac{1}{2}\omega_c y$ $\dot P_x=\frac{1}{2}\omega_cP_y-\frac{1}{4}m(\omega_c^2-2\omega_z^2)x$

$\dot y=\frac{P_y}{m}-\frac{1}{2}\omega_cx$ $\dot P_y=-\frac{1}{2}\omega_cP_x-\frac{1}{4}m(\omega_c^2-2\omega_z^2)y$

$\dot z=\frac{P_z}{m}$ $\dot P_z=-m\omega_z^2z$

$\ddot x=\omega_c\dot y+\frac{1}{2}\omega_z^2x$ ，$i\ddot y=-\omega_ci\dot x+\frac{1}{2}\omega_z^2iy$ ，$u=x+iy$ $\Rightarrow$ $x=A\cos(\omega t+\varphi_1)$，$y=B\cos(\omega t+\varphi_2)$

$\ddot z=-\omega_z^2z\Rightarrow$ 已为作简谐运动 $z=A_z\cos(\omega_zt+\varphi_z)$

4.3 泊松括号.

4.3.1 泊松括号.

力学量 $f=f(q,p,t)$ 相空间的函数.

$$\dot f=\frac{df}{dt}=\frac{\partial f}{\partial t}+\sum_{\alpha=1}^{s}\frac{\partial f}{\partial q_\alpha}\dot q_\alpha+\sum_{\alpha=1}^{s}\frac{\partial f}{\partial p_\alpha}\dot p_\alpha$$

($\dot q_\alpha=\frac{\partial H}{\partial p_\alpha}$, $\dot p_\alpha=-\frac{\partial H}{\partial q_\alpha}$)

$$\frac{df}{dt}=\frac{\partial f}{\partial t}+\sum_{\alpha=1}^{s}\left(\frac{\partial f}{\partial q_\alpha}\frac{\partial H}{\partial p_\alpha}-\frac{\partial f}{\partial p_\alpha}\frac{\partial H}{\partial q_\alpha}\right)$$

定义括号: 泊松括号. $[f,g]=\sum_{\alpha=1}^{s}\left(\frac{\partial f}{\partial q_\alpha}\frac{\partial g}{\partial p_\alpha}-\frac{\partial f}{\partial p_\alpha}\frac{\partial g}{\partial q_\alpha}\right)$

$f=f(q,p,t)$ $g=f(q,p,t)$ 泊松括号.

$$\frac{df}{dt}=\frac{\partial f}{\partial t}+[f,H]$$ f H (q,p,t) 力学量.

f 是守恒量 (运动积分) $\frac{df}{dt}=0=\frac{\partial f}{\partial t}+[f,H]=0$

$\Uparrow$ 守恒量条件. $\frac{\partial f}{\partial t}+[f,H]=0$

特例: $f=H$ 守恒量 $\frac{dH}{dt}=0=\frac{\partial H}{\partial t}+[H,H]=0$ ($\frac{\partial H}{\partial t}=0$ H 不显含时间. $[H,H]=0$ 显然)

$[f,g]=\sum_{\alpha=1}^{s}\left(\frac{\partial f}{\partial q_\alpha}\frac{\partial g}{\partial p_\alpha}-\frac{\partial f}{\partial p_\alpha}\frac{\partial g}{\partial q_\alpha}\right)$ f g (q,p,t) 力学量

性质 1. $[c,f]=0$ (c 为常数)

2. $[f,f]=0$

3. $[f,g]=-[g,f]$ (反对称性)

4. $[cf,g]=c[f,g]=[f,cg]$

5. $[f,g_1+g_2]=[f,g_1]+[f,g_2]$

6. $[f,g_1g_2]=[f,g_1]g_2+g_1[f,g_2]$

7. $\frac{\partial[f,g]}{\partial t}=[\frac{\partial f}{\partial t},g]+[f,\frac{\partial g}{\partial t}]$

8. $[q_\alpha,f]=\frac{\partial f}{\partial p_\alpha}$ $[q_\alpha,H]=\frac{\partial H}{\partial p_\alpha}=\dot q_\alpha$

($f=H$) $[p_\alpha,f]=-\frac{\partial f}{\partial q_\alpha}$ $[p_\alpha,H]=-\frac{\partial H}{\partial q_\alpha}=\dot p_\alpha$

正则方程: $\begin{cases}\dot q_\alpha=[q_\alpha,H]\\ \dot p_\alpha=[p_\alpha,H]\end{cases}$ 关于 H 对称. $\Rightarrow$ 是力学基本.

9. 基本泊松括号.

$[q_\alpha,q_\beta]=0$

$[p_\alpha,p_\beta]=0$

$[q_\alpha,p_\beta]=\delta_{\alpha\beta}=\begin{cases}\alpha=\beta & 1\\ \alpha\neq\beta & 0\end{cases}$

10. 刻画正则变换. $\overset{a}{o}(q,p,t)$ 力学量.

$[a,[b,c]]+[b,[c,a]]+[c,[a,b]]=0$

弗恩曼

2020年2月16日 19:00 分析力学(七)

正则方程 $\begin{cases}\dot{q}_\alpha = \frac{\partial H}{\partial p_\alpha} \\ \dot{p}_\alpha = -\frac{\partial H}{\partial q_\alpha}\end{cases}$

相空间 (q_α, p_α) 2s维空间

辛性

泊松括号 $f(q,p,t)$ $g(q,p,t)$

定义 $[f,g] = \sum_\alpha \left(\frac{\partial f}{\partial q_\alpha}\frac{\partial g}{\partial p_\alpha} - \frac{\partial f}{\partial p_\alpha}\frac{\partial g}{\partial q_\alpha}\right)$

$\frac{df}{dt} = \frac{\partial f}{\partial t} + [f,H]$

性质 1. $[C,f]=0$ 2. $[f,f]=0$ 3. $[f,g]=-[g,f]$

4. $[cf,g] = c[f,g] = [f,cg]$

5. $[f,g_1+g_2] = [f,g_1] + [f,g_2]$

6. $[f,g_1 g_2] = [f,g_1]g_2 + g_1[f,g_2]$

7. $\frac{\partial [f,g]}{\partial t} = \left[\frac{\partial f}{\partial t}, g\right] + \left[f, \frac{\partial g}{\partial t}\right]$

8. $[q_\alpha, f] = \frac{\partial f}{\partial p_\alpha}$ 特例 $f=H$ $[q_\alpha, H] = \frac{\partial H}{\partial p_\alpha} = \dot{q}_\alpha$

$[p_\alpha, f] = -\frac{\partial f}{\partial q_\alpha}$ $[p_\alpha, H] = -\frac{\partial H}{\partial q_\alpha} = \dot{p}_\alpha$

正则方程: $\begin{cases}\dot{q}_\alpha = [q_\alpha, H] \\ \dot{p}_\alpha = [p_\alpha, H]\end{cases}$

9. $[q_\alpha, q_\beta]=0$ $[p_\alpha, p_\beta]=0$ $[q_\alpha, p_\beta] = \delta_{\alpha\beta}$

10. $[a,[b,c]] + [b,[c,a]] + [c,[a,b]] = 0$

对任意比性质.

4.3.2. 泊松定理.

泊松定理: 力学量 $f(q,p,t)$ $g(q,p,t)$ 是运动积分(守恒量)

那么 $[f,g]$ 也是守恒量(运动积分)

证: $\frac{df}{dt} = \frac{\partial f}{\partial t} + [f,H] = 0$ $\frac{dH}{dt} = \frac{\partial H}{\partial t} + [H,H] = 0$

$\frac{dg}{dt} = \frac{\partial g}{\partial t} + [g,H] = 0$

证明: $[H,[f,g]] + [f,[g,H]) + [g,[H,f]] = 0$ ($[H,f] = -[f,H]$)

$\Rightarrow -[H,[f,g]] - [f,-\frac{\partial g}{\partial t}] - [g,+\frac{\partial f}{\partial t}] = 0$

$\Rightarrow [[f,g],H] + [f,\frac{\partial g}{\partial t}] + [\frac{\partial f}{\partial t}, g] = 0$

$\Rightarrow [[f,g],H] + \frac{\partial}{\partial t}[f,g] = 0$

$\Rightarrow \frac{d[f,g]}{dt}$ $\Rightarrow [f,g]$ 守恒量.

例: 单质点角动量 $\vec{L}$, $[L_x, L_y]$; $[L_x, L^2]$

$[L_x, L_y] = [yp_z - zp_y, zp_x - xp_z] = xp_y - yp_x = L_z$

$\vec{L} = \vec{r}\times\vec{p}$, $L_x = yp_z - zp_y$

$[L_x, L_y] = [yp_z - zp_y, zp_x] + [yp_z - zp_y, -xp_z]$

$= [yp_z, zp_x] - [zp_y, p_x] - [yp_z, xp_z] + [zp_y, xp_z]$

$[yp_z, zp_x] = y[p_z, z]p_x + [y, zp_x]p_z = -yp_x$

$= yz[p_z, p_x] + y[p_z, z]p_x + z[y, p_x]p_z + [y,z]zp_z$

$[L_x, L^2] = [L_x, L_x^2 + L_y^2 + L_z^2] = [L_x, L_x^2] + [L_x, L_y^2] + [L_x, L_z^2]$

$= L_x[L_x, L_x] + [L_x, L_x]L_x = 0$

$+ L_y[L_x, L_y] + [L_x, L_y]L_y$ $[L_x, L^2] = [L_y, L^2] = [L_z, L^2] = 0$

$+ L_z[L_x, L_z] + [L_x, L_z]L_z$

例: $H = \frac{1}{2m}(\vec{p} - e\vec{A})^2 + e\varphi$ $\varphi = \varphi(x,y,z,t)$, $A = A(x,y,z,t)$.

计算 $[x,H]$ $[p_x,H]$. $[\dot{x}, \dot{v}_x]$

$[x,H] = \frac{1}{2m}[x, (\vec{p} - e\vec{A})^2] + e[x,\varphi] = 0$ 与 p 无关 泊松定义.

$= \frac{1}{2m}(\vec{p} - e\vec{A})[x, \vec{p} - e\vec{A}] + \frac{1}{2m}[x, \vec{p} - e\vec{A}](\vec{p} - e\vec{A})$

$= \frac{1}{m}(\vec{p} - e\vec{A})[x, \vec{p}] + \frac{1}{m}(\vec{p} - e\vec{A})[x, -e\vec{A}] = 0$

弗恩曼

$[x,H]=\frac{1}{m}(\vec{p}-e\vec{A})[x,\vec{p}]=\frac{1}{m}(\vec{p}-e\vec{A})[x,p_x\vec{i}+p_y\vec{j}+p_z\vec{k}]$

$=\frac{1}{m}(\vec{p}-e\vec{A})\left(p_x[x,\vec{i}]+[x,p_x]\vec{i}+p_y[x,\vec{j}]+[x,p_y]\vec{j}+p_z[x,\vec{k}]+[x,p_z]\vec{k}\right)$

$=\frac{1}{m}(\vec{p}-e\vec{A})(\vec{i})=\frac{1}{m}(p_x-eA_x)$

$[\vec{r},H]=\frac{1}{m}(\vec{p}-e\vec{A})$

$[x,H]=\frac{\partial H}{\partial p_x}=\dot{x}\Rightarrow[\vec{r},H]=\dot{\vec{r}}=\vec{v}\Rightarrow$

$m\vec{v}=\vec{p}-e\vec{A}\Rightarrow\vec{p}=m\vec{v}+e\vec{A}$

$[p_x,H]=\frac{1}{2m}[p_x,(\vec{p}-e\vec{A})^2]+[p_x,e\varphi]$ 定义

$=\frac{1}{m}(\vec{p}-e\vec{A})[p_x,\vec{p}-e\vec{A}]-e\frac{\partial\varphi}{\partial x}$

$=\vec{v}([p_x,\vec{p}]-[p_x,e\vec{A}])-e\frac{\partial\varphi}{\partial x}$

$[p_x,\vec{p}]=0$，$[p_x,e\vec{A}]$ 由定义 $=-e\frac{\partial\vec{A}}{\partial x}$

$[p_x,H]=-e\vec{v}\cdot\frac{\partial\vec{A}}{\partial x}-e\frac{\partial\varphi}{\partial x}$

$[\vec{p},H]=-e\nabla\varphi+e\nabla(\vec{v}\cdot\vec{A})=\dot{\vec{p}}=\frac{d\vec{p}}{dt}$（$-e\nabla\varphi=\vec{E}$，$\nabla(\vec{v}\cdot\vec{A})=\vec{v}\times\vec{B}$）

$m\frac{d^2\vec{r}}{dt^2}=e\vec{E}+e\vec{v}\times\vec{B}$

例1 质量为 m 的质点在引力势场 $V=-\frac{k}{r}$ 中运动. 假定运动的轨道平面为 xy 平面, 则系统的哈密顿函数为

$$H=\frac{1}{2m}(p_x^2+p_y^2)-\frac{k}{r},$$

其中, $r=\sqrt{x^2+y^2}$. 角动量矢量沿 z 轴方向, 它的 z 分量为

$$J=xp_y-yp_x,$$

Rung-Lenz 矢量 K 位于 xy 平面内, 其分量分别为

$$K_x=p_yJ-\frac{mkx}{r},\quad K_y=-p_xJ-\frac{mky}{r}.$$

试证明: ① 角动量和 Rung-Lenz 矢量的两个分量与系统的哈密顿函数分别是对合的

$$[J,H]=0,\quad [K_x,H]=0,\quad [K_y,H]=0,$$

即 J,K_x,K_y 为系统的运动积分. ② 角动量和 Rung-Lenz 矢量的两个分量之间的 Poisson 括号为

$$[K_x,J]=-K_y,\quad [K_y,J]=K_x,\quad [K_x,K_y]=-(2mH)J.$$

如果限于考虑束缚态, 即 $H=-|E|$, 则证明 $\frac{K_x}{\sqrt{2m|E|}},\frac{K_y}{\sqrt{2m|E|}},J$ 与三维角动量的分量有相同的 Poisson 括号关系. ③ $\frac{K_x}{\sqrt{2m|E|}},\frac{K_y}{\sqrt{2m|E|}},J$ 的平方和等于 $\frac{mk^2}{2|E|}$.

例2 质量为 m 的质点在各向同性的简谐振子势场 $V=\frac{1}{2}kr^2$ 中运动. 假定运动的轨道平面为 xy 平面, 则系统的哈密顿函数为

$$H=S_0=\frac{1}{2m}(p_x^2+p_y^2)+\frac{1}{2}k(x^2+y^2).$$

定义

$$S_1=\frac{1}{2m}(p_x^2-p_y^2)+\frac{1}{2}k(x^2-y^2),\quad S_2=\frac{1}{m}p_xp_y+kxy,\quad S_3=\omega(xp_y-yp_x),$$

其中, $\omega=\sqrt{k/m}$. 证明: ① S_0 和 $S_i(i=1,2,3)$ 是分别对合的, 即它们的 Poisson 括号为

$$[S_0,S_i]=0,\quad i=1,2,3,$$

也即 S_1,S_2,S_3 是系统的运动积分. ② S_i 之间有关系

$$[S_1,S_2]=2\omega S_3,\quad [S_2,S_3]=2\omega S_1,\quad [S_3,S_1]=2\omega S_2.$$

③ S_0 与各 S_i 之间有关系

$$S_0^2=S_1^2+S_2^2+S_3^2.$$

解: $[S_1,S_2]=\left(\frac{\partial S_1}{\partial x}\frac{\partial S_2}{\partial p_x}-\frac{\partial S_1}{\partial p_x}\frac{\partial S_2}{\partial x}\right)+\left(\frac{\partial S_1}{\partial y}\frac{\partial S_2}{\partial p_y}-\frac{\partial S_1}{\partial p_y}\frac{\partial S_2}{\partial y}\right)$

$=kx\cdot\frac{p_y}{m}-\frac{p_x}{m}\cdot ky+(-ky)\cdot\frac{p_x}{m}-\left(-\frac{p_y}{m}\right)\cdot ky$

CHAPTER5 哈密顿原理(灵魂)

最重要、最普遍、最简洁、最深刻的人类(宇宙)

经典力学理论 最高阶段

↓

两套 相辅 ⇒ 理论力学 基本框架

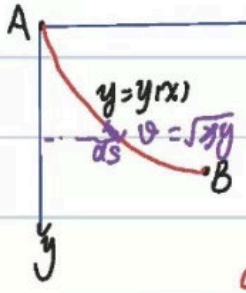
人类 基础理论 ⇒ 物理科学

5.1 变分法初步

5.1.1 变分学历史名题

1. 最速降线问题 1696 John Bernoulli 提出 (摆线)

好 Daniel Bernoulli 证明了结果



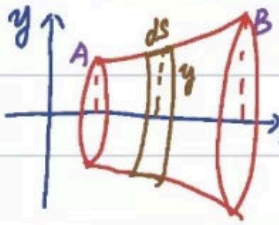
$y=y(x)$

$A(0,0)$ $B(x_B,y_B)$

$dt=\frac{ds}{v}=\frac{\sqrt{1+y'^2}\,dx}{\sqrt{2gy}}$ $y'=\frac{dy}{dx}$

$T_{min}=\int_0^{x_B}\frac{\sqrt{1+y'^2}}{\sqrt{2gy}}dx$ 我一条合适的 $y=y(x)$ 使 T_{min}

2. 回转体的最小侧面积 — 悬链线



$A(x_1,y_1)$ $B(x_2,y_2)$

我一条出镜曲线，使得 $y=y(x)$

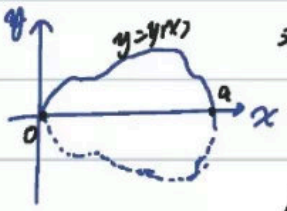
AB 依然不同 回转体侧面积小

$A=\int_{x_1}^{x_2}2\pi y\,ds=\int_{x_1}^{x_2}2\pi y\sqrt{1+y'^2}\,dx$

$A_{min}[y]=2\pi\int_{x_1}^{x_2}\sqrt{1+y'^2}\,dx$ 我一条合适的 $y=y(x)$ 使 A_{min}

3. 等周长问题 (周周)

封闭线 周长 l 固定，求所围面积最大值



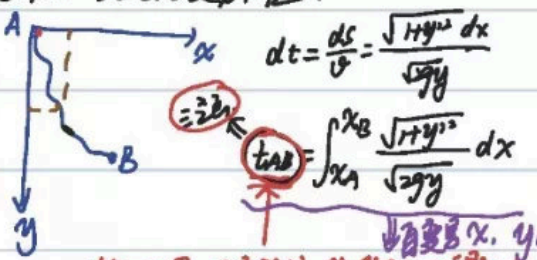
对称 $l=\int_0^a\sqrt{1+y'^2}\,dx=\int ds$

$y(0)=y(a)=0$

$A=\int_0^a y\,dx=\int_0^l y\sqrt{1-(\frac{ds}{dx})^2}\,ds$

$(dx)^2+(dy)^2=(ds)^2$ $y=y(x)$ 使得 $A=A_{max}$

5.1.2 泛函与变分问题



$dt=\frac{ds}{v}=\frac{\sqrt{1+y'^2}\,dx}{\sqrt{2gy}}$

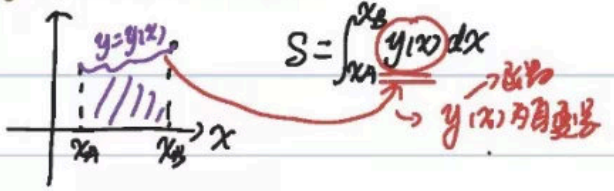
代入不同的曲线形状，$y=y(x)$ ⇒ 得到 t_{AB}

因变量 t

t_{AB} 取决于 曲线形状 $(y=y(x))$ 自变量 $y(x)$

t 称为 $y(x)$ 的泛函 (数) ⇒ 函数的函数

从 $y(x)$ 出发为自变量，得到值



一般化处理 $J=\int_{x_1}^{x_2}F(x,y,y')\,dx$; $y=y(x)$, $y'=\frac{dy}{dx}$

记作 $J=J[y(x)]$ ⇒ 变量 J 的值取决于 $y=y(x)$ 整体形态

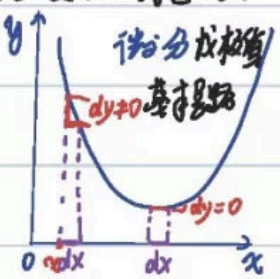
求 J 为 $y(x)$ 的泛函的 (泛函)

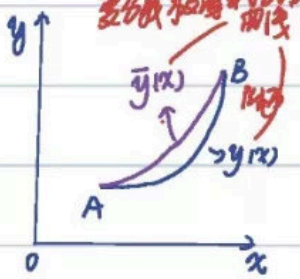
记作 $J=J[y(x)]=\int_{x_1}^{x_2}F(x,y,y')\,dx$.

不同的 $y(x)$ 代入 ⇒ 不同的 J 的值 ⇒ J 极值

泛函的极值问题 ⇒ 变分问题

5.1.3 变分的基本思路

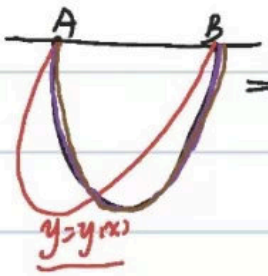




$\delta J=\bar{J}-J=0$ ⇐ $J=J[y(x)]$, $\bar{J}=\bar{J}[\bar{y}(x)]$

当 $y=y(x)$ 时，J 达到极值

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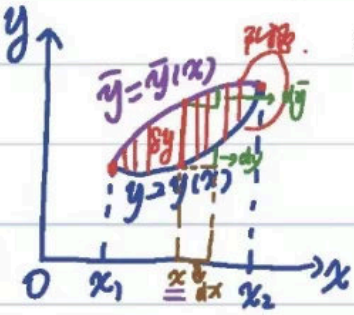
$\Rightarrow V=V[y(x)]$ V_{min} 稳定平衡.

$\delta V=V[y(x)]-\bar{V}[\bar{y}(x)]=0$

$y(x)$ 为极小.

微分极值 $\Rightarrow \frac{dy}{dx}=0$

变分极值 $\Rightarrow \boxed{\frac{\partial F}{\partial y}-\frac{d}{dx}\left(\frac{\partial F}{\partial y'}\right)}=0$



极值条件. 若 $y=y(x)$ 使 J 取极值.

$\delta J=\bar{J}[\bar{y}(x)]-J[y(x)]=0$

$=\int_{x_1}^{x_2}\bar{F}(x,\bar{y},\bar{y}')dx$

$-\int_{x_1}^{x_2}F(x,y,y')dx=0$

注意. ① 微分. 自变量 x 变. $x\to x+dx \Rightarrow y\to y+dy$

变分. 自变量 $y(x)$ 变 $y(x)\to\bar{y}(x)\Rightarrow J[y(x)]\to\bar{J}[\bar{y}(x)]$

② $\delta y=\bar{y}(x)-y(x)\Rightarrow$ 左端固定 $\delta y(x_1)=0$

右端固定 $\delta y(x_2)=0$

$\bar{J}[\bar{y}(x)]=\int_{x_1}^{x_2}\bar{F}(x,\bar{y},\bar{y}')dx$ ← 作为新的泛函. $\bar{y}\to\bar{y}(x)$ 同理, $y=y(x)$ 不动.

$=\int_{x_1}^{x_2}\left(F(x,y,y')+\frac{\partial F}{\partial y}\delta y+\frac{\partial F}{\partial y'}\delta y'\right)dx$

$\delta J=\bar{J}[\bar{y}(x)]-J[y(x)]=\int_{x_1}^{x_2}\left(\frac{\partial F}{\partial y}\delta y+\frac{\partial F}{\partial y'}\delta y'\right)dx$

$=\int_{x_1}^{x_2}\left[\frac{\partial F}{\partial y}\delta y+\frac{\partial F}{\partial y'}\cdot\frac{d(\delta y)}{dx}\right]dx$

$=\int_{x_1}^{x_2}\left[\frac{\partial F}{\partial y}\delta y+\frac{d}{dx}\left(\frac{\partial F}{\partial y'}\cdot\delta y\right)-\frac{d}{dx}\left(\frac{\partial F}{\partial y'}\right)\cdot\delta y\right]dx$

$=\int_{x_1}^{x_2}\frac{d}{dx}\left(\frac{\partial F}{\partial y'}\delta y\right)dx+\int_{x_1}^{x_2}\left[\frac{\partial F}{\partial y}-\frac{d}{dx}\left(\frac{\partial F}{\partial y'}\right)\right]\delta y\cdot dx$

$=\frac{\partial F}{\partial y'}\delta y\Big|_{x_1}^{x_2}+\int_{x_1}^{x_2}\left[\frac{\partial F}{\partial y}-\frac{d}{dx}\left(\frac{\partial F}{\partial y'}\right)\right]\delta y\cdot dx$

$\delta J=0=\underbrace{\frac{\partial F}{\partial y'}\delta y\Big|_{x_1}^{x_2}}_{①}+\int_{x_1}^{x_2}\underbrace{\left[\frac{\partial F}{\partial y}-\frac{d}{dx}\left(\frac{\partial F}{\partial y'}\right)\right]}_{=0}\delta y\cdot dx$

$①=\frac{\partial F}{\partial y'(x_2)}\cdot\delta y(x_2)-\frac{\partial F}{\partial y'(x_1)}\delta y(x_1)=0-0=0$

于是到变分问题(泛函极值问题)的步骤中.

$\boxed{\frac{\partial F}{\partial y}-\frac{d}{dx}\left(\frac{\partial F}{\partial y'}\right)=0}$ 欧拉-拉格朗日方程.

$F=F(x,y,y')$

$\frac{\partial L}{\partial q}-\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}}\right)=0\Rightarrow L=L(q,\dot{q},t)$

变换. $F=F(x,y,y')$

$\frac{dF}{dx}=\frac{\partial F}{\partial x}+\frac{\partial F}{\partial y}\cdot\frac{dy}{dx}+\frac{\partial F}{\partial y'}\cdot\frac{dy'}{dx}$

$\Rightarrow\frac{\partial F}{\partial x}=\frac{dF}{dx}-\frac{\partial F}{\partial y}\frac{dy}{dx}-\frac{\partial F}{\partial y'}\cdot\frac{dy'}{dx}$ $=-\frac{d}{dx}\left(y'\cdot\frac{\partial F}{\partial y'}\right)$ $=\frac{\partial F}{\partial y}\cdot\frac{dy}{dx}+y'\frac{d}{dx}\left(\frac{\partial F}{\partial y'}\right)$

$=\frac{d}{dx}\left(F-y'\cdot\frac{\partial F}{\partial y'}\right)-y'\left[\frac{\partial F}{\partial y}-\frac{d}{dx}\left(\frac{\partial F}{\partial y'}\right)\right]$ ← 0 E-L

E-L 第二种形式. $\frac{\partial F}{\partial x}=\frac{d}{dx}\left(F-y'\frac{\partial F}{\partial y'}\right)$

$\Rightarrow\frac{\partial F}{\partial x}+\frac{d}{dx}\left(y'\cdot\frac{\partial F}{\partial y'}-F\right)=0$

若 F 不显含 x. $\frac{\partial F}{\partial x}=0\Rightarrow F-y'\frac{\partial F}{\partial y'}=C$

L 不显含 t. $\frac{\partial L}{\partial t}=0\Rightarrow L-\dot{q}\cdot\frac{\partial L}{\partial \dot{q}}=(-H)$ 守恒量

例1. E-L 证明平面两点间为直线的是直线

解: $y=y(x)$ $ds=\sqrt{1+y'^2}dx$

① $L=\int_{x_1}^{x_2}ds=\int_{x_1}^{x_2}\sqrt{1+y'^2}dx\Rightarrow$ ② $F=F(x,y,y')$ $=\sqrt{1+y'^2}$

③ E-L 方程 $L=L_{min}\Rightarrow$ 将 F 代入 $\frac{\partial F}{\partial y}-\frac{d}{dx}\left(\frac{\partial F}{\partial y'}\right)=0$

$0-\frac{d}{dx}\left(\frac{2y'}{2\sqrt{1+y'^2}}\right)=0\Rightarrow\frac{2y'}{2\sqrt{1+y'^2}}=C_1\Rightarrow\int y'=C$

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$$0-\frac{d}{dx}\left(\frac{y'}{\sqrt{1+y'^2}}\right)=0 \Rightarrow \frac{y'}{\sqrt{1+y'^2}}=C_1$$

$$\Rightarrow \frac{dy}{\sqrt{(dy)^2+(dx)^2}}=C_1 \Rightarrow \frac{dy}{ds}=c \Rightarrow dy=c\,ds$$

第二种形式：E-L第二种形式 $\frac{\partial F}{\partial x}=\frac{d}{dx}\left(F-y'\frac{\partial F}{\partial y'}\right)$

$$\Rightarrow \frac{\partial F}{\partial x}+\frac{d}{dx}\left(y'\cdot\frac{\partial F}{\partial y'}-F\right)=0$$

若下不显含 x，$\frac{\partial F}{\partial x}=0 \Rightarrow F-y'\frac{\partial F}{\partial y'}=C$

$F=\sqrt{1+y'^2}$，F不显含x，$\frac{\partial F}{\partial x}=0$

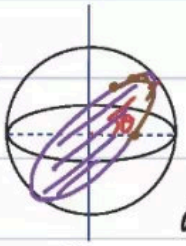
$$y'\cdot\frac{\partial F}{\partial y'}-F=C$$

$$y'\cdot\frac{y'}{\sqrt{1+y'^2}}-\sqrt{1+y'^2}=C \Rightarrow \frac{y'^2-(1+y'^2)}{\sqrt{1+y'^2}}=C$$

$$\Rightarrow \frac{-1}{\sqrt{1+y'^2}}=C \Rightarrow y'=\sqrt{\frac{1}{C^2}-1}=C_{新}$$

$y'=C_1 \Rightarrow y=C_1x+C_2$ $\begin{cases}C_1 & (x_1,y_1)\\ C_2 & (x_2,y_2)\end{cases}$ 代入确定.

例2. 求球面上两点最短路径.



$A(r,\varphi_1,\theta)$ $B(r,\varphi_2,\theta)$ θ恒.

$$ds=\sqrt{r^2(d\theta)^2+r^2\sin^2\theta(d\varphi)^2}$$

$$S=\int_A^B ds=\int_A^B \sqrt{r^2(d\theta)^2+r^2\sin^2\theta(d\varphi)^2}$$

$\begin{cases}\theta=\theta(\tau)\\ \varphi=\varphi(\tau)\end{cases}$ 参数方程 $\tau=S$

$$=\int_{\tau_A}^{\tau_B} r\sqrt{\left(\frac{d\theta}{d\tau}\right)^2+\sin^2\theta\left(\frac{d\varphi}{d\tau}\right)^2}\,d\tau$$

$$=\int_{S_A}^{S_B} \underbrace{r\sqrt{\left(\frac{d\theta}{ds}\right)^2+\sin^2\theta\left(\frac{d\varphi}{ds}\right)^2}}_{=1}\,ds$$

S θ,φ $\frac{d\theta}{ds},\frac{d\varphi}{ds}$

$$F=\sqrt{\left(\frac{d\theta}{ds}\right)^2+\sin^2\theta\left(\frac{d\varphi}{ds}\right)^2} \qquad F=F(x,y,y')$$

$$\frac{\partial F}{\partial\varphi}-\frac{d}{ds}\left(\frac{\partial F}{\partial\left(\frac{d\varphi}{ds}\right)}\right)=0 \Rightarrow 0=\frac{d}{ds}\left(\frac{2\sin^2\theta\frac{d\varphi}{ds}}{2\sqrt{\left(\frac{d\theta}{ds}\right)^2+\sin^2\theta\left(\frac{d\varphi}{ds}\right)^2}}\right)$$

$$\frac{d}{ds}\left(\frac{\sin^2\theta\frac{d\varphi}{ds}}{\sqrt{\left(\frac{d\theta}{ds}\right)^2+\sin^2\theta\left(\frac{d\varphi}{ds}\right)^2}}\right)=0 \Rightarrow \frac{d}{ds}\left(\sin^2\theta\frac{d\varphi}{ds}\right)=0$$

$$r^2\left[\left(\frac{d\theta}{ds}\right)^2+\sin^2\theta\left(\frac{d\varphi}{ds}\right)^2\right]=1$$

$$\Rightarrow \frac{d^2\varphi}{ds^2}+2\sin\theta\cos\theta\cdot\frac{d\theta}{ds}\cdot\frac{d\varphi}{ds}=0 \quad (\div\sin^2\theta)$$

$$\Rightarrow \frac{d^2\varphi}{ds^2}=-2\cot\theta\frac{d\theta}{ds}\frac{d\varphi}{ds} \quad ①$$

$$\frac{d}{ds}\left(\frac{\partial F}{\partial\left(\frac{d\theta}{ds}\right)}\right)-\frac{\partial F}{\partial\theta}=0$$

$$\Rightarrow \frac{d^2\theta}{ds^2}=\sin\theta\cos\theta\left(\frac{d\varphi}{ds}\right)^2 \quad ②$$

$$\frac{d\theta}{ds}=\frac{d\theta}{d\varphi}\cdot\frac{d\varphi}{ds}$$

$$\frac{d^2\theta}{ds^2}=\frac{d^2\theta}{d\varphi^2}\cdot\left(\frac{d\varphi}{ds}\right)^2+\frac{d\theta}{d\varphi}\cdot\frac{d^2\varphi}{ds^2}$$

$$\Rightarrow \frac{d^2\theta}{d\varphi^2}-2\cot\theta\left(\frac{d\theta}{d\varphi}\right)^2-\sin\theta\cos\theta=0$$

$$\frac{d^2\cot\theta}{d\varphi^2}+\cot\theta=0$$

$$\Downarrow$$

$$\cot\theta=A\cos(\varphi-\varphi_0)$$

$$\cot\theta=-\tan\theta_0\cos(\varphi-\varphi_0)$$

$$\Downarrow$$

$$\sin\theta\sin\theta_0\cos(\varphi-\varphi_0)+\cos\theta\cos\theta_0=0$$

$\vec{r}=(r\sin\theta\cos\varphi,\ r\sin\theta\sin\varphi,\ r\cos\theta)$

$\vec{n}=(\sin\theta_0\cos\varphi_0,\ \sin\theta_0\sin\varphi_0,\ r\cos\theta_0)$

$\vec{n}\cdot\vec{r}=0$

例3. 利用E-L方程，证明最速降线是摆线.

2020年 2月18日 13:30 分析力学(八)

CHAPTER 5 哈密顿原理

5.1 变分法初步

注意 $J=\int_{x_1}^{x_2} F(x,y,y')dx$ $y=y(x)$ $y'=\frac{dy}{dx}$

$J=J[y(x)]$

注意 a 极值问题(变分问题)

微分: $dy/dx=0$

变分问题的条件. 已之方程 $F(x,y,y')$

① $\frac{\partial F}{\partial y}-\frac{d}{dx}\left(\frac{\partial F}{\partial y'}\right)=0$ 写成 $y=y(x)$

② $\frac{\partial F}{\partial x}+\frac{d}{dx}\left(y'\frac{\partial F}{\partial y'}-F\right)=0$

⇓

若 F 不显含 x. $y'\frac{\partial F}{\partial y'}-F=$常数.

推广: $J\sim x, y_1(x), y_1'(x); y_2(x), y_2'(x)\cdots$

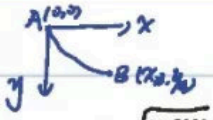
$J\sim x, y_\alpha(x), y_\alpha'(x)$, $\alpha=1,2,\cdots,s$

边界条件: $\delta y_\alpha|_{x=x_1}=0$ $\delta y_\alpha|_{x=x_2}=0$

⇓ 固定. 固定 固答.

$\frac{\partial F}{\partial y_\alpha}-\frac{d}{dt}\left(\frac{\partial F}{\partial y_\alpha'}\right)=0$, $\alpha=1,2,\cdots,s$

[例1] 最速降线



$t=\int_0^{x_2}\frac{\sqrt{1+y'^2}}{\sqrt{2gy}}dx=\frac{1}{\sqrt{2g}}\int_0^{x_2}\frac{\sqrt{1+y'^2}}{\sqrt{y}}dx$ 求 t_{min} 变分问题.

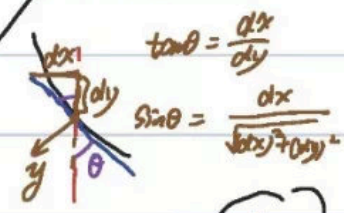
$F=F(x,y,y')=\sqrt{\frac{1+y'^2}{y}}$ ⇒ F 不显含 x.

⇒ $y'\cdot\frac{\partial F}{\partial y'}-F=C$ $\frac{\partial F}{\partial y'}=\frac{1}{\sqrt{y}}\cdot\frac{2y'}{2\sqrt{1+y'^2}}$

$y'\cdot\frac{\partial F}{\partial y'}-F=C \Rightarrow y'\cdot\frac{1}{\sqrt{y}}\cdot\frac{y'}{\sqrt{1+y'^2}}-\frac{\sqrt{1+y'^2}}{\sqrt{y}}=C$

$\Rightarrow \frac{y'^2-(1+y'^2)}{\sqrt{y}\cdot\sqrt{1+y'^2}}=\frac{-1}{\sqrt{y}\cdot\sqrt{1+y'^2}}=C$

$\frac{1}{\sqrt{2gy}\cdot\sqrt{1+y'^2}}=C'$ → $\sin\theta$



$\tan\theta=\frac{dx}{dy}$ $\sin\theta=\frac{dx}{\sqrt{(dx)^2+(dy)^2}}=\frac{1}{\sqrt{1+y'^2}}$

⇓

$\Rightarrow \frac{\sin\theta}{y}=C'$ (光的折射定律)

摆线: $y(1+y'^2)=C_1$

$y'=\cot\theta \Rightarrow y=\frac{C_1}{1+\cot^2\theta}=C_1\sin^2\theta=\frac{1}{2}C_1(1-\cos2\theta)$

$\frac{dy}{d\theta}=\frac{1}{2}C_1(+\sin2\theta)\cdot 2=2C_1\sin\theta\cos\theta=C_1\sin2\theta$

$\frac{dx}{d\theta}=\frac{dx}{dy}\cdot\frac{dy}{d\theta}=\frac{1}{y'}\cdot C_1\sin2\theta=\tan\theta\cdot C_1\sin2\theta \Rightarrow C_1(1-\cos2\theta)$

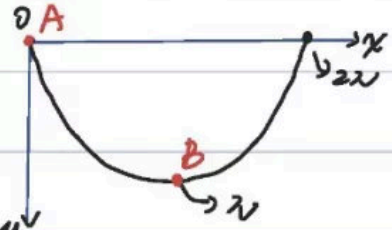
$\Rightarrow x=\frac{1}{2}C_1(\theta-\sin2\theta)+C_2$

$y=\frac{1}{2}C_1(1-\cos2\theta)$

代入 初始条件 $x=0, y=0$ ⇒ $C_2=0$, $\frac{1}{2}C_1=a$, $2\theta=\varphi$

$\Rightarrow \begin{cases} x=a(\varphi-\sin\varphi) \\ y=a(1-\cos\varphi) \end{cases}$ ⇒ 摆线 a,φ 参数

由 $B(x_2,y_2)$ 点决定 a



AB. 最速降线

[例2] 旋转体最小侧面积

$A=\int_{x_1}^{x_2}2\pi y\cdot\sqrt{1+y'^2}\,dx$ A_{min} $F=2\pi y\sqrt{1+y'^2}$

由于 F 不显含 x. 用 ②

$y'\frac{\partial F}{\partial y'}-F=C \Rightarrow y'\cdot\frac{2\pi y\cdot 2y'}{2\sqrt{1+y'^2}}-2\pi y\sqrt{1+y'^2}=C/2\pi$

$\frac{y\cdot y'^2}{\sqrt{1+y'^2}}-y\sqrt{1+y'^2}=C$

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$\frac{y\cdot y'^2}{\sqrt{1+y'^2}} - y\sqrt{1+y'^2} = C_1 \Rightarrow y' = \sqrt{y^2-C_1^2}/C_1 \Rightarrow \frac{dy}{dx} = \sqrt{y^2-C_1^2}/C_1$

$\Rightarrow x = \int \frac{dy}{\sqrt{y^2-C_1^2}/C_1} = C_1 \ln\left(\frac{y+\sqrt{y^2-C_1^2}}{C_1}\right) + C_2$

$\Rightarrow y = C_1 \cosh\left(\frac{x-C_2}{C_1}\right)$ 悬链线 — $\cosh = \frac{e^x+e^{-x}}{2}$

A、B 点坐标 (x_1,y_1) (x_2,y_2) 确定



约束下的变分问题. 四理. 共用大问题. "J"

$J = F(x, y_1, y_2, \cdots y_s; y_1', y_2', \cdots y_s')$. J 取决于 $\begin{cases} y_1(x) \\ \cdots \\ y_s(x) \end{cases}$

变分问题 $\delta J = \int_{x_1}^{x_2} F(x, y_\alpha, y_\alpha')\,dx = 0$ ① $\alpha = 1, \cdots, s$

还有 k 个积分约束 $C = \int_{x_1}^{x_2} G_j(x, y_\alpha, y_\alpha')\,dx = C_j$ ② $j = 1, 2, \cdots, k$.

若只考虑 ① 成立: $\Rightarrow \int_{x_1}^{x_2} \sum_{\alpha=1}^{s} \left[\frac{\partial F}{\partial y_\alpha} - \frac{d}{dx}\left(\frac{\partial F}{\partial y_\alpha'}\right)\right]\delta y_\alpha\,dx = 0$ ③

δy_α 独立 ⇒ 只有 $s-k$ 个.

② 取变分. $\delta C = 0 \Rightarrow \delta \int_{x_1}^{x_2} G(x, y_\alpha, y_\alpha')\,dx = 0$

$\Rightarrow \int_{x_1}^{x_2} \sum_{\alpha=1}^{s} \left[\frac{\partial G_j}{\partial y_\alpha} - \frac{d}{dx}\left(\frac{\partial G_j}{\partial y_\alpha'}\right)\right]\delta y_\alpha\,dx = 0$ ④ $j = 1, \cdots, k$ 个方程

由 k 个方程. 分别乘以 λ_j $(j = 1, \cdots, k)$ + ③

$\int_{x_1}^{x_2} \sum_{\alpha=1}^{s} \left[\frac{\partial F}{\partial y_\alpha} - \frac{d}{dx}\left(\frac{\partial F}{\partial y_\alpha'}\right)\right]\delta y_\alpha + \sum_{j=1}^{k}\lambda_j\left[\sum_{\alpha=1}^{s}\left[\frac{\partial G_j}{\partial y_\alpha} - \frac{d}{dx}\left(\frac{\partial G_j}{\partial y_\alpha'}\right)\right]\delta y_\alpha\right] dx = 0$

③

$\Rightarrow \int_{x_1}^{x_2} \sum_{\alpha=1}^{s} \left[\frac{\partial (F + \sum_{j=1}^{k}\lambda_j G_j)}{\partial y_\alpha} - \frac{d}{dx}\left(\frac{\partial (F + \sum_{j=1}^{k}\lambda_j G_j)}{\partial y_\alpha'}\right)\right]\delta y_\alpha\,dx = 0$ ⑤

$F \Rightarrow F$

⑤ 中有 s 个 ( ) $= 0$, $\alpha = 1, 2, \cdots, s$ $\Rightarrow$ $s+k$ 个

以及 ④ 中有 k 个约束方程

未知量. s 个 $y_\alpha(x)$ 和 k 个 λ_j

例: 等周问题. 封闭曲线 l. (A) 为大值



参数 $\begin{cases} x = x(t) \\ y = y(t) \end{cases}$ $t \in (t_0, t_0)$

边界 $\begin{cases} y(t_0) = y(t_0) \\ x(t_0) = x(t_0) \end{cases}$ $G(x, y, y')$ $x' = \frac{dx}{dt}$

约束 $l = \oint \sqrt{(dx)^2 + (dy)^2} = \int_0^{t_0} \sqrt{x'^2 + y'^2}\,dt$ $y' = \frac{dy}{dt}$.

格林公式: $P(x,y)$ $Q(x,y)$ 在 S 内连续 (S)

$\iint_S \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right)dx\,dy = \oint_L P\,dx + Q\,dy$

令 $P = -y$ $Q = x$

$\iint_S (1+1)\,dx\,dy = \oint_L -y\,dx + x\,dy$

$\Rightarrow A = \iint_S dx\,dy = \frac{1}{2}\oint_L (x\,dy - y\,dx)$

面积 $A = \frac{1}{2}\int_0^{t_0} x\cdot y'\,dt - y\cdot x'\,dt = \frac{1}{2}\int_0^{t_0} (xy' - yx')\,dt$ 被积函数.

$F(x, y_1, y_1', y_2, y_2')$

$F = \frac{1}{2}(xy' - yx') \Rightarrow \bar{F} = \frac{1}{2}(xy' - yx') + \lambda\sqrt{x'^2 + y'^2}$

$\frac{\partial \bar{F}}{\partial x} - \frac{d}{dt}\left(\frac{\partial \bar{F}}{\partial x'}\right) = 0 \Rightarrow \frac{1}{2}y' = \frac{d}{dt}\left(-\frac{1}{2}y + \lambda\frac{x'}{\sqrt{x'^2+y'^2}}\right)$

$\Rightarrow \frac{dy}{dt} = \frac{d}{dt}\left(\frac{\lambda x'}{\sqrt{x'^2+y'^2}}\right)$

同理 $\frac{\partial \bar{F}}{\partial y} - \frac{d}{dt}\left(\frac{\partial \bar{F}}{\partial y'}\right) = 0 \Rightarrow -\frac{dx}{dt} = \frac{d}{dt}\left(\frac{\lambda y'}{\sqrt{x'^2+y'^2}}\right)$

两式积分 $\begin{cases} y = \frac{\lambda x'}{\sqrt{x'^2+y'^2}} + C_2 \\ x = -\frac{\lambda y'}{\sqrt{x'^2+y'^2}} + C_1 \end{cases} \Rightarrow \frac{y-C_2}{x-C_1} = -\frac{x'}{y'} =$

$\Rightarrow (x-C_1)x' + (y-C_2)y' = 0$

$\Rightarrow (x-C_1)\frac{dx}{dt} + (y-C_2)\frac{dy}{dt} = 0$

$\Rightarrow (x-C_1)^2 + (y-C_2)^2 = C_3^2$

$C_3 = C_3 \Rightarrow \lambda = C_3 = l/2\pi$

圆: C_1, C_2 无关 初值. 圆心.

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5.2 哈密顿原理.

5.2.1 位形空间 哈密顿原理.

$L(t, q_\alpha, \dot{q}_\alpha)$

$S=\int_{t_1}^{t_2} L\,dt$

类比.

$F(x, y, y')$

$J=\int_{x_1}^{x_2} F(x, y, y')\,dx$

$\delta J=0$

$\frac{\partial F}{\partial y}-\frac{d}{dx}\left(\frac{\partial F}{\partial y'}\right)=0$

$L(t, q_\alpha, \dot{q}_\alpha)$

哈密顿作用量.

$S=\int_{t_1}^{t_2} L(t, q_\alpha, \dot{q}_\alpha)\,dt$

$\delta S=0$

$\frac{\partial L}{\partial q_\alpha}-\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}_\alpha}\right)=0$ 真实运动满足.

拉格朗日方程. 其本质是 $\delta\int_{t_1}^{t_2} L(t, q_\alpha, \dot{q}_\alpha)\,dt=0$ 和 E-L 方程.

哈密顿原理: 力学系统从 $t_1\to t_2$ 的一切可能的运动中, 使作用量 S 取极值的那些运动才是真实系统的运动.

位形空间 $(q_1, \cdots q_s)$ 称位空间. 任一时刻能位形 $\Rightarrow$ 位形空间点

位形空间哈密顿原理: 在一切可能的位形空间中的曲线中

使作用量 S 取极值的那个曲线代表真实的运动

地位: 比牛顿运动定律更为基本、更为普遍.

达朗贝尔原理 $\Rightarrow$ 哈密顿原理.

$\sum_{i=1}^{n}(\vec{F}_i-m_i\ddot{\vec{r}}_i)\cdot\delta\vec{r}_i=0$ 求和

$m_i\ddot{\vec{r}}_i\cdot\delta\vec{r}_i=m_i\left(\frac{d}{dt}\dot{\vec{r}}_i\right)\delta\vec{r}_i=m_i\left(\frac{d}{dt}(\dot{\vec{r}}_i\cdot\delta\vec{r}_i)-\dot{\vec{r}}_i\,\delta\dot{\vec{r}}_i\right)$

$=\frac{d}{dt}(m_i\dot{\vec{r}}_i\delta\vec{r}_i)-\delta\left(\frac{1}{2}m_i\dot{\vec{r}}_i\cdot\dot{\vec{r}}_i\right)$

$=\frac{d}{dt}(m_i\dot{\vec{r}}_i\delta\vec{r}_i)-\delta T_i$

$\int_{t_1}^{t_2} m_i\ddot{\vec{r}}_i\cdot\delta\vec{r}_i\,dt=m_i\dot{\vec{r}}_i\delta\vec{r}_i\Big|_{t_1}^{t_2}-\int_{t_1}^{t_2}\delta T_i\,dt=-\int_{t_1}^{t_2}\delta T_i\,dt$

$0=\delta\vec{r}_i(t_1)=\delta\vec{r}_i(t_2)$ 端点位置固定

$\vec{F}_i\cdot\delta\vec{r}_i=\delta W_i=\delta(\vec{F}_i\cdot\vec{r}_i)$

$\int_{t_1}^{t_2}\sum_{i=1}^{n}\vec{F}_i\cdot\delta\vec{r}_i\,dt-\int_{t_1}^{t_2}\sum_{i=1}^{n}m_i\ddot{\vec{r}}_i\cdot\delta\vec{r}_i\,dt=0$

$\Rightarrow\int_{t_1}^{t_2}\underbrace{\sum_{i=1}^{n}\delta W_i}_{\delta W}dt-\int_{t_1}^{t_2}\underbrace{\sum_{i=1}^{n}\delta T_i}_{-\delta T}dt=0$

$\Rightarrow\delta\int_{t_1}^{t_2}(W+T)\,dt=0$ 一般形式哈密顿原理.

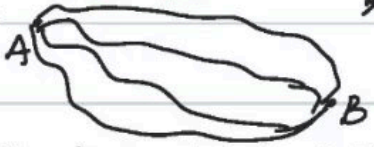
$\Rightarrow\int_{t_1}^{t_2}(\delta T+\sum_{\alpha=1}^{s}Q_\alpha\delta q_\alpha)\,dt=0$ 做功

$\Rightarrow\int_{t_1}^{t_2}\left(\delta T+\sum_{\alpha=1}^{s}\left(-\frac{\partial V}{\partial q_\alpha}\right)\delta q_\alpha\right)dt=0$

$\Rightarrow\int_{t_1}^{t_2}\delta(T-V)\,dt=0\Rightarrow\delta\int_{t_1}^{t_2}(T-V)\,dt=0$ L

$\delta\int_{t_1}^{t_2}L\,dt=0$. 哈密顿原理 $\Rightarrow$ 变分对称 $\Rightarrow$ 真实运动

问题:



$(q_1, \cdots q_s)$

真实 $\delta\int_{t_1}^{t_2}L\,dt=0$

$\delta S=0$

若两点是两种途径之间的变分问题. q_1 是循环坐标. 那么

此问题哈密顿原理解决. 意义: 两点是固定的. 端点固定

无所知. 无需considering路径.

已知 q_1 是循环坐标. $\Rightarrow p_1$ 不守恒的路径是不动的

问题: (无需considering路径 - p_1 不守恒的路径) $\Rightarrow$ 选择真动的路径.

极值. 带约束的变分问题. $p_1=C$

猜: $\delta\int_{t_1}^{t_2}(L-p_1\dot{q}_1)\,dt=0$

证明: $\delta\int_{t_1}^{t_2}L\,dt=0=\int_{t_1}^{t_2}\sum_{\alpha=1}^{s}\left(\frac{\partial L}{\partial q_\alpha}\delta q_\alpha+\frac{\partial L}{\partial\dot{q}_\alpha}\delta\dot{q}_\alpha\right)dt$

$=\int_{t_1}^{t_2}\sum_{\alpha=1}^{s}\left[\frac{\partial L}{\partial q_\alpha}\delta q_\alpha+\frac{d}{dt}\left(\frac{\partial L}{\partial\dot{q}_\alpha}\delta q_\alpha\right)-\frac{d}{dt}\left(\frac{\partial L}{\partial\dot{q}_\alpha}\right)\delta q_\alpha\right]dt$ L 对称

$=\int_{t_1}^{t_2}\sum_{\alpha=1}^{s}\left(\frac{d}{dt}\left(\frac{\partial L}{\partial\dot{q}_\alpha}\delta q_\alpha\right)\right)dt=0$.

$=\sum_{\alpha=1}^{s}\frac{\partial L}{\partial\dot{q}_\alpha}\delta q_\alpha\Big|_{t_1}^{t_2}=$

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$$\delta\int_{t_1}^{t_2}L\,dt=\sum_{\alpha=1}^{s}\frac{\partial L}{\partial\dot q_\alpha}\delta q_\alpha\Big|_{t_1}^{t_2}=0+p_t\cdot\delta q_t\Big|_{t_1}^{t_2}$$

有s项. $q_2,\cdots,q_s$ 独立. 故知 $\delta q_\alpha(t_1)=\delta q_\alpha(t_2)=0$ $\alpha=2,\cdots,s$

由于 p_t 是常数. δq_t 并不为0. 如箭所

$$\Rightarrow\delta\int_{t_1}^{t_2}(L-p_t\dot q_t)\,dt=0$$

一无所知 ✓ 勿为主观 ?

理解:

q_t 为初循环坐标的缓慢方程.

5.2.2 相空间中的哈密顿原理.

位形空间 s维: $(q_1,\cdots,q_s)$ $L(q,\dot q,t)$

相空间 2s维: $(q_1,\cdots,q_s,p_1,\cdots p_s)$ $H(q,p,t)$

位形空间: $\delta\int_{t_1}^{t_2}L\,dt=0\Rightarrow\delta\int_{t_1}^{t_2}(\sum_{\alpha=1}^{s}p_\alpha\dot q_\alpha-H)\,dt=0$

$$\Rightarrow\int_{t_1}^{t_2}\sum_{\alpha=1}^{s}\left(p_\alpha\delta\dot q_\alpha+\dot q_\alpha\delta p_\alpha-\frac{\partial H}{\partial p_\alpha}\delta p_\alpha-\frac{\partial H}{\partial q_\alpha}\delta q_\alpha\right)dt=0$$

$$\int_{t_1}^{t_2}\sum_{\alpha=1}^{s}p_\alpha\delta\dot q_\alpha\,dt=\sum_{\alpha=1}^{s}\int_{t_1}^{t_2}\left(\frac{d}{dt}(p_\alpha\delta q_\alpha)-\dot p_\alpha\delta q_\alpha\right)dt$$

$$=\sum_{\alpha=1}^{s}p_\alpha\delta q_\alpha\Big|_{t_1}^{t_2}-\sum_{\alpha=1}^{s}\int_{t_1}^{t_2}\dot p_\alpha\delta q_\alpha\,dt$$

$\Rightarrow\begin{cases}\delta q_\alpha(t_2)\\ \delta q_\alpha(t_1)\end{cases}=0$ 边界值为0

$$\sum_{\alpha=1}^{s}\int_{t_1}^{t_2}\left[\left(\dot q_\alpha-\frac{\partial H}{\partial p_\alpha}\right)\delta p_\alpha+\left(-\dot p_\alpha-\frac{\partial H}{\partial q_\alpha}\right)\delta q_\alpha\right]dt=0$$

$\dot q_\alpha-\frac{\partial H}{\partial p_\alpha}=0$, $-\dot p_\alpha-\frac{\partial H}{\partial q_\alpha}=0$

$$\Rightarrow\begin{cases}\dot q_\alpha=\dfrac{\partial H}{\partial p_\alpha}\\ \dot p_\alpha=-\dfrac{\partial H}{\partial q_\alpha}\end{cases}\quad(\alpha=1,2,\cdots,s)$$

哈密顿正则方程.

相空间的哈密顿方程.

例: 复摆的运动方程.



$$L=T-V=\frac{1}{2}I\dot\theta^2+mgl\cos\theta$$

$$S=\int_{t_1}^{t_2}L\,dt=\int_{t_1}^{t_2}\left[\frac{1}{2}I\dot\theta^2+mgl\cos\theta\right]dt$$

$$\delta S=0\Rightarrow\delta\int_{t_1}^{t_2}L\,dt=\int_{t_1}^{t_2}\left[\frac{\partial L}{\partial\theta}\delta\theta+\frac{\partial L}{\partial\dot\theta}\delta\dot\theta\right]dt=0$$

$$\delta S=\int_{t_1}^{t_2}\left[\frac{\partial L}{\partial\theta}\delta\theta+\frac{\partial L}{\partial\dot\theta}\delta\dot\theta\right]dt$$

$$=\int_{t_1}^{t_2}\left[-mgl\sin\theta\delta\theta+I\dot\theta\delta\dot\theta\right]dt$$

$$=\int_{t_1}^{t_2}\left[-mgl\sin\theta\delta\theta-I\ddot\theta\delta\theta\right]dt+I\dot\theta\delta\theta\Big|_{t_1}^{t_2}=0$$

$I\dot\theta\delta\theta=0$

$I\ddot\theta+mgl\sin\theta=0\Rightarrow$ 从理想系统求得单摆. E-L方程.

5.2.3 位形空间中的哈密顿方程. $\delta S=\int_{t_1}^{t_2}\Lambda\,dt=0$

作为s+1广义坐标.

时空 = 时间+空间. $q_1,\cdots q_s$ 平权.

$$\begin{cases}q_\alpha=q_\alpha(\tau)\Rightarrow q_\alpha'=\dfrac{dq_\alpha}{d\tau} & \dot q_\alpha=\dfrac{dq_\alpha}{dt}=\dfrac{q_\alpha'}{t'}\\ t=t(\tau)\Rightarrow t'=\dfrac{dt}{d\tau}\end{cases}$$

$$S=\int L(q_1,\cdots q_s,t,\dot q_1,\cdots\dot q_s)\,dt\qquad\Lambda=L\cdot t'$$

$$=\int L\left(q_1,\cdots,q_s,t,\frac{q_1'}{t'},\cdots\frac{q_s'}{t'}\right)t'\,d\tau\Rightarrow\Lambda\left(q,t,\frac{q'}{t'},t'\right)$$

$$=\int\Lambda\left(q,t,\frac{q'}{t'},t'\right)d\tau\Rightarrow F(x,y,y')$$

τ 为 $q\sim\tau$ 与 $t\sim\tau$ 的

位形空间中的哈密顿方程 $\delta S=0\Rightarrow\delta\int\Lambda\,d\tau=0\Rightarrow$

q_α: $\frac{d}{d\tau}\left(\frac{\partial\Lambda}{\partial q_\alpha'}\right)-\frac{\partial\Lambda}{\partial q_\alpha}=0$ $p_\alpha=$

t: $\frac{d}{d\tau}\left(\frac{\partial\Lambda}{\partial t'}\right)-\frac{\partial\Lambda}{\partial t}=0$ p_t

$$\Rightarrow\frac{\partial\Lambda}{\partial q_\alpha'}=\frac{\partial(L\cdot t')}{\partial q_\alpha'}=\frac{\partial L}{\partial\dot q_\alpha}\cdot t'+\frac{\partial t'}{\partial q_\alpha'}\cdot L=0=\frac{\partial L}{\partial\dot q_\alpha}\cdot\frac{\partial\dot q_\alpha}{\partial q_\alpha'}\cdot t'$$

$$=\frac{\partial L}{\partial\dot q_\alpha}\frac{\partial(q_\alpha'/t')}{\partial q_\alpha'}\cdot t'=\frac{\partial L}{\partial\dot q_\alpha}=p_\alpha$$

$t\Rightarrow$

$$p_t=\frac{\partial\Lambda}{\partial t'}=\frac{\partial(Lt')}{\partial t'}=L+\frac{\partial L}{\partial t}t'=L+\sum_{\alpha=1}^{s}\frac{\partial L}{\partial\dot q_\alpha}\cdot\frac{\partial(\frac{q_\alpha'}{t'})}{\partial t}\cdot t'$$

$$=L+\sum_{\alpha=1}^{s}\frac{\partial L}{\partial\dot q_\alpha}\cdot\left(-\frac{1}{t'}\right)\cdot q_\alpha'\cdot t'\Rightarrow p_\alpha$$

$$=L-\sum_{\alpha=1}^{s}p_\alpha\dot q_\alpha=-H=p_t$$

时空统一 → 时间 s+1 维.

$$\frac{d}{d\tau}\left(\frac{\partial L}{\partial\dot q_\alpha}\right)-\frac{\partial(Lt')}{\partial q_\alpha}=0\Rightarrow\frac{d}{dt}\left(\frac{\partial L}{\partial\dot q_\alpha}\right)-\frac{\partial L}{\partial q_\alpha}=0,\ \alpha=1,\cdots,s$$

$$\frac{d}{d\tau}(-H)-\frac{\partial(Lt')}{\partial t}=0\Rightarrow\frac{d}{dt}(-H)\cdot t'-\frac{\partial L}{\partial t}t'-L\frac{\partial t'}{\partial t}=0$$

$$\Rightarrow\frac{dH}{dt}=-\frac{\partial L}{\partial t}$$

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5.3 最小作用量原理(变分形式)

1. 真实运动：作用量取极值(极大/极小/恒定)

2. 自由粒子：费马原理

$\delta t = \delta\int \frac{ds}{v} = 0 \Rightarrow \delta\int \frac{n}{c}ds = 0 \Rightarrow \boxed{\delta\int n\,ds = 0}$

光程极值

3. 一般力学体系 $\Rightarrow$ 推广为作用量原理

$T = \sum_{\alpha=1}^{s}\sum_{\beta=1}^{s}\frac{1}{2}m_{\alpha\beta}\dot{q}_\alpha\dot{q}_\beta \quad m_{\alpha\beta} = \sum_{i=1}^{s} m_i \frac{\partial \vec{r}_i}{\partial q_\alpha}\cdot\frac{\partial \vec{r}_i}{\partial q_\beta}$

位形空间弧长：$(ds)^2 = \sum_{\alpha,\beta=1}^{s} m_{\alpha\beta}\,dq_\alpha\,dq_\beta$

$V =$ 势函数

$v = \frac{ds}{dt}$

$\delta\int v\,ds = 0 \Rightarrow \delta\int \sqrt{2(E-U)}\,ds = 0$

单质点，位形空间 $\Rightarrow$ 三维欧氏

$\delta\int \sqrt{E-V}\cdot ds = 0$

费马原理：$n \propto \frac{1}{y}$ $\quad \delta\int n\,ds = 0$



最速降线 $\quad ds = \sqrt{1+y'^2}\,dx$

$\delta\int \frac{1}{y}\cdot\sqrt{1+y'^2}\,dx = 0$

$F = \frac{\sqrt{1+y'^2}}{y}$

E-L $\quad \frac{d}{dx}\left(\frac{\partial F}{\partial y'}\right) - \frac{\partial F}{\partial y} = 0$

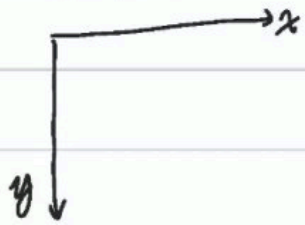
$y\cdot y'' + y'^2 + 1 = 0$

$\Rightarrow \frac{d}{dx}(yy') + 1 = 0 \Rightarrow yy' + x = C$ 一次积分

$\Rightarrow dy^2 + 2x\,dx = 2C \Rightarrow y^2 + (x-C)^2 = C^2 + d$

例：竖直面质点，仅重力作用运动

初值：(x_0, y_0)



$T = \frac{1}{2}m(\dot{x}^2+\dot{y}^2) = \frac{1}{2}(m\dot{x}^2 + 0\cdot\dot{x}\dot{y} + 0\cdot\dot{y}\dot{x} + m\dot{y}^2)$

$m_{11} = m \quad m_{12} = 0$

$m_{21} = 0 \quad m_{22} = m$

$T = \sum_{\alpha=1}^{s}\sum_{\beta=1}^{s}\frac{1}{2}m_{\alpha\beta}\dot{q}_\alpha\dot{q}_\beta$

$(ds)^2 = m_{11}(dx)^2 + m_{12}\,dx\,dy + m_{21}\,dy\,dx + m_{22}(dy)^2$

$= m(dx)^2 + m(dy)^2 = m(1+y'^2)\cdot(dx)^2$

$V = mgy$

$\delta\int \sqrt{2(E-U)}\,ds = \delta\int \sqrt{2(E-mgy)}\sqrt{m(1+y'^2)}\,dx = 0$

$F = \sqrt{2(E-mgy)m(1+y'^2)}$

$\frac{d}{dx}\left(\frac{\partial F}{\partial y'}\right) - \frac{\partial F}{\partial y} = 0$

$2(E-mgy)y'' + mg(1+y'^2) = 0$

$y' = \frac{dy}{dx} = \frac{dy'}{dx}\cdot\frac{dx}{dy} = y'\frac{dy'}{dy} = \frac{1}{2}\frac{dy'^2}{dy}$

$\frac{mg}{E-mgy}dy + \frac{1}{1+y'^2}dy'^2 = 0$

积分 $\Rightarrow 1+y'^2 = C_1(E-mgy)$

$t=0,\ x=0,\ y=0,\ \frac{dy}{dx} = \tan\alpha \Rightarrow C_1 = \frac{1}{E\cos^2\alpha}$

$y' = \frac{dy}{dx} = \sqrt{C_1(E-mgy)-1}$

$\Rightarrow \sqrt{C_1(E-mgy)-1} = -\frac{1}{2}C_1 mg x + C_2$

$t=0,\ x=0,\ y=0 \Rightarrow C_2 = \sqrt{C_1E-1}$

$C_1 = ?$

$y = x\tan\alpha - \frac{mg}{4E\cos^2\alpha}x^2 \Rightarrow$ 到斜面 $\tan\alpha$ 小角度抛

斜抛运动轨道方程

$E = T+V = T_0 + 0 = \frac{1}{2}mv_0^2$

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2020年2月20日 13:30 分析力学(九)

CHAPTER5 哈密顿原理.

5.1 变分法初步: $J=J[y(x)]$ $\delta J=0$ 求中

E-L方程: $\frac{\partial F}{\partial y}-\frac{d}{dx}\left(\frac{\partial F}{\partial y'}\right)=0$ $y'=\frac{dy}{dx}$

若F不显含x, $\frac{\partial F}{\partial x}=0 \Rightarrow y'\frac{\partial F}{\partial y'}-F=C$

5.2 哈密顿原理.

$S=\int_{t_1}^{t_2} L\,dt$ 真实 $\delta S=0$ 运动.

位形空间: $\frac{\partial L}{\partial q_\alpha}-\frac{d}{dt}\left(\frac{\partial L}{\partial \dot q_\alpha}\right)=0$, $\alpha=1,2,\cdots,s$

若部分坐标 q_β 为循环坐标: $\frac{\partial L}{\partial q_\alpha}-\frac{d}{dt}\left(\frac{\partial L}{\partial \dot q_\alpha}\right)=0$ $\alpha=2,\cdots,s$

相空间: $\dot q_\alpha=\frac{\partial H}{\partial p_\alpha}$ $(\alpha=1,2,\cdots,s)$

$\dot p_\alpha=-\frac{\partial H}{\partial q_\alpha}$

位形空间: t 作为 $s+1$ 个坐标.

$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot q_\alpha}\right)-\frac{\partial L}{\partial q_\alpha}=0$ $p_\alpha=\frac{\partial L}{\partial \dot q_\alpha}$

$\frac{\partial H}{\partial t}=-\frac{\partial L}{\partial t}$ $p_t=-H$

哈密顿原理是小作用量: $\delta\int v\,ds=\delta\int\sqrt{2(E-V)}\,ds=0$

保守系. 完整系.

$(ds)^2=\sum_{\alpha=1}^{s}\sum_{\beta=1}^{s} m_{\alpha\beta}\,dq_\alpha\,dq_\beta$

$T=\frac{1}{2}\sum_{\alpha=1}^{s}\sum_{\beta=1}^{s} m_{\alpha\beta}\dot q_\alpha\dot q_\beta$

变分原理: $\delta\int\frac{1}{v}ds=0 \Rightarrow \delta\int n\,ds=0$ 经过时间最短.

利用哈密顿原理直接证明轨迹方程.

例: 求质量为m的质点在有心力作用下的运动轨迹.

椭圆. 抛物线. 双曲线



$\dot r\to dr$ $\dot\theta\to d\theta$ $\dot q\to dq$

$T=\frac{1}{2}m\dot r^2+\frac{1}{2}mr^2\dot\theta^2=\frac{1}{2}(m\dot r\dot r+0\cdot\dot r\dot\theta+0\dot\theta\dot r+mr^2\dot\theta\dot\theta)$

$(ds)^2=m\cdot dr\cdot dr+0\cdot dr d\theta+0\,d\theta dr+mr^2 d\theta d\theta$

$=m(dr)^2+mr^2(d\theta)^2=m(r'^2+r^2)(d\theta)^2$ $r'=\frac{dr}{d\theta}$

$U=-\alpha\cdot\frac{m}{r}$ 势能. $\theta\sim t$ $r\sim t$

$F(\theta,r,r')$

$\delta\int\sqrt{2(E-V)}\,ds=\delta\int\sqrt{2(E+\frac{\alpha m}{r})}\cdot\sqrt{m(r'^2+r^2)}\cdot d\theta=0$

由于F不显含 θ, $r'\frac{\partial F}{\partial r'}-F=C$ 整理.

$\frac{dr}{d\theta}=\sqrt{\frac{2mr^2}{C^2}\left[\left(E+\frac{\alpha m}{r}\right)r^2-\frac{C^2}{2m}\right]}$ 积分.

$\theta=\frac{C}{\sqrt{2m}}\int\frac{dr}{\sqrt{Er^2+\alpha mr-\frac{C^2}{2m}}}$ 配成 $1-\square^2$ 的 arcsin

$=\arcsin\left(\frac{\alpha mr-C^2/m}{r\sqrt{\alpha^2m^2+2EC^2/m}}\right)-C'$

边界: $t=0$, $\theta=0$, $r=r_0 \Rightarrow \frac{dr}{d\theta}=0$

$C^2=2m(E+\frac{\alpha m}{r_0})r_0^2 \Rightarrow J=|C|$

$C'=\frac{\pi}{2}$

$\Rightarrow r=\frac{J^2(\alpha m)^2}{1+\left(\sqrt{1+2EJ^2/\alpha^2m^3}\right)\cdot(\cos\theta)}$ 圆锥曲线

CHAPTER 6 哈密顿正则变换

哈密顿原理✓ 哈力学✓ 构建方式

空整性质：正则变换

哈力 摆脱 拉力. 解决途径. {相空间✓ → 泊松括号. 正则变换✓

正则方程. q. p 平等地位 (相对) $\dot{p}_\alpha = \square$ $\dot{q}_\alpha = \square$

泊松括号：运动方程. 对称.

目的：求解容易 × 有立足点✓ ⇒ 拥有 ⇒ 容易

6.1 正则变换

6.1.1 位形空间点变换

拉氏量 $L = L(q_1 \cdots q_s, \dot{q}_1 \cdots \dot{q}_s, t)$ $\Rightarrow \frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}_\alpha}\right) - \frac{\partial L}{\partial q_\alpha} = 0$

旧广义坐标 q → 新广义坐标 Q 点变换

$Q_\alpha = Q_\alpha(q_1, \cdots, q_s, t)$

$L' = L'(Q_1, Q_2, \cdots Q_s, \dot{Q}_1, \cdots \dot{Q}_s, t)$

$\Rightarrow \frac{d}{dt}\left(\frac{\partial L'}{\partial \dot{Q}_\alpha}\right) - \frac{\partial L'}{\partial Q_\alpha} = 0 \Rightarrow$ 拉氏方程 $\Rightarrow$ 实验物理学.

模型：简谐摆：双摆 旧：θ_1，θ_2 $\theta_1 \sim t$ $\theta_2 \sim t$

新：$s_1 = \theta_1 + \theta_2$, $s_2 = \theta_1 - \theta_2$ $\Rightarrow$ $s_1 \sim t$ $s_2 \sim t$

目标：通过点变换寻找尽可能多地独立的积分. 直至积分完成

S 与 q，$S \Rightarrow Q$ $\Rightarrow$ 上面 S 与运动积分.

有心力. (x, y) $L = \frac{1}{2}m(\dot{x}^2 + \dot{y}^2) - \frac{k}{\sqrt{x^2+y^2}}$ 无独立积分.

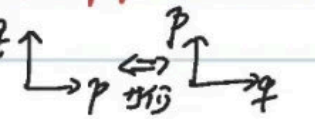
(r, θ) $L = \frac{1}{2}m(\dot{r}^2 + r^2\dot{\theta}^2) - k\frac{1}{r}$

L 中缺 θ. $p_\theta = C$ 运动积分 (角动量守恒)

6.1.2 正则变换. 相空间 (哈密顿) 李代数点

正则方程：$\begin{cases} \dot{q}_\alpha = \frac{\partial H}{\partial p_\alpha} \\ \dot{p}_\alpha = -\frac{\partial H}{\partial q_\alpha} \end{cases}$ 旧：$q, p \Rightarrow H$

$\Rightarrow$ 无法实施 解决对称性-复杂.



正则变换 p, q 地位平等.

新 $\begin{cases} Q_\alpha = Q_\alpha(q_1 \cdots q_s, p_1 \cdots p_s, t) = Q_\alpha(q, p, t) \\ P_\alpha = P_\alpha(q_1 \cdots q_s, p_1 \cdots p_s, t) = P_\alpha(q, p, t) \end{cases}$ 正则变换

若存在一个函数 $K = K(Q_1, \cdots Q_s, P_1, \cdots P_s, t)$ → 新哈密顿量

使得：$\begin{cases} \dot{Q}_\alpha = \frac{\partial K}{\partial P_\alpha} \\ \dot{P}_\alpha = -\frac{\partial K}{\partial Q_\alpha} \end{cases}$ $\Rightarrow$ 正则变换存在.

旧：q, p, H

新：Q, P, K → 新的广义哈密顿量 新的是对新的正则方程.

点变换：$q \to Q$, $L \to L'$

$(L', Q, \dot{Q})$ 满足 $\frac{d}{dt}\left(\frac{\partial L'}{\partial \dot{Q}_\alpha}\right) - \frac{\partial L'}{\partial Q_\alpha} = 0$

正则变换：$\begin{matrix} q \\ p \end{matrix} \begin{matrix} \} \\ \} \end{matrix} \begin{matrix} Q \\ P \end{matrix}$

(K, Q, P) 满足 $\dot{Q}_\alpha = \frac{\partial K}{\partial P_\alpha}$, $\dot{P}_\alpha = -\frac{\partial K}{\partial Q_\alpha}$

6.1.3 正则变换条件.

哈密顿方程：$\delta \int_{t_1}^{t_2} L\,dt = 0$

$H = L - \sum p_\alpha \dot{q}_\alpha$

$K = L - \sum P_\alpha \dot{Q}_\alpha$

相空间 p, q (旧) $\delta \int_{t_1}^{t_2} \left(\sum_{\alpha=1}^{s} p_\alpha \dot{q}_\alpha - H\right) dt = 0$ 等价 $\begin{cases} \dot{q}_\alpha = \frac{\partial H}{\partial p_\alpha} \\ \dot{p}_\alpha = -\frac{\partial H}{\partial q_\alpha} \end{cases}$

新 P, Q. $\delta \int_{t_1}^{t_2} \left(\sum_{\alpha=1}^{s} P_\alpha \dot{Q}_\alpha - K\right) dt = 0$ (+0) 等价 $\begin{cases} \dot{Q}_\alpha = \frac{\partial K}{\partial P_\alpha} \\ \dot{P}_\alpha = -\frac{\partial K}{\partial Q_\alpha} \end{cases}$

$\sum_{\alpha=1}^{s} p_\alpha \dot{q}_\alpha - H = \sum_{\alpha=1}^{s} P_\alpha \dot{Q}_\alpha - K + \frac{dU}{dt}$

$\delta \int_{t_1}^{t_2} \left(\sum P_\alpha \dot{Q}_\alpha - K + \frac{dU}{dt}\right) dt = 0 \Rightarrow \delta \int_{t_1}^{t_2} \frac{dU}{dt} dt = 0$ ✓

$\Rightarrow \delta U \Big|_{t_1}^{t_2} = \delta U(t_2) - \delta U(t_1) = 0$

$\sum_{\alpha=1}^{s} p_\alpha \dot{q}_\alpha - H = \sum_{\alpha=1}^{s} P_\alpha \dot{Q}_\alpha - K + \frac{dU}{dt} \Rightarrow$

$dU = \sum_{\alpha=1}^{s} p_\alpha dq_\alpha - \sum_{\alpha=1}^{s} P_\alpha dQ_\alpha + (K-H)dt$ ⊛

说是⊛式 $(p,q,H) \to (P,Q,K)$ 的正则变换

等价

生成出物、母出物，不同的U → 不同的变换 p,q / P,Q

6.1.4 四类正则变换母出物

$dU = \sum_{\alpha=1}^{s}\left(p_\alpha dq_\alpha + (-P_\alpha)dQ_\alpha\right) + (K-H)dt$ ⊛

$U_1 = U_1(q,Q,t)$ $U_2 = U_2(q,P,t)$ $\frac{p,q}{C^2=0}$ 旧 $\frac{P,Q}{C^2=0}$ 新

$U_3 = U_3(p,Q,t)$ $U_4 = U_4(p,P,t)$

母出物 U → 四种相同形式变换.

第一类 $U = U_1(q,Q,t) = U_1(q_1 \cdots q_s, Q_1 \cdots Q_s, t)$

$dU = dU_1 = \sum_{\alpha=1}^{s}\left(\frac{\partial U_1}{\partial q_\alpha}dq_\alpha + \frac{\partial U_1}{\partial Q_\alpha}dQ_\alpha\right) + \frac{\partial U_1}{\partial t}dt$ ①

$dU = \sum_{\alpha=1}^{s}\left(p_\alpha dq_\alpha + (-P_\alpha)dQ_\alpha\right) + (K-H)dt$ ⊛

变换 $\begin{cases} p_\alpha = \frac{\partial U_1}{\partial q_\alpha} \\ P_\alpha = -\frac{\partial U_1}{\partial Q_\alpha} \end{cases}$ $K = H + \frac{\partial U_1}{\partial t}$

第二类 $\begin{cases} p_\alpha = \frac{\partial U_2}{\partial q_\alpha} \\ Q_\alpha = \frac{\partial U_2}{\partial P_\alpha} \end{cases}$ $K = H + \frac{\partial U_2}{\partial t}$

第二类 $U_2 = U_2(q,P,t) = U$

$dU_2 = \sum_{\alpha=1}^{s}\left(\frac{\partial U_2}{\partial q_\alpha}dq_\alpha + \frac{\partial U_2}{\partial P_\alpha}dP_\alpha\right) + \frac{\partial U_2}{\partial t}dt$ ②

$dU = \sum_{\alpha=1}^{s}\left(p_\alpha dq_\alpha + (-P_\alpha)dQ_\alpha\right) + (K-H)dt$ ⊛

利用勒让德变换

$U_2 = U_2(q,P,t) = U_1(q,Q,t) + \sum_{\alpha=1}^{s} P_\alpha Q_\alpha$

$dU_2 = \sum_{\alpha=1}^{s}\left(p_\alpha dq_\alpha + (-P_\alpha)dQ_\alpha + (K-H)dt\right) + \sum_{\alpha=1}^{s}\left(P_\alpha dQ_\alpha + Q_\alpha dP_\alpha\right)$

第三类. $U_3(p,Q,t) = U_1(q,Q,t) - \sum_{\alpha=1}^{s} p_\alpha q_\alpha$

$\begin{cases} q_\alpha = -\frac{\partial U_3}{\partial p_\alpha} \\ P_\alpha = -\frac{\partial U_3}{\partial Q_\alpha} \end{cases}$ $K = H + \frac{\partial U_3}{\partial t}$

第四类 $U_4(p,P,t) = U_1(q,Q,t) - \sum_{\alpha=1}^{s} p_\alpha q_\alpha + \sum_{\alpha=1}^{s} P_\alpha Q_\alpha$

$\begin{cases} q_\alpha = -\frac{\partial U_4}{\partial p_\alpha} \\ Q_\alpha = \frac{\partial U_4}{\partial P_\alpha} \end{cases}$ $K = H + \frac{\partial U_4}{\partial t}$

举例. (1) $U = \sum_\beta f_\beta \cdot P_\beta$, $\beta = 1, \cdots, s$ (p,P,t)

第四类 $\begin{cases} q_\alpha = -\frac{\partial U}{\partial p_\alpha} = -P_\alpha \\ Q_\alpha = \frac{\partial U}{\partial P_\alpha} = +f_\alpha \end{cases} \Rightarrow \begin{cases} P_\alpha = -q_\alpha \\ Q_\alpha = +f_\alpha \end{cases}$ $K = H$ 恒等变换

检验: $\dot{Q}_\alpha = \frac{\partial K}{\partial P_\alpha} \Rightarrow \dot{f}_\alpha = -\frac{\partial H}{\partial q_\alpha}$ ✓

$\dot{P}_\alpha = -\frac{\partial K}{\partial Q_\alpha} \Rightarrow -\dot{q}_\alpha = -\frac{\partial H}{\partial f_\alpha} \Rightarrow \dot{q}_\alpha = \frac{\partial H}{\partial f_\alpha}$ ✓

(2) $U = \sum_\beta q_\beta \cdot P_\beta$ $\alpha = 1, \cdots, s$

第二类 $\begin{cases} p_\alpha = \frac{\partial U_2}{\partial q_\alpha} = P_\alpha \\ Q_\alpha = \frac{\partial U_2}{\partial P_\alpha} = q_\alpha \end{cases} \Rightarrow \begin{cases} q_\alpha = Q_\alpha \\ p_\alpha = P_\alpha \end{cases}$ 恒等变换

(3) 谐振子 $H = p^2/2m + m\omega^2 q^2/2$

母 $U = \frac{1}{2}m\omega q^2 \cot Q = U(q,Q,t)$ 第二类

$\begin{cases} p = \frac{\partial U_1}{\partial q} = m\omega q \cot Q \\ P = -\frac{\partial U_1}{\partial Q} = -\frac{1}{2}m\omega q^2 \cdot \frac{1}{-\sin^2 Q} = \frac{1}{2}m\omega q^2 \cdot \frac{1}{\sin^2 Q} \\ K = H = \frac{1}{2m}(m\omega q \cot Q)^2 + \frac{1}{2}m\omega^2 q^2 = \omega P \end{cases}$

弗恩曼

$K=\omega p$

$\dot{Q}=\frac{\partial K}{\partial p}=\omega \Rightarrow Q=\omega t+\varphi$ ⇒ 初始

$\dot{p}=-\frac{\partial K}{\partial Q}=0 \Rightarrow p=C$ 运动积分.

$p=\frac{1}{2}m\omega q^2\cdot\frac{1}{\sin^2 Q}=C$

$\Rightarrow q=A\cdot\sin Q=A\sin(\omega t+\varphi)$

6.2 哈-雅方程. (3解)

$K=K(Q_1,\cdots Q_s,\ p_1,\cdots p_s,\ t)$

$\begin{cases}\dot{Q}_\alpha=\frac{\partial K}{\partial p_\alpha}=0 \to K=0\\ \dot{p}_\alpha=-\frac{\partial K}{\partial Q_\alpha}=0\end{cases}$

目标: 尽能多的运动积分.

2S个. $\dot{Q}_\alpha=0$, $\dot{p}_\alpha=0$

一类母题方法. 目标 $K=0$

正则变换 $K=H+\frac{\partial S}{\partial t}$ ⇒ $H+\frac{\partial S}{\partial t}=0$

旧 H 新 K ⇒ 选择合适 $K=0$ 时的 S

新一个坐标 Q, 称为 S

$H+\frac{\partial S}{\partial t}=0$ 解出来. $H(q,p,t)+\frac{\partial S}{\partial t}=0$

设 $S=S(q,P,t)$ 第二类 ⇒ $\begin{cases}p_\alpha=\frac{\partial S}{\partial q_\alpha}\\ Q_\alpha=\frac{\partial S}{\partial P_\alpha}\end{cases}$

$H(q_\alpha,\frac{\partial S}{\partial q_\alpha},t)+\frac{\partial S}{\partial t}=0$ ⇒ 哈密顿-雅可比方程.

解: $S=S(q,P,t)$

S 是 $(S+1)$ 个积分常数 P_α 与 t. 一个微分方程.

$S=C_\alpha+S(q_1,\cdots,q_s,C_1,\cdots C_s,t)$

另名称 $P_1\to C_1$ 此时没有 $K=0$ 的条件

$P_s\to C_s$

$S=S(q_1,\cdots q_s,C_1,\cdots C_s,t)$

$\frac{dS}{dt}=\frac{\partial S}{\partial t}+\sum_{\alpha=1}^{s}\frac{\partial S}{\partial q_\alpha}\dot{q}_\alpha=-H+\sum_{\alpha=1}^{s}\frac{\partial S}{\partial q_\alpha}\dot{q}_\alpha$

$\frac{\partial S}{\partial t}+H(q_\alpha,\frac{\partial S}{\partial q_\alpha},t)=0$ $=\sum_{\alpha=1}^{s}p_\alpha\dot{q}_\alpha-H=L$

$p_\alpha=\frac{\partial S}{\partial q_\alpha}$

S是作用量. 沿真实运动路径运动的作用量.

例: 利用哈-雅法解 $H=\frac{1}{2}p^2/m+\frac{1}{2}kx^2$

$\frac{\partial S}{\partial t}+H(q,\frac{\partial S}{\partial q},t)=0$

$\Rightarrow \frac{\partial S}{\partial t}+\frac{1}{2}kx^2+\frac{1}{2m}\cdot(\frac{\partial S}{\partial x})^2=0$

由于 H 不显含 t. $S=W(x)+T(t)$

$\frac{1}{2}kx^2+\frac{1}{2m}(\frac{dW}{dx})^2=-\frac{dT}{dt}=E$ (常数)

空间 时间

时间 $\frac{dT}{T}=-E \Rightarrow T=-Et \Rightarrow \frac{\partial T}{\partial E}=-t$

空间: $\frac{1}{2m}(\frac{dW}{dx})^2+\frac{1}{2}kx^2=E$ $P=?$

$\Rightarrow W=\sqrt{2m}\int\sqrt{E-\frac{1}{2}kx^2}\,dx$ $\Rightarrow \omega=\sqrt{k/m}$

$S=-Et+\sqrt{2m}\int\sqrt{E-\frac{1}{2}kx^2}\,dx$

变换: $p=\frac{\partial S}{\partial x}=\frac{dW}{dx}$

$Q=\frac{\partial S}{\partial E}=-t+\frac{\partial W}{\partial E}$

$\frac{\partial W}{\partial E}=\int m\omega\frac{\partial}{\partial E}\sqrt{\frac{2E}{m\omega^2}-x^2}\,dx$

$=\int\frac{1}{\omega\sqrt{\frac{2E}{m\omega^2}-x^2}}dx=\frac{1}{\omega}\arcsin\frac{x}{\sqrt{\frac{2E}{m\omega^2}}}$

$t-t_0=\frac{1}{\omega}\arcsin\frac{x}{\sqrt{\frac{2E}{m\omega^2}}}$ $\Rightarrow x=\sqrt{\frac{2E}{m\omega^2}}[\sin\omega(t-t_0)]$

弗恩曼

目录:

CHAPTER 1 分析力学引言 } 准备.
CHAPTER 2 达朗贝尔原理
CHAPTER 3 拉格朗日力学 例题.

理论.
CHAPTER 4 哈密顿力学
CHAPTER 5 哈密顿原理 理论分析.
CHAPTER 6 正则变换. 哈密顿-雅可比理论.

拉格朗日力学: ⟹ 例题
哈密顿力学: 例题不太如何 / 理论分析.
目标: 熟练掌握两大力学体系 { 拉 / 哈

弗恩曼